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Transparent end-to-end
Packet-switched Streaming Service (PSS);
Progressive Download and Dynamic
Adaptive Streaming over HTTP (3GP-DASH)
(3GPP TS 26.247 version 19.0.0 Release 19)



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Foreword

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The 3GPP transparent end-to-end packet-switched streaming service (PSS) specification consists of seven 3GPP TSs: 3GPP TS 22.233 [1], 3GPP TS 26.233 [2], 3GPP TS 26.234 [3], 3GPP TS 26.244 [4], 3GPP TS 26.245 [5], 3GPP TS 26.246 [6], and the present document.

The TS 22.233 contains the service requirements for the PSS. The TS 26.233 provides an overview of the PSS. The TS 26.234 provides the details of the protocols and codecs used by the PSS. The TS 26.244 defines the 3GPP file format (3GP) used by the PSS and MMS services. The TS 26.245 defines the Timed text format used by the PSS and MMS services. The TS 26.246 defines the 3GPP SMIL language profile. The present document defines Progressive Download and Dynamic Adaptive Streaming over HTTP.

The TS 26.244, TS 26.245 and TS 26.246 start with Release 6. Earlier releases of the 3GPP file format, the Timed text format and the 3GPP SMIL language profile can be found in TS 26.234.

The TS 26.247 starts with Release 10. Earlier releases of Progressive Download and Dynamic Adaptive Streaming over HTTP can be found in TS 26.234.

Introduction

Progressive Download and Dynamic Adaptive Streaming over HTTP (3GP-DASH) collects a set of technologies how progressive download and adaptive streaming of continuous media may be carried out exclusively over HTTP.

1 Scope

The present document specifies Progressive Download and Dynamic Adaptive Streaming over HTTP (3GP-DASH). This specification is part of Packet-switched Streaming Service (PSS) and 5G Media Streaming. HTTP-based progressive download and dynamic adaptive streaming had initially been separated from TS 26.234 to differentiate from RTP-based streaming that is maintained in TS 26.234. HTTP-based progressive download and dynamic adaptive streaming may be deployed independently from RTP-based PSS, for example by using standard HTTP/1.1 servers for hosting data formatted as defined in the present document, and in particular together with 5G Media Streaming.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document.*

[1] 3GPP TS 22.233: "Transparent End-to-End Packet-switched Streaming Service; Stage 1".

[2] 3GPP TS 26.233: "Transparent end-to-end Packet-switched Streaming service (PSS); General description".

[3] 3GPP TS 26.234: "Transparent end-to-end packet switched streaming service (PSS); Protocols and codecs".

[4] 3GPP TS 26.244: "Transparent end-to-end packet switched streaming service (PSS); 3GPP file format (3GP)".

[5] 3GPP TS 26.245: "Transparent end-to-end packet switched streaming service (PSS); Timed text format".

[6] 3GPP TS 26.246: "Transparent end-to-end packet switched streaming service (PSS); 3GPP SMIL Language Profile".

[7] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

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[13] IETF RFC 5646: "Tags for Identifying Languages", A. Phillips, M. Davis, September 2009.

[14] Void.

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[20] Void.

[21] Void.

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[30] IETF RFC 3023, "XML Media Types," January 2001.

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[32] 3GPP TS 29.213: "Policy and Charging Control signalling flows and Quality of Service (QoS) parameter mapping".

[33] 3GPP TS 29.214: "Policy and Charging Control over Rx reference point".

[34] Void.

[35] ITU-T Recommendation H.264 (04/2013): "Advanced video coding for generic audiovisual services".

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[37] IETF RFC 3629: "UTF-8, a transformation format of ISO 10646," November 2003.

[38] IETF RFC 4288: "Media Type Specifications and Registration Procedures," December 2005.

[39] IETF RFC 4648: "The Base16, Base32, and Base64 Data Encodings," October 2006.

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[44] ISO/IEC 23009-3 "Information technology — Dynamic adaptive streaming over HTTP (DASH) — Part 3: Implementation and Deployment Guidelines".

[45] ISO/IEC 23009-2 " Information technology — Dynamic adaptive streaming over HTTP (DASH) — Part 2: Conformance and Reference Software".

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[53] 3GPP TS 25.331 "Radio Resource Control (RRC); Protocol specification".

[54] ISO/IEC 23009-5:2017: "Information Technology — Dynamic adaptive streaming over HTTP (DASH) — Part 5: Server and network assisted DASH (SAND)".

[55] 3GPP TR 26.957: "Study on Server And Network-assisted DASH (SAND) for 3GPP Multimedia Services".

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[58] 3GPP TS 26.116: "Television (TV) over 3GPP services; Video profiles".

[59] 3GPP TS 36.331: "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification".

[60] 3GPP TS 26.347: " Multimedia Broadcast/Multicast Service (MBMS); Application Programming Interface and URL".

[61] 3GPP TS 27.007: " Technical Specification Group Core Network and Terminals; AT command set for User Equipment (UE)".

[62] DASH Industry Forum: "DASH Player’s Application Events and Timed Metadata Processing Model and APIs", see under https://dashif.org/guidelines/.

[63] 3GPP TS 28.405; "Management of Quality of Experience (QoE) measurement collection; Control and configuration"

[64] 3GPP TS 26.501: "5G Media Streaming (5GMS); General description and architecture".

[65] 3GPP TS 26.511: "5G Media Streaming (5GMS); Profiles, codecs and formats".

[66] 3GPP TS 26.512: " 5G Media Streaming (5GMS); Protocols".

[67] ISO/IEC 23000-19: "Information Technology Multimedia Application Format (MPEG-A) – Part 19: Common Media Application Format (CMAF) for segmented media".

[68] DASH Industry Forum – Guidelines for Interoperability: "DASH Low Latency Modes", see https://dashif.org/guidelines/.

[69] 3GPP TS 24.526: "User Equipment (UE) policies for 5G System (5GS); Stage 3".

[70] 3GPP TS 38.331: "NR; Radio Resource Control (RRC); Protocol specification".

[71] 3GPP TS 38.300: "NR; NR and NG-RAN Overall description; Stage-2".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [7], in ISO/IEC 23009-1 [43], clause 3.1, and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [7].

frame-packed stereoscopic 3D video: a video consisting of two views in which both views were packed into a single stream before compression.

multiview stereoscopic 3D video: a video consisting of two views packed into a single stream during compression.

interactivity event: Time interval, whose occurrence may be either pre-scheduled or unscheduled during the presentation of a main program, within which the user may engage with and/or consume interactive content.

rebuffering: a condition occurring in the media buffer when the buffer fill level is sufficiently depleted and buffer exhaust is imminent, typically resulting in suspended media playout.

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [7] and the following apply.

An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [7].

3GP	3GPP file format
3GP-DASH	3GPP Dynamic Adaptive Streaming over HTTP
5GMS	5G Media Streaming
AHS	Adaptive HTTP Streaming
A/V	Audio/Video
AVC	Advanced Video Coding
CMAF	Common Media Application Format
DANE	DASH-Aware Network Element
DM	Device Management
DNS	Domain Name System
DRM	Digital Rights Management
FQDN	Fully Qualified Domain Name
HSD	HTTP Streaming and Download
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IDR	Instantaneous Decoding Refresh
MPD	Media Presentation Description
MPEG-2 TS	Moving Picture Experts Group Transport Stream
MIME	Multipurpose Internet Mail Extensions
MOS	Mean Opinion Score
NAS	Non-Access Stratum
OMA	Open Mobile Alliance
PDCF	Packetized DRM Content Format
PER	Parameters Enhancing Reception
PSS	Packet-switched Streaming Service
QMC	QoE Measurement Collection
QoE	Quality-of-Experience
RFC	Request For Comments
RRC	Radio Resource Control
RTP	Real-time Transport Protocol

SAND	Server and Network Assisted DASH
SMIL	Synchronised Multimedia Integration Language
TLS	Transport Layer Security
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
URN	Uniform Resource Name
UTF-8	Unicode Transformation Format (the 8-bit form)
XML	eXtensible Markup Language

3.3 Conventions

The naming conventions of ISO/IEC 23009-1 [43], clause 3.3 apply in the present document.

4 Overview

The present document specifies Progressive Download and Dynamic Adaptive Streaming over HTTP (3GP-DASH) for continuous media. The features are separated from the umbrella specification TS 26.234 [3] to differentiate from RTP-based streaming that is specified and maintained in TS 26.234. Services relying exclusively on these features may be deployed independently from RTP-based PSS servers, for example by using standard HTTP/1.1 servers for hosting the services and in particular also for 5G Media Downlink Streaming when content is hosted on 5GMSd ASs as defined in the stage 2 specification in TS 26.501 [64] as well as in the stage 3 specifications in TS 26.511 [65] and TS 26.512 [66].

The specification covers the following aspects:

- System description (clause 5): describes the relationship to the PSS architecture and refines the architecture, interfaces and protocols that are defined in the present document.
- Progressive Download over HTTP (clause 6).
- 3GPP Dynamic Adaptive Streaming over HTTP (3GP-DASH) (clause 7)provides an overview of the architecture, the formats and the models that build the basis for 3GP-DASH. Also, 3GP-DASH Profiles provide identifiers and refers to a set of specific restrictions in this or other specifications.
- DASH: media presentation (clause 8) describes the data model of a Media Presentation. It also provides an overview on elements and attributes that may be used to describe components and properties of a media presentation in a Media Presentation Description (MPD).
- DASH: usage of the 3GP file format (clause 9) defines how segments can be formed based on the 3GP file format.
- Quality of Experience for Progressive Download and 3GP-DASH (clause 10).
- Server And Network-assisted DASH (SAND) (clause 13) introduces messages between DASH clients and network elements or between various network elements for the purpose to improve efficiency of streaming sessions by providing information about real-time operational characteristics of networks, servers, proxies, caches as well as DASH client's performance and status.
- Normative annexes for MPD schema (annex B), Descriptor Scheme Definitions (annex C), OMA DM QoE Management Object (annex F), File format extensions for 3GPP DASH support (annex G) and MIME type registration for MPD (annex H).
- Informative annexes for client behaviour (annex A), MPD examples (annex D), and mapping MPD structure and semantics to SMIL (annex E).

NOTE: Several of the annexes refer partially or exclusively to ISO/IEC 23009-1 [43].

5 System description

5.1 Overview

Progressive Download and Dynamic Adaptive Streaming over HTTP (3GP-DASH) enables to provide services to deliver continuous media content over Hypertext Transfer Protocol (HTTP) in a sense that all resources that compose the service are accessible through HTTP-URLs and the HTTP/1.1 protocol as specified in RFC 9112 [9] may be used to deliver the metadata and media data composing the service. This enables that standard HTTP servers and standard HTTP caches can be used for hosting and distributing continuous media content. Figure 1 shows the architecture for services using progressive download and figure 2 shows the architecture for services using 3GP-DASH.

The present document deals with the specification of interfaces between the client and the server. Specifically, it defines the formats that may be delivered exclusively over the HTTP interface to enable progressive download and streaming services.

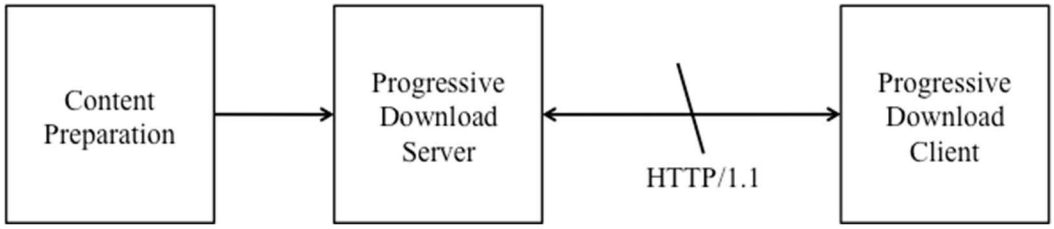


Figure 1: Architecture for Progressive Download over HTTP

Services using the features described in the present document may be deployed within PSS as specified in TS 26.233 [2] and TS 26.234 [3]. In this case the Progressive Download/3GP-DASH Server may be a sub-function of the PSS server and the Progressive Download/3GP-DASH client may be a sub-function of the PSS client.

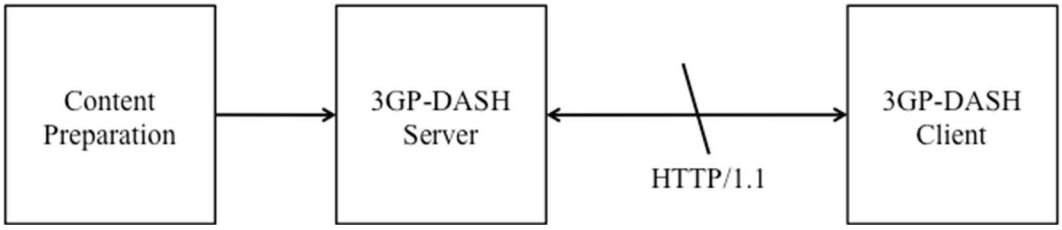


Figure 2: Architecture for 3GP-DASH

Services using the features defined in the present document may also be deployed independent of the PSS servers and clients. In this case the Progressive Download/3GP-DASH client shall support the formats and codecs according to the present document.

Access to services based on the features defined in the present document is introduced in clause 5.2.

The protocol support for services using the features defined in the present document is provided in clause 5.3.

Clients supporting progressive download-based services shall support the features and formats as specified in clause 6 of the present document.

Clients supporting 3GP-DASH shall support the features and formats as specified in clause 7 of the present document.

Clients supporting QoE Metrics and Reporting shall support the features as specified in clause 10 of the present document.

Clients supporting SAND functionality shall support the features as specified in clause 13 of the present document. Figure 2a shows the extended architecture for services for 3GP-DASH using SAND. The new interface between the DASH-Aware Network Element (DANE) and 3GP-DASH enables the SAND functionality.

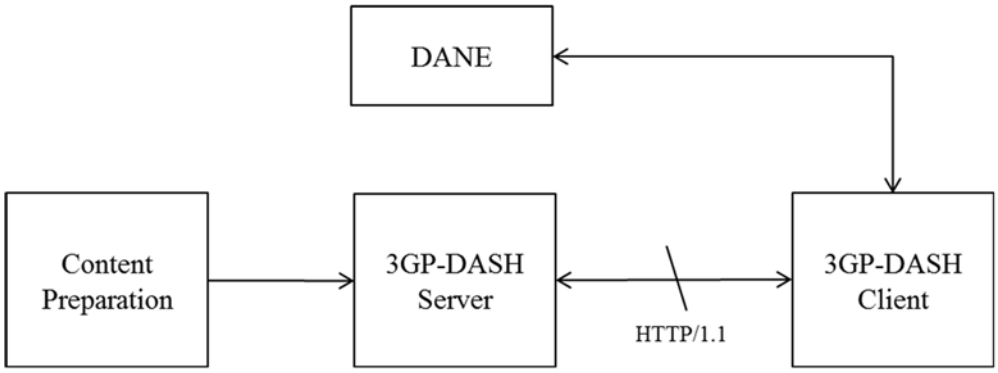


Figure 2a: Extended architecture for 3GP-DASH using SAND functionality

5.2 Service access

Service access refers to the method by which a Client initially accesses the service. Service access for services based in the specification can be achieved e.g. by a Media Presentation Description or a URL to the media file.

The service access URL can be made available to a client in many different ways. Clients supporting services based on the features in the present document shall be able to access services that are provided through an HTTP-URL. However, it is out of the scope of the present document to mandate any specific mechanism. A preferred way may be to embed URLs for service establishment within HTML pages.

5.3 Protocols

5.3.1 General

Progressive Download and 3GP-DASH clients shall comply with a *client* as specified in RFC 9112 [9]. The resource hosting the 3GP files and DASH Segments shall comply with a *server* as specified in RFC 9112 [9].

Progressive Download and 3GP-DASH clients should use the HTTP GET method or the HTTP partial GET method, as specified in RFC 9112 [9], clause 9.3, to access media offered at HTTP-URLs.

Figure 3 shows a protocol stack for services in the context of the present document. 3GP Files in progressive download as well as Segments based on the 3GPP File Format shall be accessible through HTTP.

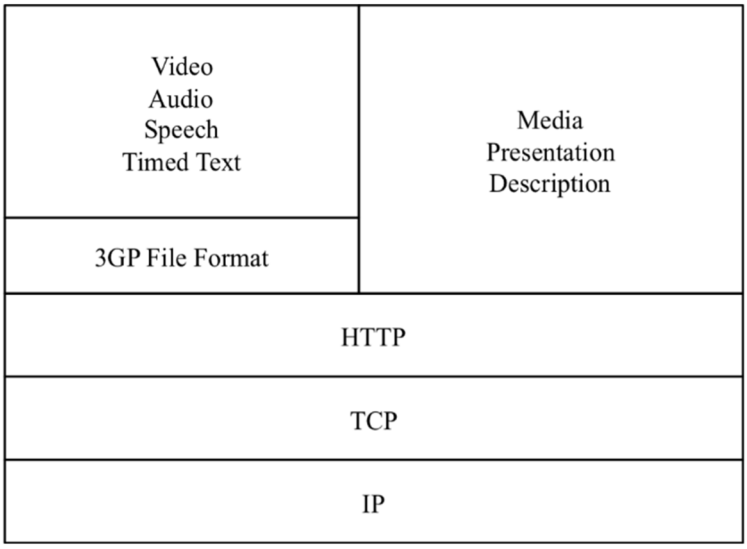


Figure 3: Overview of the protocols stack

Transport security in Progressive Download and Dynamic Adaptive Streaming over HTTP (3GP-DASH) is achieved using the HTTPS (Hypertext Transfer Protocol Secure) specified in RFC 2818 [12] and TLS as specified in TLS profile of annex E in TS 33.310 [23]. In case secure delivery is desired, HTTPS should be used to authenticate the server and to ensure secure transport of the content from server to client.

NOTE 1: The use of HTTPS for delivering Media Segments may inhibit caching at proxies and add overhead at the server and the client.

5.3.2 Partial file handling

In certain circumstances, for example in the case of MBMS download delivery of 3GP-DASH content, a file or segment may only be partially available on the HTTP server.

If the application, i.e. the 3GP file format client or the 3GP-DASH client supports the handling of partial files, it shall signal its capability using the partial-file-accept request defined in TS 26.346 [42], clause 7.9.2.1, i.e. using the Accept header `application/3gpp-partial` in combination with e.g. an HTTP GET request. If the application uses a partial-file-accept request, then the application shall also be able to handle the request response as defined in TS 26.346 [42], clause 7.9.2.2.

If the 3GP file format server or the 3GP-DASH server supports the handling of partial files, it should support the HTTP response format as defined in TS 26.346 [42], clause 7.9.2.2.

6 Progressive Download over HTTP

6.1 General

As an alternative to conventional streaming, a client may download, typically through HTTP, a media file that encapsulates continuous media and may play the media from the local storage. A PSS client shall support progressive download and playout of 3GP files [4] as specified in the remainder of this clause.

The media file encapsulating the continuous media is accessed directly by issuing one or more HTTP GET or partial GET requests to the referenced media file. An example of a valid URL is http://example.com/morning_news.3gp.

6.2 Progressive Download

Progressive Download uses normal HTTP download using HTTP GET or partial GET requests. The differences between regular download and Progressive Download are that 1) the content may be authored as progressively downloadable, and 2) the terminal recognises that the content is suitable for progressive download.

A client downloading continuous media may decide to start playout of the encapsulated media data before the download of the media file is completed.

6.3 3GPP file format profiles

The following profiles of the 3GPP file format in TS 26.244 [4] shall be supported by clients supporting Progressive Download over HTTP:

- Basic profile, and
- Progressive-download profile.

7 3GPP Dynamic Adaptive Streaming over HTTP

7.1 System description

The 3GPP Dynamic Adaptive Streaming over HTTP (3GP-DASH) specified in the present document provides streaming services over HTTP. 3GP-DASH is a set of profiles of ISO/IEC 23009-1 [43], also known as MPEG-DASH, with some extensions. For this it specifies XML and binary formats that enable delivering content from standard HTTP servers to an HTTP-Streaming client and enables caching content by standard HTTP caches.

The specification for 3GP-DASH primarily defines two formats:

1) The Media Presentation Description (MPD) describes a *Media Presentation*, i.e. a bounded or unbounded presentation of media content. In particular, it defines formats to announce resource identifiers for *Segments* and to provide the context for these identified resources within a Media Presentation. For 3GP-DASH, the resource identifiers are exclusively HTTP-URLs possibly combined with a byte range.

2) The Segment formats specify the formats of the entity body of the HTTP response to an HTTP GET request or an HTTP partial GET request with the indicated byte range through HTTP/1.1 as defined in RFC 9112 [9] to a resource identified in the MPD. Segments typically contain efficiently coded media data and metadata according to or aligned with common media formats.

The MPD provides sufficient information for a client to provide a streaming service to the user by accessing the Segments through the protocol specified in the scheme of the defined resources, in the context of the present document exclusively HTTP/1.1. Such a client is referred to as a 3GP-DASH client in the remainder of the present document. However, the present document does not provide a normative definition for such a client. An informative client model to illustrate the formats defined in the present document is provided in clause 7.2. An informative example client behaviour description is provided in annex A of ISO/IEC 23009-1 [43].

Figure 7-1 shows an architecture in which the formats defined in the present document are typically used. Boxes with solid lines indicate devices that are mentioned in the present document as they host or process the formats defined in the present document whereas dashed boxes are conceptual or transparent. This specification deals with the definition of formats that are accessible on the interface to the 3GP-DASH client, indicated by the solid lines. Any other formats or interfaces are not in scope of the present document. In the considered deployment scenario, it is assumed that the 3GP-DASH client has access to an MPD. The MPD provides sufficient information for the 3GP-DASH client to provide a streaming service to the user by requesting Segments from an HTTP server and demultiplexing, decoding and rendering the included media streams.

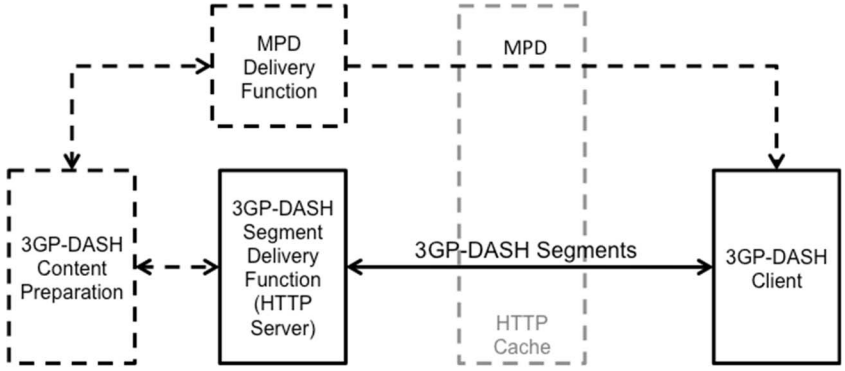


Figure 7-1: System Architecture for 3GP-DASH

The normative aspects of 3GP-DASH formats are defined by

- the profiles defined in clause 7.3.
- the DASH Media Presentation as defined in clause 8.
- the usage of the 3GPP file format for DASH as defined in clause 9.

The clauses mentioned above may refer to normative aspects in clause 10 on Quality of Experience, normative annexes B and H as well as ISO/IEC 23009-1 [43].

7.2 3GP-DASH client model

The design of the formats defined in the present document is based on the informative client model as shown in figure 7-2. The figure illustrates the logical components of a conceptual 3GP-DASH client model. In this figure the 3GP-DASH Access Engine receives the Media Presentation Description (MPD), constructs and issues requests and receives Segments or parts of Segments. In the context of this standard, the output of the DASH Access Engine consists of media in container formats according to the ISO/IEC 14496-12 ISO Base Media File Format [11] and specifically the 3GP file format [4]. In addition, timing information is provided that maps the internal timing of the media to the time line of the Media Presentation.

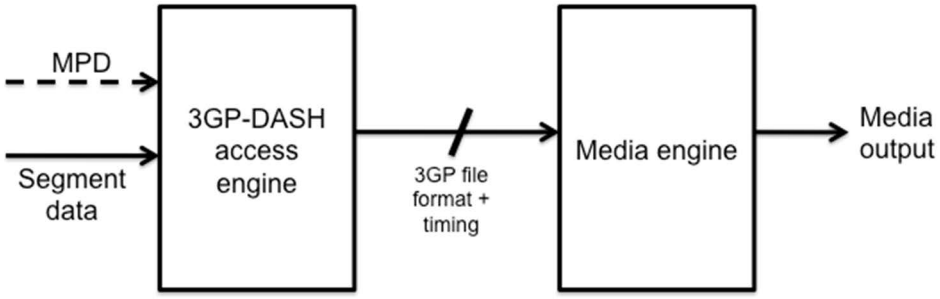


Figure 7-2: 3GP-DASH client model

7.3 3GP-DASH profiles

7.3.1 General

Profiles of 3GP-DASH are defined so as to enable interoperability and the signalling of the use of features etc. A profile refers to a set of specific restrictions. Those restrictions might be on features of the MPD as defined in clause 8 of the present document, Segment formats as for example defined in clause 9 of the present document, usage of the network, codec(s) used, content protection formats, or on quantitative measures such as bit rates, segment lengths, screen size, and so on. Profiles defined in the present document define restrictions on features of the present document, but may additionally impose restrictions on other aspects of media delivery.

NOTE A profile can also be understood as permission for 3GP-DASH clients that only implement the features required by the profile to process the Media Presentation. However, as 3GP-DASH client operation is not specified normatively, it is also unspecified how a 3GP-DASH client conforms to a particular profile. Hence, profiles merely specify restrictions on MPD and Segments rather than DASH client behaviour.

For details on the use of profiles, refer to ISO/IEC 23009-1 [43], clause 8.1.

7.3.2 3GPP Adaptive HTTP Streaming (Release-9 AHS)

Release-9 Adaptive HTTP Streaming as defined in TS 26.234 [3] Release-9, clause 12 is not a profile of the present document. Rel-9 AHS uses a different namespace "`urn:3GPP:ns:PSS:AdaptiveHTTPStreamingMPD:2009`" and a different MIME type signalling "`application/3gpp-ahs+xml`" for the MPD. However, a Media Presentation may be defined such that segments complying with the segment formats in TS 26.234 [3] Release-9, clause 12, also comply with segment formats for the present document.

7.3.3 3GP-DASH Release-10 Profile

7.3.3.1 Introduction

The 3GP-DASH Release-10 profile is identified by the URN "`urn:3GPP:PSS:profile:DASH10`".

This profile includes all features defined in the Release-10 version of the present document in clauses 7.3.6 (media codecs), 7.3.7 (content protection), 8 (Media Presentation Description), 9 (File Format) and 10 (QoE).The `@mimeType` attribute of each Representation shall be provided according to RFC4337. Additional parameters may be added according to RFC6381 [26].

7.3.4 3GP-DASH Release 11 multiview stereoscopic 3D video profile

The 3GP-DASH Release 11 multiview stereoscopic 3D video profile is identified by the URN "`urn:3GPP:PSS:profile:DASH11:MS3D`".

The `@mimeType` attribute of each Representation shall be provided according to RFC4337. Additional parameters may be added according to RFC6381 [26].

This profile includes all features defined in clauses 7.3.7, 8, 9 and 10.

Clients that support 3GP-DASH Release 11 multiview stereoscopic 3D video profile shall support multiview stereoscopic 3D video as specified in clause 7.4 of TS 26.234 [3]. For any other particular continuous media type, the corresponding media decoders are specified in TS 26.234 [3], clause 7.2 for speech, 7.3 for audio, 7.4 for video, 7.9 for timed text and 7.11 for timed graphics. Additionally, the following constraints apply for multiview stereoscopic 3D video bitstreams, if present in a media presentation:

The DASH multiple views scheme as defined in 5.8.5.6 of ISO/IEC 23009-1 [43] shall be used in the MPD.

One of the following shall be true:

- The base view of the stereoscopic multiview bitstream shall be a complementary representation and the non-base view of the bitstream shall be a dependent representation. The @dependencyId attribute as specified in 5.3.5.2 of ISO/IEC 23009-1 [43] shall be used to indicate the complementary and dependent representations.

- The base view and the non-base view of the stereoscopic multiview bitstream shall reside in the same representation. The SubRepresentation element shall be used for the representation, and the base view and the non-base view shall form separate sub-representations. The @level and @dependencyLevel attributes within the SubRepresentation element shall be used. The Level Assignment box shall be used. For each leaf segment index, that is, each Segment Index box that indexes only subsegments but not other Segment index boxes, there shall be exactly one Subsegment Index box.

7.3.5 3GP-DASH Release 11 frame-packed stereoscopic 3D video profile

The 3GP-DASH Release 11 frame-packed stereoscopic 3D video profile is identified by the URN "`urn:3GPP:PSS:profile:DASH11:FPS3D`".

The `@mimeType` attribute of each Representation shall be provided according to RFC4337. Additional parameters may be added according to RFC6381 [26].

This profile includes all features defined in clauses 7.3.7, 8, 9 and 10.

Clients that support 3GP-DASH Release 11 frame-packed stereoscopic 3D video profile shall support frame-packed stereoscopic 3D video as specified in clause 7.4 of TS 26.234 [3]. For any other particular continuous media type, the corresponding media decoders are specified in TS 26.234 [3], clause 7.2 for speech, 7.3 for audio, 7.4 for video, 7.9 for timed text and 7.11 for timed graphics. Additionally, the following constraints apply for frame-packed stereoscopic 3D video bitstreams, if present in a media presentation:

The FramePacking element as defined in clause 8.4.3.2 shall be used in the MPD.

7.3.6 Media codecs

For 3GP-DASH clients supporting a particular continuous media type, media decoders are specified in TS 26.234 [3], clause 7.2 for speech, 7.3 for audio, 7.4 for video, 7.9 for timed text and 7.11 for timed graphics.

NOTE: If 3GP-DASH formatted content is delivered outside the scope of the present document, additional constraints may apply on which media codecs can be used. For example, TS 26.346 [42] defines further restrictions when delivered over MBMS.

7.3.7 Content protection

3GP-DASH clients content protection may support OMA DRM 2.0 [15] or OMA DRM 2.1 [16]. Other content protection schemes may be supported. The ContentProtection element in the MPD should be used to convey content protection information.

When using OMA DRM V2.0 or OMA DRM V2.1 scheme for content protection, the non-streamable Packetized DRM Content Format (PDCF) shall be used. An OMA-DRM encrypted Representation shall include the brand "opf2". OMA-DRM [15] [16] defines the procedures for acquiring the Rights Object from the Rights Issuer to decrypt PDCF protected content. The scheme is identified by a **`ContentProtection`**`@schemeIdUri` set to "`urn:mpeg:dash:mp4protection`" and the **`ContentProtection`**`@value` shall include the version number; it starts with "`odkm`", which is the scheme_type contained in the Scheme Type Box of the PDCF file, followed by a ":" and the scheme_version from the Scheme Type Box of the PDCF file, encoded as up to 8 hexadecimal digits, where the leading '0's may be omitted. For example, for OMA DRM2.0 the value could be "odkm:200".

7.3.8 Partial file handling

3GP-DASH clients should support partial-file-accept requests and partial file responses as defined in clause 5.3.2. If 3GP-DASH clients support partial file handling they shall use partial-file-accept requests as defined in TS 26.346 [42], clause 7.9.2.1.

Without excluding other response options, as a response to a partial-file-accept request using a regular HTTP GET request a 3GP-DASH client may typically receive one of the following responses:

1) 200 OK with Content-Type set to the Media Type of the requested object
2) 200 OK with the Content-Type set to `application/3gpp-partial` and the message format according to the definition in clause 7.9.2.2 of TS 26.346 [42].
3) `416 Requested Range Not Satisfiable` with the additional information according to the definition in clause 7.9.2.2 of TS 26.346 [42].
4) `404 Not Found`

If the 3GP-DASH server supports the handling of partial files, then it should implement the HTTP response format as defined in TS 26.346 [42], clause 7.9.2.2. Consequently, a 3GP-DASH client may receive a response indicating 404 Not Found along with the `Content-Type` header set to '`application/3gpp-partial`' as indication of partial file availability at the server.

Case 1 is the regular response.

Guidelines for handling request responses according to case 4 from above are provided in clause A.7.

Guidelines for handling request responses 2 and 3 from above are provided in clause A.9.

7.3.9 3GP-DASH enhanced interoperability point

The 3GP-DASH Enhanced interoperability point (IOP) is identified by the URN "`urn:3GPP:PSS:iop:DASH-enhanced`".

This interoperability point includes all features defined in the Release-13 version of the present document in clauses 7.3.6 (media codecs), 7.3.7 (content protection), 8 (Media Presentation Description), 9 (File Format), 10 (QoE), 11 (simple live) and 12 (ad insertion).

A DASH client conforms to the IOP by supporting at least the following features:

- All DASH-related features as defined in clause.8 of this document.
- The file format related aspects defined in clause 9 of this document.
- The QoE related aspects defined in clause 10 of this document.
- The requirements and guidelines in clause 11 for simple live operation.

- The requirements and guidelines in clause 12 for server-based ad insertion.
- The following additional requirements:
  - Segment formats are based on ISO BMFF with fragmented movie files, i.e. (Sub)Segments are encoded as movie fragments containing a track fragment as defined in ISO/IEC 14496-12, plus the following constraints to make each movie fragment independently decodable:
    - Default parameters and flags shall be stored in movie fragments ('`tfhd`' or '`trun`' box) and not track headers ('`trex`' box)
    - The '`moof`' boxes shall not use external data references, the flag '`default-base-is-moof`' shall also be set (aka movie-fragment relative addressing) and `data-offset` shall be used, i.e. `base-data-offset-present` shall not be used (follows ISO/IEC 23009-1).
  - Withing each Adaptation Set the following applies
    - Fragmented movie files are used for encapsulation of media data
    - (Sub)Segments are aligned to enable seamless switching
  - The following additional restrictions are applied.
    - IDR-like SAPs (i.e., SAPs type 2 or below) at the start of each (Sub)Segment for simple switching.
    - Segments should have almost equal duration.
    - only non-multiplexed Representations should be used, i.e. each Representation only contains a single media component.
  - Addressing schemes are restricted to
    - templates with number-based addressing
    - Subsegments with Segment Index. In this case either the `@indexRange` attribute shall be present or the **`RepresentationIndex`** element shall be present. Only a single `sidx` box shall be present.
  - For DRM purposes, suitable DRM may be used together with common encryption as defined in ISO/IEC 23001-7 [47]. If used, only the AES-128 CTR mode shall be used. No specific requirements on a specific DRM system are added.

Content shall only be authored claiming conformance to this IOP if such a client can properly play the content. In addition, the content shall follow the mandatory aspects and should take into account the recommendations and guidelines for content authoring documented in clauses 7.3.6 (media codecs), 7.3.7 (content protection), 8 (Media Presentation Description), 9 (File Format), 10 (QoE), 11 (simple live) and 12 (ad insertion).

The `@mimeType` attribute of each Representation shall be provided according to RFC4337. Additional parameters may be added according to RFC6381 [26].

7.3.10 DASH in TeleVision (TV) over 3GPP services

If used for TeleVision (TV) over 3GPP Services, 3GP-DASH clients shall support the DASH features defined in clause 5 of TS 26.116 [58].

7.3.11 5G Media Streaming DASH interoperability point

If DASH [43] is used in 5G Media Streaming as defined in TS 26.501 [64] and TS 26.512 [66], then the 5GMSd AS takes the role of a DASH Server and the Media Player in the 5GMSd Client takes the role of a DASH Client. A detailed definition of the Media Player is provided in TS 26.512 [66] separating the DASH Access Client and a CMAF-based playback platform.

This clause defines a 5G Media Streaming DASH Interoperability Point for the DASH Access Client, in particular the processing requirements for the MPD and Segment formats. An interoperability point following the requirements in this

clause is identified by the URN "`urn:3GPP:5GMS:iop:DASH`". This profile is targeted to support the playback of segmented media content according to CMAF as defined in ISO/IEC 23000-19 [66].

The requirements for playback of codecs and formats for a 5GMSd Client are documented in TS 26.511 [12].

The Media Presentation shall conform to a DASH profile for CMAF Content as defined in ISO/IEC 23009-1 [43] with the following additional restrictions and extensions:

- Exactly one of the following Segment and Subsegment Information Modes shall be used within one Subset of one Period:
  - The **`SegmentTemplate`** element with `@media` containing a `$Number$` template and `@duration` is present.
  - The **`SegmentTemplate`** element with `@media` containing a `$Number$` template and **`SegmentTimeline`** is present.
  - The **`SegmentTemplate`** element with `@media` containing a `$Time$` template and **`SegmentTimeline`** is present.
  - The **`SegmentBase`** element with the Segment Index signalling is present.

The following extensions may apply for the 5G Media Streaming DASH Interoperability Point:

- The DASH Media Presentation may conform to DASH-IF low latency Media Presentation with the `@profiles` parameter set to `'http://www.dashif.org/guidelines/low-latency-live-v5'` as defined in DASH-IF IOP Low-Latency Modes [68].
- The DASH Media Presentation may contain one or several **`ServiceDescription`** elements.
- The DASH Media Presentation may contain one or several **`Subset`** elements. If the value of the `@id` of the Subset is identical to the value of the `@id` of the **`ServiceDescription`** element, then this Subset defines a restriction of Adaptation Sets being available for playback in case this Service Description is selected.

The DASH Access Client shall support playback and handling of Media Presentations conforming to the 5G Media Streaming DASH Interoperability Point as defined in this clause. Specifically, this includes support for:

- The playback of CMAF Content and the DASH profiles for CMAF Content as defined in ISO/IEC 23009-1 [43] with the restrictions of Segment and Subsegment Information modes as documented above.
- The requirements of a Low-Latency client as defined in DASH-IF IOP Low-Latency Modes [68].

8 DASH: media presentation

8.1 Introduction

MPEG had initiated a standardization process to provide specifications to enable scalable and flexible video distribution that addresses fixed and mobile networks. The work had been in close coordination with a parallel effort in 3GPP such that the two standards are aligned for broad industry support across different access networks. 3GPP's Release-9 specification on Adaptive HTTP Streaming (AHS) [3], clause 12 completed in 2010 served as a baseline for MPEG's DASH [43] (MPEG-DASH) as well as for 3GP DASH specification in this document.

In addition to the format specification, MPEG provides additional supporting material as part of MPEG-DASH, namely:

- ISO/IEC 23009-2: Conformance and Reference software [44]
- ISO/IEC 23009-3: Implementation and Deployment Guidelines [45]

Due to the close coordination in the development, 3GP-DASH can be viewed as a set of profiles of MPEG-DASH, MPEG DASH for the segment format uses the ISO base media file format [7]. In 3GPP, compatibility is achieved with the 3GP file format [5]. Dynamic Adaptive Streaming over HTTP (DASH) [43] specifies XML and binary formats that

enable delivery of media content from standard HTTP servers to HTTP clients and enable caching of content by standard HTTP caches.

ISO/IEC 23009-1 [43] primarily defines two formats:

- The Media Presentation Description (MPD) describes a *Media Presentation*, i.e. a bounded or unbounded presentation of media content. In particular, it defines formats to announce resource identifiers for *Segments* and to provide the context for these identified resources within a Media Presentation. These resource identifiers are HTTP-URLs possibly combined with a byte range.

- The Segment formats specify the formats of the entity body of the HTTP response to an HTTP GET request or a partial HTTP GET with the indicated byte range using HTTP/1.1 as defined in RFC 9112 to a resource identified in the MPD. Segments typically contain efficiently coded media data and metadata conforming to or at least closely aligned with common media formats.

For more details on DASH System description, please refer to ISO/IEC 23009-1 [43], clause 4.1.

For more details on DASH Client model, please refer to ISO/IEC 23009-1 [43], clause 4.2.

For more details on DASH Data model overview, please refer to ISO/IEC 23009-1 [43], clause 4.3.

For more details on DASH Media Stream and Representation properties, please refer to ISO/IEC 23009-1 [43], clause 4.5.

The format of the Media Presentation Description in 3GP-DASH is defined in clause 8.2.

The format of the Segments in 3GP-DASH is defined in clause 9.

Figure 8.1: Void

8.2 Media Presentation Description

8.2.1 General

The Media Presentation Description shall conform to a DASH Media Presentation as defined in ISO/IEC 23009-1 [43], clause 5.2.Updates may also be done using MPD delta files as defined in clause 8.5.2. The MIME type of an MPD delta file is defined in annex H.2.

8.2.2 Schema and 3GPP extension

The overview of the XML schema of the MPD is provided in below. Specific types, elements and attributes are introduced in the remainder of this clause. The complete MPD schema is provided in annex B of the present document. In case of any inconsistencies the schema in annex B takes precedence over the XML-syntax snippets provided in this clause. For the normative schema refer to the schema in annex B.

The main schema is provided in table 8-1 with the namespace "`urn:mpeg:dash:schema:mpd:2011`". The 3GPP extension namespace is provided in table 8-2 with namespace "`urn:3GPP:ns:DASH:MPD-ext:2011`". An extension schema for 3GPP in the context of the specification is referred to as "3gpp-2011.xsd". Elements and attributes in the extension namespace are preceded with "`x3gpp:`" throughout the present document.

The MPD shall be authored such that, after unrecognized XML attributes or elements are removed, the result is a valid XML document formatted according to the XML schema provided in annex B and that complies with the present document. Namespaces may be used to extend functionalities. Therefore, all extended elements and attributes added to a **`Representation`** in particular shall be such that they can be safely ignored by 3GP-DASH clients.

NOTE 1: Based on this if DASH clients remove all XML attributes and elements from the MPD in the DASH namespace (`urn:mpeg:dash:schema:mpd:2011`) and in other namespaces that are not in the XML schema documented in annex B, the MPD results in a valid XML document which complies with the present document. The DASH client can use such a resulting MPD for presentation of a conforming Media Presentation.

Example for valid MPDs are provided in annex D.

Table 8-1: Overview of XML schema of the MPD

```
<?xml version="1.0"?>
<xs:schema targetNamespace="urn:mpeg:dash:schema:mpd:2011"
    attributeFormDefault="unqualified" elementFormDefault="qualified"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xlink="http://www.w3.org/1999/xlink"
    xmlns:x3gpp="urn:3GPP:ns:DASH:MPD-ext:2011"
    xmlns="urn:mpeg:dash:schema:mpd:2011">
    <xs:annotation>
        <xs:appinfo>Media Presentation Description</xs:appinfo>
    </xs:annotation>

    <xs:import namespace="http://www.w3.org/1999/xlink" schemaLocation="xlink.xsd"/>
    <xs:import namespace="urn:3GPP:ns:DASH:MPD-ext:2011" schemaLocation="3gpp-2011.xsd"/>

    <!-- MPD: main element -->
    <xs:element name="MPD" type="MPDtype"/>
    ...

</xs:schema>
```

Table 8-2: Overview of XML schema for 3GPP MPD extensions

```
<?xml version="1.0"?>
<xs:schema targetNamespace="urn:3GPP:ns:DASH:MPD-ext:2011"
    attributeFormDefault="unqualified" elementFormDefault="qualified"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns="urn:3GPP:ns:DASH:MPD-ext:2011">
    <xs:annotation>
        <xs:appinfo>Extensions to Media Presentation Description for 3GPP</xs:appinfo>
    </xs:annotation>

    ...

</xs:schema>
```

8.2.3 Void

8.2.4 Void

8.3 MPD assembly

3GP-DASH inherits the ability from ISO/IEC 23009-1 [43] to enable to distribute an MPD not as a fully contained document, but to assemble the document by dereferencing certain links by using subset of W3C XLINK [20].

An MPD in 3GP DASH may reference a *remote element entities* from within a local MPD as defined in ISO/IEC 23009-1 [43], clause 5.5.

8.4 Hierarchical data model

8.4.1 General

A Media Presentation is described in the **MPD** element that is contained in an MPD document formatted as defined in clause 8.2.

A Media Presentation consists of:

- A sequence of one or more Periods described in 8.4.2.
- Each Period contains one or more Adaptation Sets that itself contains one or more Representations as described in clause 8.4.3. clause 8.4.3 also defines media content components and Sub-Representations.
- Each Representation consists of one or more Segments. Segment Information is introduced in clause 8.4.4. Segments contain media data and/or metadata to access, decode and present the included media content.

Beyond the mandatory elements and attributes in ISO/IEC 23009-1 [43], the **MPD** element used in 3GP-DASH may contain the following attributes and elements:

- **MPD**@id
- **MPD**@type
- **MPD**@availabilityStartTime
- **MPD**@availabilityEndTime
- **MPD**@mediaPresentationDuration
- **MPD**@minimumUpdatePeriod
- **MPD**@timeShiftBufferDepth
- **MPD**@suggestedPresentationDelay
- **MPD**@maxSegmentDuration
- **MPD**@maxSubsegmentDuration
- **MPD.ProgramInformation**
- **MPD.UTCTiming**
- **MPD.BaseURL**
- **MPD.x3gpp:DeltaSupport**
- **MPD.Location**
- **MPD.Metrics**

Table 8-5: Void

Table 8-6: Void

8.4.2 Period

8.4.2.1 General

A Media Presentation consists of one or more Periods as defined in ISO/IEC 23009-1 [43], clause 5.3.2. A Period is defined by **Period** element in the **MPD** element.

Beyond the mandatory elements and attributes in ISO/IEC 23009-1 [43], the **Period** element used in 3GP-DASH may contain the following attributes and elements:

- **Period**@xlink:href
- **Period**@xlink:actuate
- **Period**@id
- **Period**@start
- **Period**@duration
- **Period**@bitstreamSwitching
- **Period.BaseURL**
- **Period.SegmentBase**
- **Period.SegmentList**

- **Period.SegmentTemplate**
- **Period.AdaptationSet**
- **Period.EventStream**
- **Period.AssetIdentifier**

Table 8-7: Void

Table 8-8: Void

8.4.2.2 Content offering with multiple Periods

NOTE: This text was adopted for Amd.3 of ISO/IEC 23009-1 [43], which is not yet published. It is expected that the below text will be replaced with a reference to ISO/IEC 23009-1 once Amd.3 is published.

Content with multiple Periods may be created for different reasons, for example:

- to enable splicing of content, for example for ad insertion,
- to provide a synchronization point to avoid drift in segment numbering,
- to remove or add certain Representations in an Adaptation Set,
- to remove or add certain Adaptation Sets,
- to add or remove content offering on certain CDNs,
- to enable signalling of shorter segments, if produced by the encoder

Periods provide opportunities for resync, for ad insertion, for adding and removing Representations, but if they are not used or only used for minor changes, then continuous playout of the client is expected.

In certain circumstances the Media Presentation author offers content in the next Period that is a continuation of the content in the previous Period, possibly in the immediately following Period or in a later Period. The latter case applies for example after an advertisement Period had been inserted. The content provider may provide period-continuous Adaptation Sets as follows:

- The Adaptation Set identifiers are the same across two Periods.
- The sum of the value of the `@presentationTimeOffset` and the presentation duration of all Representations in one Adaptation Set are identical to the value of the `@presentationTimeOffset` of the associated Adaptation Set in the next Period.
- If Representations in both Adaptation Sets have the same value for `@id`, then they should have functionally equivalent Initialization Segments, i.e. the Initialization Segment may be used to continue the play-out the Representation.

From a client perspective, at Period boundary typically no continuity in terms of content offering is ensured. The content may be offered with different codecs, language attributes, different protection and so on. The client should play the content continuously across Periods, but there may be implications in terms of implementation to provide fully continuous and seamless playout. It may be the case that at Period boundaries, the presentation engine needs to be reinitialised, for example due to changes in formats, codecs or other properties. This may result in a re-initialisation delay. Such a re-initialisation delay should be minimized.

If the client presents media components of a certain Adaptation Set in one Period, and if the following Period has an identical Adaptation Set identifier, then the client is suggested to continue playing the content in the associated Adaptation Set.

8.4.3 Adaptation Sets and Representations

8.4.3.1 Overview

Periods are further subdivided as follows:

- Each Period contains one or more groups. Groups consist of Adaptation Sets as described in clause 8.4.3.3.
- In case an Adaptation Set contains multiple media content components, then each media content component is described individually as defined in clause 8.4.3.6.
- Each Adaptation Set contains one or more Representations as described in clause 8.4.3.4.
- A Representation may contain one or more Sub-Representations as described in clause 8.4.3.5.
- Adaptation Sets, Representations and Sub-Representations share common attributes and elements that are described in clause 8.4.3.2.

8.4.3.2 Common attributes and elements

The elements **`AdaptationSet`**, **`Representation`** and **`SubRepresentation`** have assigned common attributes and elements.

Beyond the mandatory elements and attributes in ISO/IEC 23009-1 [43], the following common attributes and elements may be used in 3GP-DASH:

- @profiles
- @width
- @height
- @frameRate
- @audioSamplingRate
- @mimeType
- @codecs
- @maximumSAPPeriod
- @startWithSAP
- @maxPlayoutRate
- @codingDependency
- **FramePacking**
- **AudioChannelConfiguration**
- **ContentProtection**
- **EssentialProperty**
- **SupplementalProperty**
- **InbandEventStream**

Table 8-9: Void

Table 8-10: Void

8.4.3.3 Adaptation Set

An Adaptation Set is described by an **`AdaptationSet`** element. **`AdaptationSet`** elements are contained in a **`Period`** element. An Adaptation Set contains alternate Representations, i.e. only one Representation within an Adaptation Set is expected to be presented at a time. All Representations contained in one Adaptation Set represent the same media content components and therefore contain media streams that are considered to be perceptually equivalent.

The Adaptation Set shall conform to the definition in ISO/IEC 23009-1 [43], clause 5.3.3.

Beyond the mandatory elements and attributes in ISO/IEC 23009-1 [43], the **`AdaptationSet`** element used in 3GP-DASH may contain the following attributes and elements:

- @xlink:href
- @xlink:actuate
- @id
- @group
- ***CommonAttributesElements***
- @lang
- @contentType
- @minBandwidth
- @maxBandwidth
- @minWidth
- @maxWidth
- @minHeight
- @maxHeight
- @minFrameRate
- @maxFrameRate
- @segmentAlignment
- @bitStreamSwitching
- @subsegmentAlignment
- @subsegmentStartsWithSAP
- **Accessibility**
- **Role**
- **Rating**
- **Viewpoint**
- **ContentComponent**
- **BaseURL**
- **SegmentBase**

- **SegmentList**
- **SegmentTemplate**
- **Representation**

Table 8-11: Void

Table 8-12: Void

8.4.3.4 Representation

Representations are described by the **`Representation`** element. **`Representation`** elements are contained in an **`AdaptationSet`** element.

A Representation is one of the alternative choices of the complete set or subset of media content components comprising the media content during the defined Period.

A Representation shall conform to the definition in ISO/IEC 23009-1 [43], clause 5.3.4.

Beyond the mandatory elements and attributes in ISO/IEC 23009-1 [43], the **`Representation`** element used in 3GP-DASH may contain the following attributes and elements:

- Representation@qualityRanking
- Representation@mediaStreamStructureId
- Representation.CommonAttributesElements
- Representation.BaseURL
- Representation.SubRepresentation
- Representation.SegmentBase
- Representation.SegmentList
- Representation.SegmentTemplate

Table 8-13: Void

Table 8-14: Void

8.4.3.5 Sub-representation

Sub-representations are embedded in regular Representations and are described by the **`SubRepresentation`** element. **`SubRepresentation`** elements are contained in a **`Representation`** element.

The **`SubRepresentation`** element describes properties of one or several media content components that are embedded in the Representation. It may for example describe the exact properties of an embedded audio component (language, codec, etc.), an embedded sub-title (language) or it may describe some embedded lower quality video layer (e.g. some lower frame rate, etc.).

A Sub-representation shall conform to the definition in ISO/IEC 23009-1 [43], clause 5.3.5.

Beyond the mandatory elements and attributes in ISO/IEC 23009-1 [43], the **`SubRepresentation`** element used in 3GP-DASH may contain the following attributes and elements:

- SubRepresentation@level
- SubRepresentation@dependencyLevel
- SubRepresentation@bandwidth
- SubRepresentation@contentComponent

- *CommonAttributesElements*

Table 8-15: Void

Table 8-16: Void

8.4.3.6 Content Component

Each Adaptation Set contains one or more media content components. The properties of each media content component are described by a **`ContentComponent`** element or may be described directly on the **`AdaptationSet`** element if only one media content component is present in the Adaptation Set. **`ContentComponent`** elements are contained in an **`AdaptationSet`** element.

A Content Component shall conform to the definition in ISO/IEC 23009-1 [43], clause 5.3.6.

Beyond the mandatory elements and attributes in ISO/IEC 23009-1 [43], the **`ContentComponent`** element used in 3GP-DASH may contain the following attributes and elements:

- ContentComponent@id
- ContentComponent@lang
- ContentComponent@contentType
- ContentComponent.Accessibility
- ContentComponent.Role
- ContentComponent.Rating
- ContentComponent.Viewpoint

Table 8-17: Void

Table 8-18: Void

8.4.4 Segments and Segment Information

A Segment is the smallest addressable unit described by an MPD and has a defined format. Segment formats are defined in clause 9. This clause defines the MPD information for Segments.

Representations are assigned *Segment Information* through the presence of the elements **`BaseURL`**, **`SegmentBase`**, **`SegmentTemplate`** and/or **`SegmentList`**. The *Segment Information* provides information on the location, availability and properties of all Segments contained in one Representation. Specifically, information on the presence and location of Initialization, Media, Index and Bitstream Switching Segments is provided.

Segment Information shall be provided as defined in ISO/IEC 23009-1 [43], clause 5.3.9 using the elements **`BaseURL`**, **`SegmentBase`**, **`SegmentTemplate`** and/or **`SegmentList`**.

- The duration of Segments shall be described by the `@duration` attribute.
- If the **`SegmentTemplate`** is used, then only the following identifiers as defined in ISO/IEC 23009-1 [43], table 16 may be used: *$$*, *$RepresentationID$*, *$Number$*, and *$Bandwidth$*. The identifier *$Time$* shall not be used.

Table 8-19: Void

Table 8-20: Void

Table 8-21: Void

Table 8-22: Void

Table 8-23: Void

Table 8-24: Void

Table 8-25: Void

Table 8-26: Void

Table 8-27: Void

8.5 MPD update

8.5.1 General

If the **MPD**@type is set to 'dynamic', the MPD may be updated during the Media Presentation. Updates typically extend the accessible Segment list for each Representation, introduce a new Period, update Segment locations or terminate the Media Presentation. The MPD update functionality in 3GP-DASH shall use the MPD update functionality in ISO/IEC 23009-1 [43], clause 5.4.

8.5.2 Media Presentation Description delta

If the **x3gpp:DeltaSupport** element is present in the **MPD** element, the content provider indicates that MPD delta files, as defined in this clause, are supported on the server. The URI of the MPD delta is provided in **x3gpp:DeltaSupport** @sourceURL. The **x3gpp:DeltaSupport** @availabilityDuration element, if present, indicates that the MPD delta file referenced by the URI is available for at least the value of the @availabilityDuration attribute (after this time, the server may redirect the client to the full MPD). If **x3gpp:DeltaSupport** @availabilityDuration is not present, then no information is conveyed about the availability of the MPD delta. If a client request for an MPD delta file results in an error, the client should request a full MPD.

The semantics of the attributes within the **x3gpp:DeltaSupport** element are provided in table 8-28. The XML-syntax of **x3gpp:DeltaSupport** element is provided in table 8-29.

Table 8-28: Semantics of x3gpp:DeltaSupport element

Element or attribute name	Use	Description
x3gpp:DeltaSupport		If present, this element indicates that MPD delta files are supported by the server.
@sourceURL	M	The source string providing the URL of the MPD delta. The URL may be relative to any **BaseURL** on MPD level and reference resolution according to clause 8.2.3 shall be applied.
@availabilityDuration	O	When provided, indicates the duration that the server guarantees the availability of the MPD delta file referenced in @sourceURL after the MPD has been updated. After that the client may be redirected to the full MPD.
Legend: For attributes: M=Mandatory, O=Optional, OD=Optional with Default Value, CM=Conditionally Mandatory. For elements: <minOccurs>...<maxOccurs> (N=unbounded) Elements are **bold**; attributes are non-bold and preceded with an @.		

Table 8-29: XML-Syntax of `x3gpp:DeltaSupport` element

```
    <!--DeltaSupport for the MPD -->
    <xs:complexType name="DeltaSupportType">
        <xs:sequence>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute name="sourceURL" type="xs:anyURI" use="required"/>
        <xs:attribute name="availabilityDuration" type="xs:duration"/>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>
```

An MPD delta is a text file that shall include the delta between the MPD that references it and the latest provided MPD. Note that the value of `@sourceURL` in successive MPDs is necessarily different because it is impossible for the delta between two different MPDs and the most recent MPD to be the same.

The output format consists of one or more structures, each corresponding to a change. The changes are in decreasing line number order. The structure format looks like:

```
change-command
to-file-line
to-file-line...
.
```

There are three types of change commands `change-command`. Each consists of a line number or comma-separated range of lines in the first file and a single character indicating the kind of change to make. All line numbers are the original line numbers in the file. The types of change commands and the instructions are provided in table 8-30.

Table 8-30: Change commands and the instructions for delta MPD files

Change command	Instruction	Example
*l*a	Add text from the second file after line *l* in the first file.	'8a' means to add the following lines after line 8 of file 1
*r*c	Replace the lines in range *r* in the first file with the following lines. Like a combined add and delete, but more compact.	'5,7c' means change lines 5–7 of file 1 to read as the text file 2.
*r*d	Delete the lines in range *r* from the first file.	'5,7d' means delete lines 5–7 of file 1.
NOTE: This is the format supported by the GNU diff utilities, see http://www.gnu.org/software/diffutils/manual/#Detailed-ed		

Regardless of the presence of a **`x3gpp:DeltaSupport`** element, the full MPD shall always be available to clients for regular MPD updates as defined in clause 8.5.1. MPD Delta related procedures are optional at the client.

8.6 Additional Media Presentation Information

8.6.1 Introduction

The MPD, Periods, Adaptation Sets, Representations and Sub-representations may have assigned descriptors for describing the content or other elements in the MPD. This clause specifies this descriptive information.

8.6.2 Program Information

Descriptive information on the program may be provided for each period within the **`ProgramInformation`** element.

Program Information shall conform to the definition in ISO/IEC 23009-1 [43], clause 5.7.

Table 8-31: Void

Table 8-32: Void

8.6.3 Descriptors

The MPD may contain descriptors that are all in the same format as defined in this clause. The elements of type `DescriptorType` provide a flexible mechanism for DASH content authors to annotate and extend the **`MPD`**, **`Period`**, **`AdaptationSet`** and **`Representation`** elements.

Descriptors shall conform to the definition in ISO/IEC 23009-1 [43], clause 5.8.1.

The following descriptors may be used in 3GP-DASH:

- Content Protection as defined in ISO/IEC 23009-1 [43], clause 5.8.4.1
- Role as defined in ISO/IEC 23009-1 [43], clause 5.8.4.3. The DASH Role scheme as defined in ISO/IEC 23009-1 [43], clause 5.8.5.4 may be used.
- Rating as defined in ISO/IEC 23009-1 [43], clause 5.8.4.4
- ViewPoint as defined in ISO/IEC 23009-1 [43], clause 5.8.4.5
- Accessibility as defined in ISO/IEC 23009-1 [43], clause 5.8.4.2
- Audio Channel Configuration as defined in ISO/IEC 23009-1 [43], clause 5.8.4.7
- Frame Packing as defined in ISO/IEC 23009-1 [43], clause 5.8.4.6
- Essential Property as defined in ISO/IEC 23009-1 [43], clause 5.8.4.8
- Supplemental Property as defined in ISO/IEC 23009-1 [43], clause 5.8.4.9
- UTC Timing Descriptor as defined in ISO/IEC 23009-1 [43], clause 5.8.4.10. The DASH UTC Timing Schemes as defined in ISO/IEC 23009-1 [43], clause 5.8.5.10 may be used.

Table 8-33: Void

Table 8-34: Void

8.7 Base URL processing

The **`BaseURL`** element may be used to specify one or more common locations for Segments and other resources. Base URL Processing shall conform to ISO/IEC 23009-1 [43], clause 5.6.

Table 8-35: Void

Table 8-36: Void

8.8 Event Streams

Events may be signalled in the MPD or within a Representation in order to signal aperiodic information to the DASH client or to an application. Events are timed, i.e. each event starts at a specific media presentation time and typically has a duration. Events may include DASH specific signalling or application-specific events. In the latter case, a proper scheme identifier identifies the application such that the DASH client can forward the event to the proper application.

Events of the same type are clustered in Event Streams. This enables a DASH client to subscribe to an Event Stream of interest and ignore Event Streams that are of no relevance or interest.

Two ways of signalling events are provided, namely:

- Events signalled in the MPD as defined in 5.10.2 of ISO/IEC 23009-1 [34]. Event streams for MPD events are signalled by **`Period.EventStream`** element.
- Events signalled in band of the Representation as defined in clause 5.10.3 of ISO/IEC 23009-1 [34]. In-band event streams are signalled by the element **`InbandEventStream`** for a specific Representation or all Representations in an Adaptation Set.

This specification does not provide any specific information on how to use Event Streams. It is up to the application that employs DASH formats to instantiate the description elements with appropriate scheme information.

If the DASH client is interested in a specific Event Stream, it is expected to properly extract each event from an event stream and pass it to the application.

9 DASH: usage of 3GPP file format

9.1 Introduction

3GPP Dynamic Adaptive Streaming over HTTP uses many elements of fragmented 3GP files to define the Segment formats. This provides Segments according to the requirements defined in clause 8.4.4.1 and enables reuse of existing content, easy encoding and recording, etc. This clause introduces how to use the 3GPP file format as specified in TS 26.244 [4] for DASH Segment formats.

9.2 Segment types and formats

9.2.1 Introduction

3GP-DASH defines a Segment format that is used in the delivery of media data over HTTP. A Segment shall contain one or more boxes in accordance with the boxed structure of the ISO Base Media File Format [11].

For 3GP-DASH the following applies:

- In all cases for which a Representation contains more than one Media Segment, the following applies:
  - The Initialization Segment as defined in clause 9.2.2 shall be present. The Initialization Segment shall be available for the 3GP-DASH client before any Media Segment is processed within the Representation.
  - Media Segments shall not be self-initializing. The Media Segment format is defined in clause 9.2.3.
- In case a Representation contains only a single Media Segment, then either one of the following two options is used:
  1) An Initialization Segment as defined in clause 9.2.2 and one Media Segment as defined in clause 9.2.3.
  2) One Self-Initializing Media Segment as defined in clause 9.2.4.

9.2.2 Initialization Segment

The Initialization Segment is conformant with the 3GPP file format, adaptive streaming profile and shall carry "`3gh9`" as compatibility brand.

The Initialization Segment shall conform to the Initialization Segment in ISO/IEC 23009-1 [43], clause 6.3.3.

9.2.3 Media Segment

9.2.3.1 General

A Media Segment contains and encapsulates media streams that are either described within this Media Segment or described by the Initialization Segment of this Representation or both.

In addition, a Media Segment shall conform to the general format type in ISO/IEC 23009-1 [43], clause 6.2.3.1.

9.2.3.2 Subsegments and Segment Index

Media Segments may contain multiple Subsegments documented by the Segment Index.

Subsegments and Segment Index shall follow the definition in ISO/IEC 23009-1 [43], clause 6.2.3.2.

9.2.3.3 Subsegment Index

Media Subsegments may be indexed further to enable accessing different levels of Subsegments in a Media Subsegment. This Subsegment Index may also be provided in separate Index Segments together with the Segment Index.

Subsegment Index shall follow the definition in ISO/IEC 23009-1 [43], clause 6.2.3.3.

9.2.3.4 3GP-DASH Media Segment format

A Media Segment conforming to the Media Segment Format for 3GP DASH shall carry '`3gmA`' as a compatible brand and shall conform to the General format type in ISO/IEC 23009-1 [43], clause 6.3.4.2.

9.2.4 Self-initializing Media Segment

A Self-Initializing Media Segment shall conform to the Self-Initializing Media Segment in ISO/IEC 23009-1 [43], clause 6.3.5.1.

9.2.5 Media Stream and Segment properties

9.2.5.1 Media Stream Access Points

To be able to access a Representation, each of the media streams that are contained in the Representation requires Media Stream Access Points (SAPs). For the definition refer to ISO/IEC 23009-1 [43], clause 4.5.1.

9.2.5.2 Non-overlapping Segments and Subsegments

The property of "non-overlapping" Segments (respectively Subsegments) is used to define the terms Segment alignment and Subsegment alignment. The definition is provided in ISO/IEC 23009-1 [43], clause 4.5.1.

9.2.5.3 Bitstream concatenation

The definition for bitstream concatenation is provided in ISO/IEC 23009-1 [43], clause 4.5.2.

9.3 Usage on server and client

3GP-DASH uses 3GP files according to the 3GP Adaptive-Streaming profile as specified in TS 26.244 [4]. Content may be prepared as 3GP files according to the 3GP Adaptive-Streaming profile. Initialization Segments and Media Segments may be generated by segmenting such 3GP files. Segment Index '`sidx`' boxes may be pre-contained in 3GP files or may be generated during the segmentation process. Clients may store a concatenation of a received Initialization Segment and a sequence of Media Segments from the same Representation to create a compliant 3GP file according to the Adaptive Streaming profile without accessing any media samples.

NOTE: As specified in TS 26.244, the MPD may be linked or embedded in the '`meta`' box of the '`moov`' box. This enables clients to access the MPD from a 3GP file that was made available from other means than 3GP-DASH (e.g. progressive download).

9.4 Segment properties with MPD constraints

9.4.1 General

9.4.1.1 Introduction

The content, especially the Segments across Representations at the same media time may have been prepared in a joint or at least coordinated manner. To expose these properties to the client, certain flags in the MPD can be set to true to indicate such coordinated content preparation. Clients consuming 3GP-DASH formatted media presentations may benefit from properly authored content when switching between or presenting Representations.

9.4.1.2 Media Presentation timeline

One of the key features in DASH is that encoded versions of different media components share a common timeline. The presentation time of access unit within the media content is mapped to the global common presentation timeline for synchronization of different media components and to enable seamless switching of different coded versions of the same media components.

The requirements and recommendations in ISO/IEC 23009-1 [43], clause 7.2.1 and clause 7.3.2 shall apply for 3GP-DASH.

9.4.1.3 Segment Index

If a Segment Index is present in a Media Segment of one Representation within an Adaptation Set, then the constraints in ISO/IEC 23009-1 [43], clause 7.2.2 shall apply for 3GP-DASH.

9.4.2 Segment alignment

No additional requirements beyond those stated in clause 8.4.3.3 are defined.

9.4.3 Bitstream switching

If the `@bitstreamSwitching` is set to '`true`' for a set of Representations within an Adaptation Set, the conditions in ISO/IEC 23009-1 [43], clause 7.3.3.2 shall apply.

9.4.4 Sub-representation

If a **`SubRepresentation`** element is present in a Representation in the MPD and the **`SubRepresentation`** `@level` is present, then the conditions in ISO/IEC 23009-1 [43], clause 7.3.4 shall apply.

10 QoE for Progressive Download and DASH

10.1 General

A progressive download or 3GP-DASH client supporting Quality of Experience (QoE) shall report QoE metrics according to the QoE configuration. QoE reporting is optional, but if a 3GP-DASH client reports DASH metrics, it shall report all requested metrics.

The quality metrics are defined in clause 10.2.

The quality metrics applicable for progressive download are specified in clause 10.3. In this case the activation and configuration of QoE reporting framework is achieved by a corresponding OMA DM QoE Management Object as specified in annex F, or by the QMC functionality as specified in annex L.

The quality metrics for DASH are specified in clause 10.4. In this case, QoE reporting may be triggered using the MPD (i.e. when the **`Metrics`** element is present in the MPD) or using OMA DM QoE Management Object as specified in Annex F, or by the QMC functionality as specified in annex L. When QoE reporting is triggered via any of the above methods, the 3GP-DASH client is expected to collect quality metrics according to the QoE configuration. When using the MPD or the QMC functionality, the Quality Reporting scheme as defined in clause 10.5 may be used.

The QoE configuration shall only be evaluated by the client at the start of a QoE measurement and reporting session ("QoE session") associated with a streaming session. This includes evaluation of any filtering criteria such as by geographical area. Client evaluation of all measurement and reporting criteria for an ongoing QoE session shall be unaffected by any QoE configuration changes received during that session – i.e., any changes to the QoE configuration shall only affect QoE sessions started after these configuration changes have been received.

The quality metric reporting protocol is defined in clause 10.6. This protocol shall be used when QoE reporting is triggered via the MPD or OMA DM QoE Management Object. When QoE reporting is triggered via the QMC functionality, the reporting is specified in annex L.

The usage of ITU-T P.1203 [49] Audio/Video Mean Opinion Score (A/V MOS) estimation is defined in annex K.

10.2 QoE metric definitions

10.2.1 Introduction

This clause provides the general QoE metric definitions and measurement framework.

The semantics are defined using an abstract syntax. clause 10.6 provides a mapping to an XML schema. Items in this abstract syntax have one of the following primitive types (`Integer`, `Real`, `Boolean`, `Enum`, `String`) or one of the following compound types:

- `Objects`: an unordered sequence of (`key`, `value`) pairs, where the key always has string type and is unique within the sequence.
- `List`: a ordered list of items.
- `Set`: an unordered set of items.

Additionally, there are two kinds of timestamp defined, i.e. *real time* (wall-clock time) and *media time*.

10.2.2 Void

This clause is void.

Table 25: Void

10.2.3 Representation switch events

Clause D.4.4 in ISO/IEC 23009-1 [43] defines the metrics for representation switch events.

Table 26: Void

10.2.4 Average throughput

This metric in table 27 indicates the average throughput that is observed by the client during the measurement interval.

Table 27: Average throughput

Key		Type	Description
AvgThroughput		Object	Average throughput that is observed by the client during the measurement interval.
	numbytes	Integer	The total number of the content bytes, i.e. the total number of bytes in the body of the HTTP responses, received during the measurement interval.
	activitytime	Integer	The activity time during the measurement interval in milliseconds. The activity time during the measurement interval is the time during which at least one GET request is still not completed (i.e. excluding inactivity time during the measurement interval).
	t	Real Time	The real time of the start of the measurement interval.
	duration	Integer	The time in milliseconds of the measurement interval.
	accessbearer	String	Access bearer for the TCP connection for which the average throughput is reported.
	inactivitytype	Enum	Type of the inactivity, if known and consistent throughout the reporting period: User request (e.g. pause) Client measure to control the buffer Error case

If the client requests the media Segments from the server separately over multiple non-competing parallel TCP connections established over separate access network bearers named as `accessbearer`, then the average throughput values should be reported as a list of events with average throughput for each access network and associated access network bearer information reported separately, following the same guidelines as described above.

10.2.5 Initial playout delay

This metric in table 28 signals the initial playout delay at the start of the streaming of the presentation.

The metric is only logged at the time point when the playout of streaming video begins.

Table 28: Initial playout delay

Key	Type	Description
InitialPlayoutDelay	Integer	The initial playout delay is measured as the time in milliseconds from the fetch of the first media Segment (or sub-segment) and the time at which media is retrieved from the client buffer.

10.2.6 Buffer level

Clause D.4.5 in ISO/IEC 23009-1 [43] defines the metrics for buffer level status events.

Table 29: Void

10.2.7 Playlist

Decoded samples are generally rendered in presentation time sequence, each at or close to its specified presentation time. A compact representation of the information flow can thus be constructed from a list of time periods during which samples of a single representation were continuously rendered, such that each was presented at its specified presentation time to some specific level of accuracy (e.g. +/-10 ms).

Such a sequence of periods of continuous delivery is started by a user action that requests playout to begin at a specified media time (this could be a "play", "seek" or "resume" action) and continues until playout stops either due to a user action, the end of the content, or a permanent failure.

Table 30 defines the playlist event metric.

Table 30: Playlist

Key	Type	Description
PlayList	List	A list of playback periods. A playback period is the time interval between a user action and whichever occurs soonest of the next user action, the end of playback or a failure that stops playback.
Entry	Object	A record of a single playback period.
start	Real Time	Timestamp of the user action that starts the playback period.
mstart	Media Time	The presentation time at which playout was requested by the user action.
starttype	Enum	Type of user action which triggered playout - New playout request (e.g. initial playout or seeking) - Resume from pause - Other user request (e.g. user-requested quality change) - Start of a metrics collection period (hence earlier entries in the play list not collected)
Trace	List	List of periods of continuous rendering of decoded samples.
Traceentry	Objects	Single entry in the list.
representationid	String	The value of Representation@id from which the samples were taken. This is an optional parameter and should not be reported in case of progressive download.
subreplevel	Integer	If not present, this metric concerns the Representation as a whole. If present, subreplevel indicates the greatest value of any SubRepresentation@level being rendered. This is an optional parameter and should not be reported in case of progressive download.
start	Real Time	The time at which the first sample was rendered.
sstart	Media Time	The presentation time of the first sample rendered.
duration	Integer	The time in milliseconds of the duration of the continuously presented samples (which is the same in real time and media time). "Continuously presented" means that the media clock continued to advance at the playout speed throughout the interval.
playbackspeed	Real	The playback speed relative to normal playback speed (i.e., normal forward playback speed is 1.0).
stopreason	Enum	The reason why continuous presentation of this representation was stopped. Either: - representation switch (not relevant in case of progressive download) - rebuffering - user request - end of period - end of content - end of a metrics collection period - failure - other
stopreasonoth	String	The stopreasonother attribute shall be included only if stopreason attribute is included and has the enum value *other*. In this release of the specification, the sender of this string shall set its value to one of the following: - switch from unicast to broadcast - switch from broadcast to unicast The receiver of this attribute shall ignore this attribute if its string is set to different value than the values listed above.

NOTE: The trace may include entries for different representations that overlap in time, because multiple representations are being rendered simultaneously, for example one audio and one video representation.

The playlist includes user actions about start/stop, but also other non-user actions such as adaptation and rebuffering. Thus, the playlist may be used to derive many other metrics, and an example calculation of a few stalling-related metrics is shown below.

Assume a user at wallclock time hh:mm:ss = 09:00:00 clicks to start a 60-second video with the following playout characteristics:

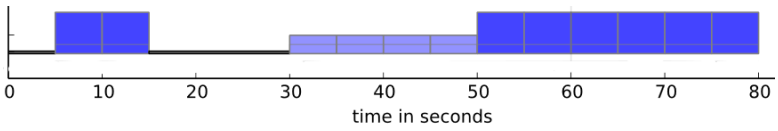


Figure 10-1 Example for Playout

- 5 seconds of initial stalling
- 10 seconds playing with representation 1
- 15 seconds of stalling
- 20 seconds playing with representation 2
- 30 seconds playing with representation 1

This would result in the following (somewhat simplified in this example) playout list being reported by the client:

Playlist
  Entry#1
    start = 09:00:00
    mstart = 00:00:00
    starttype = New playout request
    Trace
      Traceentry#1
        representationid = 1
        start = 09:00:05
        sstart = 00:00:00
        duration = 10 seconds
        stopreason = rebuffering
      Traceentry#2
        representationid = 2
        start = 09:00:30
        sstart = 00:00:10
        duration = 20 seconds
        stopreason = representation switch
      Traceentry#3
        representationid = 1
        start = 09:00:50
        sstart = 00:00:30
        duration = 30 seconds
        stopreason = end of content

The number of stalling occurrences may be calculated by counting how many times a stop reason is specified as "rebuffering".

The time duration for a stalling event may be calculated based on the time difference between the end time of a trace entry with stopreason equal to "rebuffering", and the start time of the next trace entry. In the example above the stalling starts at "Traceentry#1, (start + duration)" = 09:00:05 + 10 secs = 09:00:15, and ends at "Traceentry#2, start" = 09:00:30. Thus the length of the stalling is 15 seconds.

10.2.8 MPD information

This metric can be used to report Representation information from the MPD, so that reporting servers without direct access to the MPD can understand the used media characteristics.

The metric is reported whenever the client sends any other quality metrics report containing references to a Representation which MPD information has still not been reported.

Table 31 defines the MPD information for quality reporting.

Table 31: MPD Information for quality reporting

Key		Type	Description
MPDInformation		Object	
	representationid	String	Value of **Representation**@id for the representation addressed by the QoE metrics report.
	subreplevel	Integer	If present, value of SubRepresentation@level for the subrepresentation addressed by the QoE metrics report. If not present, the QoE metrics report concerns the representation as a whole.
	Mpdinfo	RepresentationType	Provides the MPD information for the representation or subrepresentation identified by representationid and subreplevel, if present. The following attributes and elements shall be present within mpdinfo if they are present for the identified representation or subrepresentation and their values shall be identical to those presented in the MPD: @bandwidth, @qualityRanking, @width, @height, @mimeType, and @codecs.

10.2.9 Playout delay for media start-up

This metric in table 31a indicates the waiting time that the user experiences for media start-up.

The metric is only logged at the time point when the media start-up happens.

Table 31a: Playout delay for media start-up

Key	Type	Description
PlayoutDelayforMediaStartup	Integer	The playout delay for media start-up is measured as the time in milliseconds from the time instant of DASH player receives play-back-start trigger to the instant of media playout. - If the MPD has been delivered earlier before the user clicks, it may include the process time of MPD, the fetch time of some media segments which are required for media presentation, the process time of segments, and the time for media decode and render to the user. - If no MPD has been fetched earlier, it also needs to add the fetch time of MPD.

10.2.10 Device information

This metric contains information about the displayed video resolution as well as the physical screen characteristics. If the video is rendered in full-screen mode, the video resolution usually coincides with the characteristics of the full physical display. If the video is rendered in a smaller sub-window, the characteristics of the actual video window shown shall be logged.

If known by the DASH client, the physical screen width and the horizontal field-of-view shall also be logged.

The metric is logged at the start of each QoE reporting period, and whenever the characteristics changes during the session (for instance if the UE is rotated from horizontal to vertical orientation, or if the video sub-window size is changed).

Table 31b defines the device information metrics. If an individual metric cannot be logged, its value shall be set to 0 (zero).

Table 31b: Device information

Key			Type	Description
DeviceInformationList			List	A list of device information objects.
	Entry		Object	A single object containing new device information.
		start	Real-Time	Timestamp when the device information was logged.
		mstart	Media-Time	The presentation time at which the device information was logged.
		videoWidth	Integer	The width of the displayed video, in screen pixels (not encoded video pixels).
		videoHeight	Integer	The height of the displayed video, in screen pixels (not encoded video pixels).
		screenWidth	Integer	The total width of the screen, in screen pixels.
		screenHeight	Integer	The total height of the screen, in screen pixels.
		pixelWidth	Float	The width of a screen pixel, in mm.
		pixelHeight	Float	The height of a screen pixel, in mm.
		fieldOfView	Float	The actual or estimated horizontal angle subtended at the eye by the screen, measured in degrees.

10.3 Quality metrics for Progressive Download

The following metrics shall be supported by Progressive Download clients supporting the QoE reporting feature:

- Average Throughput (clause 10.2.4),
- Initial Playout Delay (clause 10.2.5),
- Buffer Level (clause 10.2.6),
- Play List (clause 10.2.7), and
- Device information (clause 10.2.10).

10.4 Quality metrics for DASH

The following metrics shall be supported by 3GP-DASH clients supporting the QoE reporting feature:

- List of Representation Switch Events (clause 10.2.3),
- Average Throughput (clause 10.2.4),
- Initial Playout Delay (clause 10.2.5),
- Buffer Level (clause 10.2.6),
- Play List (clause 10.2.7),
- MPD Information (clause 10.2.8),
- Playout Delay for Media Start-up (clause 10.2.9),
- Device information (clause 10.2.10).

The `@metrics` attribute contains a list of quality metric keys listing all metrics that the DASH shall collect and report.

The semantics of the attributes within the **`Metrics`** element are provided in table 32. The XML-syntax of a **`Metrics`** element is provided in table 33.

Table 32: Semantics of `Metrics` element

Element or attribute name	Use	Description
`Metrics`		DASH metric element
`@metrics`	M	This attribute lists all quality metrics (as a list of quality metric keys as defined in clause 10.2, separated by a whitespace) that the client shall report. Certain keys allow specifying a measurement interval or period over which a single value of the metric is derived and potentially also other parameters controlling the collection of the metrics. The parameters, if any, are included in parentheses after the key and their semantics are specified in clause 10.2 with the metric definition itself.
`Range`	0..N	When specified, it indicates the time period during which quality metric collection is requested. When not present, quality metric collection is requested for the whole duration of the content.
`@starttime`	O	When specified, it indicates the start time of the quality metric collection operation. When not present, quality metric collection is requested from the beginning of content consumption. For services with **`MPD`**`@type "Live"`, the start time of quality metric collection can be obtained in wallclock time by adding the value of this attribute indicated in media time to the value of the **`MPD`**`@availabilityStartTime` attribute. For services with **`MPD`**`@type "OnDemand"`, the start time is indicated in media time and is relative to the *PeriodStart* time of the first period in this MPD.
`@duration`	O	When specified, it indicates the duration of the quality metric collection period. The value of this attribute is expressed in media time.
`Reporting`	1...N	Descriptors that provide information about the requested Quality Reporting method and formats, and Auxiliary Reporting method and format. See clause 10.5 for the 3GP-DASH quality reporting schemes, and clause 14 for the 3GP-DASH auxiliary reporting scheme.
Legend: For attributes: M=Mandatory, O=Optional, OD=Optional with Default Value, CM=Conditionally Mandatory. For elements: <minOccurs>...<maxOccurs> (N=unbounded) Elements are **bold**; attributes are non-bold and preceded with an @.		

Table 33: XML syntax of `Metrics` element

```
    <!-- QoE Collection and Reporting -->
    <xs:complexType name="MetricsType">
        <xs:sequence>
            <xs:element name="Reporting" type="DescriptorType" maxOccurs="unbounded"/>
            <xs:element name="Range" type="RangeType" minOccurs="0" maxOccurs="unbounded"/>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute name="metrics" type="xs:string" use="required"/>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>

    <xs:complexType name="RangeType">
        <xs:sequence>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute name="startTime" type="xs:duration" use="optional"/>
        <xs:attribute name="duration" type="xs:duration" use="required"/>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>
```

10.5 Quality reporting scheme for DASH

This clause specifies a 3GP-DASH quality reporting scheme.

The quality reporting scheme is signalled using in the **`Reporting`** element in the **`Metrics`** element. The URN to be used for the **`Reporting`**`@schemeIdUri` shall be "`urn:3GPP:ns:PSS:DASH:QM10`".

The reporting scheme shall use the quality reporting protocol defined in clause 10.6.

The semantics and XML syntax of the scheme information for the 3GP-DASH quality reporting scheme are specified in table 34 and table 35, respectively. The filename of this schema is "TS26247_QualityReportingScheme Information.xsd".

Table 34: Semantics of quality reporting scheme information

Element or attribute name	Use	Description
`@apn`	O	This attribute gives the access point that should be used for sending the QoE reports.
`@format`	O	This field gives the requested format for the reports. Possible formats are: "uncompressed" and "gzip".
`@samplepercentage`	O	Percentage of the clients that should report QoE. The client uses a random number generator with the given percentage to find out if the client should report or not.
`@reportingserver`	M	The reporting server URL to which the reports will be sent.
`@reportinginterval`	O	Indicates the time(s) reports should be sent. If not present, then the client should send a report after the streaming session has ended. If present, `@reportingInterval=n` indicates that the client should send a report every n-th second provided that new metrics information has become available since the previous report. For each report sent, only the newly collected information since the previous report shall be reported.
`LocationFilter`	0..1	When present, this element indicates the geographic area(s) or location(s) where quality metric collection is requested. When not present, quality metric collection is requested regardless of the device's location. The LocationFilter element comprises one or more instances of any combination of targeted cell-IDs, polygons and circular areas. Each cell-ID entry in LocationFilter is announced in cellList, and each polygon and circular area entry is announced in the polygonList or and circularAreaList elements, respectively.
`cellList`	0..N	This element specifies a list of cells identified by E-UTRAN-CGI or CGI.
`shape`		Geographic area comprising one or more instances of polygonList and/or circularAreaList elements.
`polygonList`	0..N	This element, when present, comprises a list of 'Polygon' shapes as defined by OMA MLP [51].
`@confLevel`	O	This attribute indicates the probability in percent that the DASH client is located in the corresponding polygon area. It is defined as 'lev_conf' by OMA MLP. If not present, it has default value of 60.
`circularAreaList`	0..N	This element, when present, comprises a list of 'CircularArea' shapes as defined by OMA MLP [51].
`@confLevel`	O	This attribute indicates the probability in percent that the DASH client is located in the corresponding circular area. It is defined as 'lev_conf' by OMA MLP. If not present, it has default value of 60.
`@sliceScope`	O	When present, this attribute indicates a list of network slices in which the collection and reporting of QoE metrics is requested. When not present, quality metric collection is requested for all network slices. The value is a list of S-NSSAIs.

@communicationServiceType	O	When present, this attribute indicates a list of communication service type(s) for which communication service type the collection and reporting of QoE metrics is requested, and shall contain one or more of the following values: - The value `unicast` refers to the common unicast communication type carrying media streaming services. - The value `mbsMulticast` refers to the *MBS Multicast communication service* per clause 21.1 of TS 38.300 [71]. - The value `mbsBroadcast` refers to the *MBS Broadcast communication service* per clause 21.1 of TS 38.300 [71]. - The value `all` refers to all communication service types. When absent, quality metrics collection is requested for all communication service types.
Legend: For attributes: M=Mandatory, O=Optional, CM=Conditionally Mandatory. For elements: <minOccurs>…<maxOccurs> (N=unbounded) Elements are **bold**; attributes are non-bold and preceded with an @		

Table 35: Syntax of quality reporting scheme information

```
<?xml version="1.0"?>
<xs:schema targetNamespace="urn:3GPP:ns:PSS:AdaptiveHTTPStreaming:2009:qm"
    attributeFormDefault="unqualified"
    elementFormDefault="qualified"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xlink="http://www.w3.org/1999/xlink"
    xmlns="urn:3GPP:ns:PSS:AdaptiveHTTPStreaming:2009:qm">

    <xs:annotation>
        <xs:appinfo>3GPP DASH Quality Reporting</xs:appinfo>
        <xs:documentation xml:lang="en">
            This Schema defines the quality reporting scheme information for 3GPP DASH.
        </xs:documentation>
    </xs:annotation>

    <xs:element name="ThreeGPQualityReporting" type="SimpleQualityReportingType"/>

    <xs:complexType name="SimpleQualityReportingType">
        <xs:sequence>
            <xs:element name="LocationFilter" type="LocationFilterType" minOccurs="0"/>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute name="apn" type="xs:string" use="optional"/>
        <xs:attribute name="format" type="FormatType" use="optional"/>
        <xs:attribute name="samplePercentage" type="xs:double" use="optional"/>
        <xs:attribute name="reportingServer" type="xs:anyURI" use="required"/>
        <xs:attribute name="reportingInterval" type="xs:unsignedInt" use="optional"/>
        <xs:attribute name="sliceScope" type="UnsignedIntVectorType" use="optional"/>
        <xs:attribute name="communicationServiceType" type="CommunicationServiceTypeType"
use="optional" default="all"/>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>

    <xs:simpleType name="FormatType">
        <xs:restriction base="xs:string">
            <xs:enumeration value="uncompressed" />
            <xs:enumeration value="gzip" />
        </xs:restriction>
    </xs:simpleType>

    <xs:simpleType name="CommunicationServiceTypeType">
        <xs:list>
            <xs:simpleType>
                <xs:restriction base="xs:NMTOKEN">
                    <xs:enumeration value="all" />
                    <xs:enumeration value="unicast" />
                    <xs:enumeration value="mbsBroadcast" />
                    <xs:enumeration value="mbsMulticast" />
                </xs:restriction>
            </xs:simpleType>
        </xs:list>
    </xs:simpleType>
```

```
    <xs:complexType name="LocationFilterType">
        <xs:sequence>
            <xs:element name="cellID" type="xs:unsignedLong" minOccurs="0" maxOccurs="unbounded"/>
            <xs:element name="shape" type="ShapeType" minOccurs="0"/>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>

    <xs:complexType name="ShapeType">
        <xs:sequence>
            <xs:element name="PolygonList" type="PolygonListType" minOccurs="0"/>
            <xs:element name="CircularAreaList" type="CircularAreaListType" minOccurs="0"/>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>

    <xs:complexType name="PolygonListType">
        <xs:annotation>
            <xs:documentation> see [OMA MLP] </xs:documentation>
        </xs:annotation>
        <xs:sequence>
            <xs:element name="Polygon" minOccurs="0" maxOccurs="unbounded"/>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute name="ConfLevel" type="xs:unsignedInt" use="optional"/>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>

    <xs:complexType name="CircularAreaListType">
        <xs:annotation>
            <xs:documentation> see [OMA MLP] </xs:documentation>
        </xs:annotation>
        <xs:sequence>
            <xs:element name="CircularArea" minOccurs="0" maxOccurs="unbounded"/>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute name="ConfLevel" type="xs:unsignedInt" use="optional"/>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>
    <xs:simpleType name="UnsignedIntVectorType">
        <xs:list itemType="xs:unsignedInt"/>
    </xs:simpleType>
</xs:schema>
```

10.6 Quality reporting protocol

10.6.1 General

The quality reporting protocol consists of:

- The XML-based report format defined in clause 10.6.2.
- The reporting protocol defined in clause 10.6.3.

The MIME type of an XML-formatted QoE report shall be "`application/3gpdash-qoe-report+xml`" as defined in annex J.

10.6.2 Report format

The QoE report is formatted as an XML document that complies with the XML schema in listing 10.6.2-1. The filename of this schema is "TS 26247_HSDReceptionReport.xsd".

Listing 10.6.2-1: QoE report XML schema

```
<?xml version="1.0"?>
<xs:schema version="TSG105-Rel18" xmlns:xs="http://www.w3.org/2001/XMLSchema"
    targetNamespace="urn:3gpp:metadata:2017:HSD:receptionreport"
    xmlns:sup="urn:3gpp:metadata:2016:PSS:SupplementQoEMetric"
    xmlns:rtc="urn:3gpp:metadata:2023:RTC:QoEMetrics"
    xmlns:src="urn:3gpp:metadata:2024:RTC:SR_MSEQoEMetrics"
    xmlns:evex="urn:3gpp:metadata:2024:PSS:SupplementalEventExposureReportingFields"
    xmlns:sv="urn:3gpp:metadata:2016:PSS:schemaVersion"
    xmlns="urn:3gpp:metadata:2017:HSD:receptionreport" elementFormDefault="qualified">

    <xs:import namespace="urn:3gpp:metadata:2016:PSS:SupplementQoEMetric"
schemaLocation="TS 26247_SupplementalQoEMetric.xsd"/>
    <xs:import namespace="urn:3gpp:metadata:2023:RTC:QoEMetrics"
schemaLocation="TS 26113_QoEMetrics.xsd"/>
    <xs:import namespace="urn:3gpp:metadata:2024:RTC:SR_MSEQoEMetrics"
schemaLocation="TS 26565_SR_MSEQoEMetrics.xsd"/>
    <xs:import namespace="urn:3gpp:metadata:2024:PSS:SupplementalEventExposureReportingFields"
schemaLocation="TS 26247_SupplementalEventExposureReporting.xsd"/>
    <xs:import namespace="urn:3gpp:metadata:2016:PSS:schemaVersion"
schemaLocation="TS 26247_SchemaVersion.xsd"/>

    <xs:element name="ReceptionReport" type="ReceptionReportType"/>

    <xs:complexType name="ReceptionReportType">
        <xs:sequence>
            <xs:element ref="sv:schemaVersion"/>
            <xs:element name="QoeReport" type="QoeReportType" minOccurs="0"
maxOccurs="unbounded"/>
            <xs:element ref="sv:delimiter"/>
            <xs:any namespace="##other" processContents="skip" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute name="contentURI" type="xs:anyURI" use="required"/>
        <xs:attribute name="clientID" type="xs:string" use="optional"/>
    </xs:complexType>

    <xs:complexType name="QoeReportType">
        <xs:sequence>
            <xs:element name="QoeMetric" type="QoeMetricType" minOccurs="1"
maxOccurs="unbounded"/>
            <xs:element ref="sup:supplementQoEMetric" minOccurs="0" maxOccurs="1"/>
            <xs:element ref="sv:delimiter"/>
            <xs:element ref="rtc:QoeMetric" minOccurs="0"/>
            <xs:element ref="sv:delimiter"/>
            <xs:element ref="src:QoeMetric" minOccurs="0"/>
            <xs:element ref="sv:delimiter"/>
            <xs:element ref="evex:SliceInfo" minOccurs="0" maxOccurs="1"/>
            <xs:element ref="evex:DNN" minOccurs="0" maxOccurs="1"/>
            <xs:element ref="evex:Locations" minOccurs="0" maxOccurs="1"/>
            <xs:element ref="sv:delimiter"/>
            <xs:any namespace="##other" processContents="skip" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute name="periodID" type="xs:string" use="required"/>
        <xs:attribute name="reportTime" type="xs:dateTime" use="required"/>
        <xs:attribute name="reportPeriod" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="qoeReferenceId" type="xs:hexBinary" use="optional"/>
        <xs:attribute name="recordingSessionId" type="xs:hexBinary" use="optional"/>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>
    <xs:complexType name="QoeMetricType">
        <xs:choice>
            <xs:element name="RepSwitchList" type="RepSwitchListType"/>
            <xs:element name="AvgThroughput" type="AvgThroughputType" maxOccurs="unbounded"/>
            <xs:element name="InitialPlayoutDelay" type="xs:unsignedInt"/>
            <xs:element name="BufferLevel" type="BufferLevelType"/>
            <xs:element name="PlayList" type="PlayListType"/>
            <xs:element name="MPDInformation" type="MpdInformationType" maxOccurs="unbounded"/>
            <xs:element name="PlayoutDelayforMediaStartup" type="xs:unsignedInt"/>
```

```
        </xs:choice>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:complexType name="RepSwitchListType">
        <xs:choice>
            <xs:element name="RepSwitchEvent" type="RepSwitchEventType" maxOccurs="unbounded"/>
        </xs:choice>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:complexType name="RepSwitchEventType">
        <xs:attribute name="to" type="xs:string" use="required"/>
        <xs:attribute name="mt" type="xs:duration" use="optional"/>
        <xs:attribute name="t" type="xs:dateTime" use="optional"/>
        <xs:attribute name="lto" type="xs:unsignedInt" use="optional"/>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:complexType name="AvgThroughputType">
        <xs:attribute name="numBytes" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="activityTime" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="t" type="xs:dateTime" use="required"/>
        <xs:attribute name="duration" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="accessbearer" type="xs:string" use="optional"/>
        <xs:attribute name="inactivityType" type="InactivityType" use="optional"/>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:simpleType name="InactivityType">
        <xs:restriction base="xs:string">
            <xs:enumeration value="Pause"/>
            <xs:enumeration value="BufferControl"/>
            <xs:enumeration value="Error"/>
        </xs:restriction>
    </xs:simpleType>
    <xs:complexType name="BufferLevelType">
        <xs:choice>
            <xs:element name="BufferLevelEntry" type="BufferLevelEntryType"
maxOccurs="unbounded"/>
        </xs:choice>

        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:complexType name="BufferLevelEntryType">
        <xs:attribute name="t" type="xs:dateTime" use="required"/>
        <xs:attribute name="level" type="xs:unsignedInt" use="required"/>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:complexType name="PlayListType">
        <xs:choice>
            <xs:element name="Trace" type="PlayListEntryType" maxOccurs="unbounded"/>
        </xs:choice>

        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:complexType name="PlayListEntryType">
        <xs:choice>
            <xs:element name="TraceEntry" type="PlayListTraceEntryType" maxOccurs="unbounded"/>
        </xs:choice>
        <xs:attribute name="start" type="xs:dateTime" use="required"/>
        <xs:attribute name="mstart" type="xs:duration" use="required"/>
        <xs:attribute name="startType" type="StartType" use="required"/>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:complexType name="PlayListTraceEntryType">
        <xs:attribute name="representationId" type="xs:string" use="optional"/>
        <xs:attribute name="subrepLevel" type="xs:unsignedInt" use="optional"/>
        <xs:attribute name="start" type="xs:dateTime" use="required"/>
        <xs:attribute name="sstart" type="xs:duration" use="required"/>
        <xs:attribute name="duration" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="playbackSpeed" type="xs:double" use="optional"/>
        <xs:attribute name="stopReason" type="StopReasonType" use="optional"/>
```

```
        <xs:attribute name="stopReasonOther" type="xs:string" use="optional"/>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:simpleType name="StartType">
        <xs:restriction base="xs:string">
            <xs:enumeration value="NewPlayoutRequest"/>
            <xs:enumeration value="Resume"/>
            <xs:enumeration value="OtherUserRequest"/>
            <xs:enumeration value="StartOfMetricsCollectionPeriod"/>
        </xs:restriction>
    </xs:simpleType>
    <xs:simpleType name="StopReasonType">
        <xs:restriction base="xs:string">
            <xs:enumeration value="RepresentationSwitch"/>
            <xs:enumeration value="Rebuffering"/>
            <xs:enumeration value="UserRequest"/>
            <xs:enumeration value="EndOfPeriod"/>
            <xs:enumeration value="EndOfContent"/>
            <xs:enumeration value="EndOfMetricsCollectionPeriod"/>
            <xs:enumeration value="Failure"/>
            <xs:enumeration value="Other"/>
        </xs:restriction>
    </xs:simpleType>

    <xs:complexType name="MpdInformationType">
        <xs:choice>
            <xs:element name="Mpdinfo" type="RepresentationType" maxOccurs="unbounded"/>
        </xs:choice>
        <xs:attribute name="representationId" type="xs:string" use="required"/>
        <xs:attribute name="subrepLevel" type="xs:unsignedInt" use="optional"/>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:complexType name="RepresentationType">
        <xs:attribute name="codecs" type="xs:string" use="required"/>
        <xs:attribute name="bandwidth" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="qualityRanking" type="xs:unsignedInt" use="optional"/>
        <xs:attribute name="frameRate" type="xs:double" use="optional"/>
        <xs:attribute name="width" type="xs:unsignedInt" use="optional"/>
        <xs:attribute name="height" type="xs:unsignedInt" use="optional"/>
        <xs:attribute name="mimeType" type="xs:string" use="required"/>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:simpleType name="DoubleVectorType">
        <xs:list itemType="xs:double"/>
    </xs:simpleType>

    <xs:simpleType name="StringVectorType">
        <xs:list itemType="xs:string"/>
    </xs:simpleType>

    <xs:simpleType name="UnsignedIntVectorType">
        <xs:list itemType="xs:unsignedInt"/>
    </xs:simpleType>

</xs:schema>
```

The schema in listing 10.6.2-2 is an extension to allow additional QoE metrics. The filename of this schema is "TS 26247_SupplementalQoEMetric.xsd".

Listing 10.6.2-2: Additional QoE metrics XML schema

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
    xmlns="urn:3gpp:metadata:2016:PSS:SupplementQoEMetric"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    targetNamespace="urn:3gpp:metadata:2016:PSS:SupplementQoEMetric"
    elementFormDefault="qualified">

    <xs:element name="supplementQoEMetric" type="SupplementQoEMetricType"/>

    <xs:complexType name="SupplementQoEMetricType">
        <xs:sequence>
            <xs:element name="deviceinformation" type="DeviceInformationType" minOccurs="0"/>
```

```
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
    </xs:complexType>

    <xs:complexType name="DeviceInformationType">
        <xs:choice>
            <xs:element name="Entry" type="DeviceInformationEntryType" maxOccurs="unbounded"/>
        </xs:choice>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:complexType name="DeviceInformationEntryType">
        <xs:attribute name="start" type="xs:dateTime" use="required"/>
        <xs:attribute name="mstart" type="xs:duration" use="required"/>
        <xs:attribute name="videoWidth" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="videoHeight" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="screenWidth" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="screenHeight" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="pixelWidth" type="xs:double" use="required"/>
        <xs:attribute name="pixelHeight" type="xs:double" use="required"/>
        <xs:attribute name="fieldOfView" type="xs:double" use="required"/>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>
</xs:schema>    <xs:complexType name="DeviceInformationEntryType">
        <xs:attribute name="start" type="xs:dateTime" use="required"/>
        <xs:attribute name="mstart" type="xs:duration" use="required"/>
        <xs:attribute name="videoWidth" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="videoHeight" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="screenWidth" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="screenHeight" type="xs:unsignedInt" use="required"/>
        <xs:attribute name="pixelWidth" type="xs:double" use="required"/>
        <xs:attribute name="pixelHeight" type="xs:double" use="required"/>
        <xs:attribute name="fieldOfView" type="xs:double" use="required"/>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>
</xs:schema>
```

The schema in listing 10.6.2-2a is an extension to allow additional fields to be included in QoE metrics reports to support event exposure. The filename of this schema is "TS 26247_SupplementalEventExposureReporting.xsd".

Listing 10.6.2-2a: Supplemental event exposure reporting fields XML schema

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema version="TSG104-Rel18"
    xmlns="urn:3gpp:metadata:2024:PSS:SupplementalEventExposureReportingFields"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    targetNamespace="urn:3gpp:metadata:2024:PSS:SupplementalEventExposureReportingFields"
    elementFormDefault="qualified">

    <!-- Slice identification -->
    <xs:simpleType name="SliceDifferentiatorType">
        <xs:restriction base="xs:string">
            <xs:pattern value="[A-Fa-f0-9]{6}"/>
        </xs:restriction>
    </xs:simpleType>

    <xs:complexType name="SliceInfoType">
        <xs:annotation>
            <xs:documentation>Information about a network slice.</xs:documentation>
        </xs:annotation>
        <xs:attribute name="sst" type="xs:unsignedByte" use="required">
            <xs:annotation>
                <xs:documentation>Slice/Service Type, expressed as an integer between 0 and 255
inclusive.</xs:documentation>
            </xs:annotation>
        </xs:attribute>
        <xs:attribute name="sd" type="SliceDifferentiatorType" use="optional">
            <xs:annotation>
                <xs:documentation>Slice Differentiator, expressed as a 6-digit hexadecimal
string.</xs:documentation>
            </xs:annotation>
        </xs:attribute>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:element name="SliceInfo" type="SliceInfoType"/>
```

```
    <!-- Data Network naming -->
    <xs:element name="DNN" type="xs:string" nillable="false"/>

    <!-- Location identification -->
    <xs:simpleType name="CGIType">
        <xs:restriction base="xs:string">
            <xs:pattern value="\d{2,3}-^\d{3}-[A-Fa-f0-9]{4}-[A-Fa-f0-9]{4}"/>
        </xs:restriction>
    </xs:simpleType>

    <xs:simpleType name="ECGIType">
        <xs:restriction base="xs:string">
            <xs:pattern value="\d{2,3}-^\d{3}$-[A-Fa-f0-9]{7}(-[A-Fa-f0-9]{11})+"/>
        </xs:restriction>
    </xs:simpleType>

    <xs:simpleType name="NCGIType">
        <xs:restriction base="xs:string">
            <xs:pattern value="\d{2,3}-^\d{3}-[A-Fa-f0-9]{9}(-[A-Fa-f0-9]{11})+"/>
        </xs:restriction>
    </xs:simpleType>

    <xs:complexType name="UserLocationsType">
        <xs:annotation>
            <xs:documentation>A set of UE locations.</xs:documentation>
        </xs:annotation>
        <xs:sequence minOccurs="1" maxOccurs="unbounded">
            <xs:choice>
                <xs:element name="CGI" type="CGIType">
                    <xs:annotation>
                        <xs:documentation>A string of four numeric codes separated by hyphen characters that
encode a Cell Global Identification per clause 4.3.1 of 3GPP TS 23.003:
-- Mobile Country Code (MCC) part of the PLMN ID, as defined in clause 9.3.3.5 of 3GPP TS 38.413, encoded using a
string of 2 or 3 decimal digits.
- Mobile Network Code (MNC) part of the PLMN ID, as defined in clause 9.3.3.5 of 3GPP TS 38.413, encoded using a
string of exactly 3 decimal digits.
- A Location Area Code (LAC) encoded using a string of 4 hexadecimal digits with the most significant nybble
appearing first, and padded with leading zeroes as necessary.
- A Cell Identification (CI) encoded using a string of 4 hexadecimal digits with the most significant nybble
appearing first, and padded with leading zeroes as necessary.</xs:documentation>
                    </xs:annotation>
                </xs:element>
                <xs:element name="ECGI" type="ECGIType">
                    <xs:annotation>
                        <xs:documentation>A string of four numeric codes separated by hyphen characters that
encode a E-UTRAN Cell Global Identification per clause 19.6 of 3GPP TS 23.003:
- Mobile Country Code (MCC) part of the PLMN ID, as defined in clause 9.3.3.5 of 3GPP TS 38.413, encoded using a
string of 2 or 3 decimal digits.
- Mobile Network Code (MNC) part of the PLMN ID, as defined in clause 9.3.3.5 of 3GPP TS 38.413, encoded using a
string of exactly 3 decimal digits.
- E-UTRAN Cell Identity (ECI), as specified in clause 9.3.1.9 of 3GPP TS 38.413, encoded as a string of 7
hexadecimal digits with the most significant nybble appearing first, and padded with leading zeroes as necessary.
- Optional Network Identifier (NID), as specified in 3GPP TS 23.003 and clause 5.30.2.1 3GPP TS 23.501, encoded
as a string of 11 hexadecimal digits with the most significant nybble appearing first, and padded with leading
zeroes as necessary.</xs:documentation>
                    </xs:annotation>
                </xs:element>
                <xs:element name="NCGI" type="NCGIType">
                    <xs:annotation>
                        <xs:documentation>A string of four numeric codes separated by hyphen characters that
encode a E-UTRAN Cell Global Identification per clause 19.6 of 3GPP TS 23.003:
- Mobile Country Code (MCC) part of the PLMN ID, as defined in clause 9.3.3.5 of 3GPP TS 38.413, encoded using a
string of 2 or 3 decimal digits.
- Mobile Network Code (MNC) part of the PLMN ID, as defined in clause 9.3.3.5 of 3GPP TS 38.413, encoded using a
string of exactly 3 decimal digits.
- NR Cell Identity (NCI), as specified in clause 9.3.1.7 of 3GPP TS 38.413, encoded as a string of 9 hexadecimal
digits with the most significant nybble appearing first, and padded with leading zeroes as necessary.
- Optional Network Identifier (NID), as specified in 3GPP TS 23.003 and clause 5.30.2.1 of 3GPP TS 23.501,
encoded as a string of 11 hexadecimal digits with the most significant nybble appearing first, and padded with
leading zeroes as necessary.</xs:documentation>
                    </xs:annotation>
                </xs:element>
            </xs:choice>
            <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>
```

```
    <xs:element name="Locations" type="UserLocationsType"/>
</xs:schema>
```

The schema in listing 10.6.2-3 is used by XML instance documents to declare their compatibility with a particular version of the QoE reporting schema specified in listing 10.6.2-1. The filename of this schema is "TS 26247_SchemaVersion.xsd".

Listing 10.6.2-3: XML Schema for the version

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema  xmlns="urn:3gpp:metadata:2016:PSS:schemaVersion"
            xmlns:xs="http://www.w3.org/2001/XMLSchema"
            targetNamespace="urn:3gpp:metadata:2016:PSS:schemaVersion"
            elementFormDefault="qualified">
    <xs:element name="schemaVersion" type="xs:string"/>
    <xs:element name="delimiter" type="xs:byte"/>
</xs:schema>
```

If a `supplementQoEMetric` needs to be sent when no ordinary `QoEMetric` is due, a dummy `MPDInformation` metric shall be sent with `codecs="none"`, `bandwidth=0, mimeType="none"`, `representationId="none"`.

If the attribute `qoeReferenceId` was defined in the QMC configuration (see clause L.2), the value shall be copied into each QoE report, to facilitate network-side correlation (see [63]). If this attribute was defined the attribute `recordingSessionId` shall also be returned for each QoE report. When metrics are reported via the QMC functionality (see annex L) the `recordingSessionId` is a two-byte numeric value defined by the client. It shall remain the same for all QoE reports belonging to the same streaming session, and it should be different for QoE reports belonging to different streaming sessions.

For the QMC scheme, if the **`SliceScope`** element is included in the QoE configuration and the slice associated with the streaming service is within the **`SliceScope`**, the DASH client should execute the QoE collection and include the S-NSSAI and DNN that correspond to the report data for support of per-slice QoE reporting and evaluation in OAM. This information may be retrieved via the AT Command +CGDCONT [61]) or the specific traffic mapping with URSP rule [69].

10.6.3 Reporting protocols

For configuration done via the QMC functionality (see annex L), the client shall also send QoE reports via the QMC functionality. For MPD or OMA-DM configuration, if a specific metrics server has been configured, the client shall send QoE reports using the HTTP (RFC 9112) [9] `POST` request carrying XML-formatted metadata in its body.
An example QoE reporting based on HTTP POST request signalling is shown below:

Listing 10.6.3-1: Example QoE reporting request

```
POST http://www.exampleserver.com HTTP/1.1
Host: 192.68.1.1
User-Agent: Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 6.1; Trident/4.0)
Content-Type: text/xml; charset=utf-8
Content-Length: 4408

<?xml version="1.0"?>
<ReceptionReport contentURI="http://www.example.com/content/content.mpd" clientID="35848574673"
xmlns="urn:3gpp:metadata:2011:HSD:receptionreport"
    xmlns:evex="urn:3gpp:metadata:2024:PSS:SupplementalEventExposureReporting"
    xmlns:sv="urn:3gpp:metadata:2016:PSS:schemaVersion"
    xsi:schemaLocation="urn:3gpp:metadata:2011:HSD:receptionreport DASH-QoE-Report.xsd"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
    <sv:schemaVersion>TSG105-Rel18</sv:schemaVersion>
    <QoeReport recordingSessionId="7a25" periodID="Period1" reportTime="2011-02-16T09:00:00"
reportPeriod="500">
        <QoeMetric>
            <HttpList>
                <HttpListEntry type="MPD" url="http://www.example.com/content/content.mpd"
trequest="2011-02-16T08:59:30" tresponse="2011-02-16T08:59:31" interval="50">
                    <Trace s="2011-02-16T08:59:30Z" d="171" b="2367 1990 2463 1254"/>
                </HttpListEntry>
```

```
                <HttpListEntry type="InitializationSegment"
url="http://www.example.com/content/initRep1.3gp" trequest="2011-02-16T08:59:40"
tresponse="2011-02-16T08:59:41" interval="200">
                    <Trace s="2011-02-16T08:59:40.5Z" d="159" b="9345"/>
                </HttpListEntry>
                <HttpListEntry type="InitializationSegment"
url="http://www.example.com/content/initRep2.3gp" trequest="2011-02-16T08:59:41"
tresponse="2011-02-16T08:59:42" interval="200">
                    <Trace s="2011-02-16T08:59:41.5Z" d="123" b="6723"/>
                </HttpListEntry>
                <HttpListEntry type="InitializationSegment"
url="http://www.example.com/content/initRep3.3gp" trequest="2011-02-16T08:59:42"
tresponse="2011-02-16T08:59:43" interval="200">
                    <Trace s="2011-02-16T08:59:42.5Z" d="195" b="9786"/>
                </HttpListEntry>
            </HttpList
        </QoeMetric>
        <QoeMetric>
            <InitialPlayoutDelay>10000</InitialPlayoutDelay>
        </QoeMetric>
        <sv:delimiter>0</sv:delimiter>
        <sv:delimiter>0</sv:delimiter>
        <sv:delimiter>0</sv:delimiter>
        <evex:SliceInfo sst="42" sd="a1b2c3"/>
        <evex:UserLocations>
            <evex:NCGI>234-015-000000001</evex:NCGI>
        </evex:UserLocations>
        <evex:DNN>broadband</evex:DNN>
        <sv:delimiter>0</sv:delimiter>
    </QoeReport>
    <QoeReport periodID="Period1" reportTime="2011-02-16T09:08:20" reportPeriod="500">
        <QoeMetric>
            <BufferLevel>
                <BufferLevelEntry t="2011-02-16T09:08:19" level="84673"/>
                <BufferLevelEntry t="2011-02-16T09:08:20" level="93874"/>
            </BufferLevel>
        </QoeMetric>
        <QoeMetric>
            <RepSwitchList>
                <RepSwitchEvent to="Rep2"/>
                <RepSwitchEvent to="Rep3"/>
            </RepSwitchList>
        </QoeMetric>
        <sv:delimiter>0</sv:delimiter>
        <sv:delimiter>0</sv:delimiter>
        <sv:delimiter>0</sv:delimiter>
        <evex:SliceInfo sst="42" sd="a1b2c3"/>
        <evex:UserLocations>
            <evex:NCGI>234-015-000000002</evex:NCGI>
        </evex:UserLocations>
        <evex:DNN>broadband</evex:DNN>
        <sv:delimiter>0</sv:delimiter>
    </QoeReport>
    <sv:delimiter>0</sv:delimiter>
</ReceptionReport>
```

11 Live services

11.1 Overview of dynamic and live media presentations

DASH Media Presentations with **MPD**`@type` set to `"dynamic"` enable that media is made available over time and its availability may also be removed over time. This has two major effects, namely:

1) The content creator can announce a DASH Media Presentation for which not all content is yet available, but only becomes available over time.

2) Clients are forced into a timed schedule for the playout, such that they follow the schedule as desired by the content author.

Dynamic services may be used for different types of services:

1) *Dynamic distribution of available content:* Services, for which content is made available as dynamic content, but the content is entirely generated prior to distribution. In this case the details of the Media Presentation, especially the Segments (duration, URLs) are known and can be announced in a single MPD without MPD updates.

2) *MPD-controlled live service:* Services for which the content is typically generated on the fly, and the MPD needs to be updated occasionally to reflect changes in the service offerings, e.g. announcing a new Period with new media or providing the end of the Media Presentation. The DASH client operates solely on information in the MPD.

Dynamic and Live services are typically controlled by different client transactions and server-side signalling.

For initial access to the service and joining the service, an MPD is required. MPDs may be accessed at join time or may have been provided earlier, for example along with an Electronic Service Guide. The initial MPD or join MPD is accessed and processed by the client and the client is synchronized with the server can analyse the MPD and extract suitable information in order to initiate the service. This includes, but is not limited to:

- identifying the currently active Periods in the service and the Period that expresses the live edge (for more details see below)

- selecting the suitable media components by selecting one or multiple Adaptation Sets. Within each Adaptation Set selecting an appropriate Representation and identifying the live edge segment in each Representations. The client then issues requests for the Segments.

The MPD may be updated on the server based on certain rules and clients consuming the service are expected to update MPDs based on certain triggers.

This clause provides requirements and recommendations for different functions, namely:

- for the dynamic segment download in clause 11.2.

- for services with MPD updates in clause 11.3.

- for offering live generated services in on-demand mode in clause 11.4.

- for client-server timing synchronization in clause 11.5.

- for robust service offerings and corresponding clients in clause 11.6.

11.2 Dynamic segment download

11.2.1 Background and assumptions

The dynamic segment download function is a key component of live services, In addition, the dynamic segment download function may also be used for scheduling playout of on-demand content by the content author. In this clause, it is assumed that the client has access to a single instance of an MPD and all information of the entire Media Presentation is contained in the MPD.

The timing model of dynamic services, especially the generation of Segment availabilities is a relevant component of a live services. This forms the basis for live services and explains the key concepts and rules for Segment availabilities.

11.2.2 MPD information and timing model

11.2.2.1 MPD information

If the Media Presentation is of type dynamic, then Segments have different Segment availability times, i.e. the earliest time for which the service provider permits the DASH client to issue a request to the Segment and guarantees, under regular operation modes, that the client gets a 200 OK response for the Segment. The Segment availability times for each Representation can be computed based on the information in an MPD.

For a dynamic service the MPD should at least contain information as available in table 11-1. Information included there may be used to compute a list of announced Segments, Segment Availability Times and URLs.

Assume that an MPD is available to the DASH client at a specific wall-clock time *NOW*. It is assumed that the client and the DASH server providing the Segments are synchronized to wall-clock, either through external means or through a specific client-server synchronization. Details on synchronization are provided in clause 11.6.

Assuming synchronization, the information in the MPD can then be used by the client at time *NOW* to derive the availability (or non-availability) of Segments on the server.

Table 11-1: Information related to Segment Information and Availability Times for a dynamic service

MPD Information	Status	Comment
MPD@type	mandatory, set to "dynamic"	The type of the Media Presentation is dynamic, i.e. Segments get available over time.
MPD@availabilityStartTime	mandatory	The start time is the anchor for the MPD in wall-clock time. The value is denoted as *AST* in the following.
MPD@mediaPresentationDuration	mandatory (for the considered use cases)	Duration of the Media Presentation.
MPD@suggestedPresentationDelay	optional, but recommended	Suggested presentation delay as delta to segment availability start time. The value is denoted as *SPD*. Details on the setting and usage of the parameter is provided in the following.
MPD@minBufferTime	mandatory	Minimum buffer time, used in conjunction with the @bandwidth attribute of each Representation. The value is denoted as *MBT*. Details on the setting and usage of the parameter is provided in the following.
MPD@timeShiftBufferDepth	optional, but recommended	Time shift buffer depth of the media presentation. The value is denoted as *TSB*. Details on the setting and usage of the parameter is provided in the following.
Period@start	Mandatory for the first Period in the MPD	Start time of the Period relative to the MPD availability start time.
Representation@availabilityTimeOffset	Optional default	Offset in availability time for this Representation. It may also be available on a Base URL or default. NOTE: the value of "INF" implies availability of all segments starts at **MPD**@availabilityStartTime
SegmentTemplate@media	mandatory	Template for the Media Segment assigned to a Representation.
SegmentTemplate@startNumber	optional default	Number of the first segment in the Period assigned to a Representation
SegmentTemplate@timescale	optional default	Timescale for this Representation.
SegmentTemplate@duration	mandatory	Duration of each Segment in units of a time.

11.2.2.2 Segment information derivation

11.2.2.2.1 Introduction

Based on an MPD including information as documented in table 11-1 and available at time *NOW* on the server, a synchronized DASH client derives the information of the list of Segments for each Representation in each Period. This clause only describes the information that is expressed by the values in the MPD. The generation of the information on the server and the usage of the information in the client is provided in clause 11.2.3 and 11.2.4 respectively.

11.2.2.2.2 Definitions

The following definitions are relevant and aligned with ISO/IEC 23009-1:

- available Segment is a Segment that is accessible at its assigned HTTP-URL. This means that a request with an HTTP GET to the URL of the Segment results in a reply of the Segment and 2xx status code.

- valid Segment URL is an HTTP-URL that is promised to reference a Segment during its Segment availability period.

- *NOW* is a time that is expressing the time on the content server as wallclock time. All information in the MPD related to wallclock is expressed as a reference to the time *NOW*.

11.2.2.2.3 MPD information

For a dynamic service without MPD updates, the following information shall be present and not present in the MPD (also please refer to table 11-1):

- The **MPD**@type shall be set to "dynamic".
- The **MPD**@mediaPresentationDuration shall be present, or the **Period**@duration of the last Period shall be present.
- The **MPD**@minimumUpdatePeriod shall not be present.

Furthermore, it is recommended to provide values for **MPD**@timeShiftBufferDepth and **MPD**@suggestedPresentationDelay.

11.2.2.2.4 Period information

Each Period is documented by a **Period** element in the MPD. An MPD may contain one or more Periods. In order to document the use of multiple Periods, the sequence of Period elements is expressed by an index *i* with *i* increasing by 1 for each new Period element.

- Each regular Period *i* in the MPD is assigned a
  - Period start time *PSwc*[*i*] in wall-clock time,
  - Period end time *PEwc*[*i*], in wall-clock time.

NOTE: An MPD update may extend the Period end time of the last regular Period. For details refer to clause 11.3.

The Period start time *PSwc*[*i*] for a regular Period *i* is determined according to clause 5.3.2.1 of ISO/IEC 23009-1:

- If the attribute @start is present in the **Period**, then *PSwc*[*i*] is the sum of *AST* and the value of this attribute.
- If the @start attribute is absent, but the previous **Period** element contains a @duration attribute then the start time of the Period is the sum of the start time of the previous Period *PSwc*[*i*] and the value of the attribute @duration of the previous Period. Note that if both are present, then the @start of the new Period takes precedence over the information derived from the @duration attribute.

The Period end time *PEwc*[*i*] for a regular Period *i* is determined as follows:

- If the Period is the last one in the MPD, the time *PEwc*[*i*] is obtained as
  - the sum of *AST* and Media Presentation Duration *MPDur*, with *MPDur* the value of **MPD**@mediaPresentationDuration if present, or the sum of *PSwc*[*i*] of the last Period and the value of **Period**@duration of the last Period.
- Else
  - the time *PEwc*[*i*] is obtained as the Period start time of the next Period, i.e. *PEwc*[*i*] = *PSwc*[*i*+1].

11.2.2.2.5 Representation information

Based on such an MPD at a specific time *NOW*, a list of Segments contained in a Representation in a Period *i* with Period start time *PSwc*[*i*] and Period end time *PEwc*[*i*] can be computed.

Let:

- N_S=1,
- *ato* is the value of the @availabilityTimeOffset attribute, if present. Otherwise, it is zero.

- *ts* the value of the `@timescale` attribute
- *t*[*s*] is 0,
- the *d*[*s*] is the value of `@duration` attribute
- *r*[*s*] is the ceiling of (*PEwc*[*i*] − *PSwc*[*i*] − *t*[*s*]/*ts*) × *ts*/*d*[*s*])

11.2.2.2.6 Media time information of segment

Each Media Segment at position *k=1,2,* ... for each Representation has assigned an earliest media presentation time *EPT*[*k,r,i*] and an accurate segment duration *SDUR[k,r,j]*, all measured in media presentation time.

The earliest presentation time may be estimated from the MPD using the segment availability start time minus the segment duration announced in the MPD. The earliest presentation time may be accurately determined from the Segment itself.

11.2.2.2.7 Segment List parameters

For each Period *i* with Period start time *PSwc*[*i*] and Period end time *PEwc*[*i*] and each Representation *r* in the Period the following information can be computed:

- the presentation time offset described in the MPD, o[*i,r*]
- the availability time offset of this Representation, ato[*r*]
- the number of the first segment described in the MPD, k1[*i,r*]
- the number of the last segment described in the MPD, k2[*i,r*]
- segment availability start time of the initialization segment SAST[*0,i,r*]
- segment availability end time of the initialization segment SAET[*0,i,r*]
- segment availability start time of each media segment SAST[*k,i,r*], *k*=k1, ..., k2
- segment availability end time of each media segment SAET[*k,i,r*], *k*=k1, ..., k2
- adjusted segment availability start time ASAST[*0,i,r*], *k*=0, k1, ..., k2
- segment duration of each media segment SD[*k,i,r*], k=k1, ..., k2
- the URL of each of the segments, URL[*k,i,r*]

In addition,

- the latest available Period *i*[*NOW*] and the latest segment available at the server k[*NOW*] can be computed. This segment is also referred to as *live edge segment*.
- the earliest available Period *i* × [*NOW*] and the earliest segment available at the server *k* × [*NOW*] can be computed.

Based on the above information, for each Representation *r* in a Period *i*, the segment availability start time *SAST*[*k,i,r*], the segment availability end time of each segment *SAET*[*k,i,r*], the segment duration of each segment *SD*[*k,i,r*], and the URL of each of the segments, *URL*[*k,i,r*] within one Period *i* be derived as follows using the URL Template function `URLTemplate`(`ReplacementString`, Address):

- k=0
- SAST[0,i,r] = PSwc[i]
- ASAST[0,i,r] = PSwc[i] - ato
- for s=1, ... N_s [i,r]
  - *k* = *k* + 1

- *SAST*[*k,i,r*] = *PSwc*[*i*] + (*t*[*s,i,r*] + *d*[*s,i,r*] - o[*i,r*])/*ts*
- *ASAST*[*k,i,r*] = *SAST*[*k,i,r*] – *ato*
- *SD*[*k,i,r*] = *d*[*s,i,r*]/*ts*
- *SAET*[*k,i,r*] = SAST[*k,i,r*] + *TSB* + *d*[*s,i,r*]/*ts*
- if **`SegmentTemplate`**`@media` contains `$Number$`
  - *Address*=@startNumber
  - *URL*[*k,i,r*] = URLTemplate ($Number$, *Address*)
- for *j* = 1, ..., *r*[*s,i,r*]
  - *k* = *k* + 1
  - *SAST*[*k,i,r*] = SAST[*k-1,i,r*] + *d*[*s,i,r*]/*ts*
  - *ASAST*[*k,i,r*] = *SAST*[*k,i,r*] – *ato*
  - *SAET*[*k,i,r*] = SAST[*k,i,r*] + *TSB* + *d*[*s,i,r*] /*ts*
  - *SD*[*k,i,r*] = *d*[*s,i,r*] /*ts*
  - if **`SegmentTemplate`**`@media` contains `$Number$`
    - Address = Address + 1
    - URL[k,i,r] = `URLTemplate` (`$Number$`, Address)
- k2[*i,r*] = *k*
- *SAET*[*0,i,r*] = *SAET*[k2[*i,r*],*i,r*]

Note that not all segments documented above may necessarily be accessible at time *NOW*, but only those that are within the segment availability time window. Hence, the number of the first media segment described in the MPD for this Period, k1[*i,r*], is the smallest *k*=1, 2, ... for which *SAST*[*k,i,r*] >= *NOW*.

The latest available Period *i*[*NOW*] is the Period *i* with the largest *PEwc*[*i*] and *PEwc*[*i*] is smaller than or equal to *NOW*.

The latest available segment *k*[*NOW*] available for a Representation of Period *i*[*NOW*] (also the live edge segment) is the segment with the largest *k*=0,1,2,... such that *SAST*[*k,i,r*] is smaller than or equal to *NOW*. Note that this contains the Initialization Segment with *k*=0 as not necessarily any media segment may yet be available for Period *i*[*NOW*]. In this case, last media segment k2[*i*[*NOW*]-1,*r*], i.e., the last media segment of the previous Period is the latest accessible media Segment.

However, if the `@availabilityTimeOffset` is present, then the segments for this Representation are available earlier than the nominal segment availability start time, namely at *ASAST*[*k,i,r*].

11.2.2.2.8 URL generation with segment template

The function URL Template function `URLTemplate`(`ReplacementString`, Address) generates a URL. For details refer to ISO/IEC 23009-1, clause 5.3.9.4. Once the Segment is generated, processing of the Base URLs that apply on this segment level is done as defined in ISO/IEC 23009-1, clause 5.6.

11.2.3 Service offering requirements and guidelines

11.2.3.1 General service offering requirements

For dynamic service offerings, the MPD shall conform to the MPD in clause 8 and shall at least contain the mandatory information as documented in table 11-1.

If such an MPD is accessible at time *NOW* at the location **`MPD.Location`**, then

- all Segments for all Representations in all Periods as announced in an MPD shall be available latest at the announced segment availability start time *SAST*[*k,i,r*] at all *URL*[*k,i,r*] as derived in clause 11.2.2.2;
- all Segments for all Representations in all Periods as announced in an MPD shall at least be available until the announced segment availability end time *SAET*[*k,i,r*] at all *URL*[*k,i,r*] as derived in clause 11.2.2.2;
- for all Media Segments for all Representations in all Periods as announced in an MPD the Segment in this Period is available prior to the sum of Period start, earliest presentation time and segment duration, i.e. *SAST*[*k,i,r*] <= *PSwc*[*i*] + *SD*[*k,r,i*] + *EPT*[*k,r,i*];
- if a Media Segments with segment number *k* is delivered over a constant bitrate channel with bitrate equal to value of the `@bandwidth` attribute then each presentation time *PT* is available at the client latest at time with a delay of at most *PT* + *MBT*.

11.2.3.2 Dynamic service offering guidelines

11.2.3.2.1 Introduction

In order to offer a simple dynamic service for which the following details are known in advance,

- start at wall-clock time `START`,
- exact duration of media presentation `PDURATION`,
- location of the segments for each Representation at "`http://example.com/$RepresentationID$/$Number$`",

a service provide may offer an MPD as follows:

Table 11-2: Basic service offering

MPD Information	Value
`MPD``@type`	`dynamic`
`MPD``@availabilityStartTime`	`START`
`MPD``@mediaPresentationDuration`	`PDURATION`
`MPD``@suggestedPresentationDelay`	`SPD`
`MPD``@minBufferTime`	`MBT`
`MPD``@timeShiftBufferDepth`	`TSB`
`MPD.BaseURL`	`"http://example.com/"`
`Period``@start`	`PSTART`
`Representation``@bandwidth`	`BW`
`SegmentTemplate``@media`	`"$RepresentationID$/$Number$"`
`SegmentTemplate``@startNumber`	`1`
`SegmentTemplate``@duration`	`SDURATION`
NOTE: The setting of capitalized parameters is discussed in clause 11.2.3.2.2.	

Based on the details in clause 11.2.2.2, the Segment Information is derived as:

- k1 = 1
- k2 = `ceil(PDURATION/SDURATION)`
- for k = 1, ..., k2
  - SAST[*k*] = START + PSTART + *k**SDURATION
  - SAET[*k*] = SAST[*k*] + TSB + SDURATION
  - SD[*k*] = SDURATION
  - URL[*k*] = `http://example.com/$RepresentationID$/k`

- The segment availability times of the Initialization Segment are as follows:
  - SAST[*0*] = START + PSTART
  - SAET[0] = SAET[k2]

11.2.3.2.2 Basic parameter settings

In the following recommendations are provided for the

- Time Shift Buffer Depth (TSB):
  - If the content should be consumed at the live edge, then the time shift buffer depth should be set short. However, the TSB should not be smaller than the recommended value of 4*SDURATION and 6 seconds in media time in order for the client to do some prebuffering in more difficult network conditions.
  - If no restrictions on the accessibility of the content are provided, then the TSB may be set to a large value that even exceeds PDURATION.
- Suggested Presentation Delay (SPD)
  - If synchronized play-out with other devices adhering to the same rule is desired and/or the service provider wants to define the typical live edge of the program, then this value should be provided. The service provider should set the value taking into account at least the following:
    - the desired end-to-end latency
    - the typical required buffering in the client, for example based on the network condition
    - the segment duration SDURATION
    - the time shift buffer depth TSB
  - A reasonable value may be 2 to 4 times of the segment duration SDURATION, but the time should not be smaller than 4 seconds in order for the client to maintain some buffering.
- Segment Duration (SDURATION)
  - The segment duration typically influences the end-to-end latency, but also the switching and random access granularity as in DASH-264/AVC each segment starts with a stream access point which can also be used as s switch point. The service provider should set the value taking into account at least the following:
    - the desired end-to-end latency
    - the desired compression efficiency
    - the start-up latency
    - the desired switching granularity
    - the desired amount of HTTP requests per second
    - the variability of the expected network conditions
  - Reasonable values for segment durations are between 1 second and 10 seconds.
- Minimum Buffer Time (MBT) and bandwidth (BW)
  - the value of the minimum buffer time **does not provide any instructions to the client on how long to buffer the media**. The value describes how much buffer a client should have under ***ideal*** network conditions. As such, MBT is not describing the burstiness or jitter in the network, it is describing the burstiness or jitter in the **content encoding**. Together with the BW value, it is a property of the content. Using the "leaky bucket" model, it is the size of the bucket that makes BW true, given the way the content is encoded.

- The minimum buffer time provides information that for each Stream Access Point (and in the case of DASH-IF therefore each start of the Media Segment), the property of the stream: If the Representation (starting at any segment) is delivered over a constant bitrate channel with bitrate equal to value of the BW attribute then each presentation time *PT* is available at the client latest at time with a delay of at most *PT + MBT*.

- In the absence of any other guidance, **the** MBT **should be set** to the maximum GOP size (coded video sequence) of the content, which quite often is identical **to the maximum segment duration**. The *MBT* may be set to a smaller value than maximum segment duration, but should not be set to a higher value.

11.2.3.3 Content offering with Periods

11.2.3.3.1 General

For content offered within a Period, and especially when offered in multiple Periods, then the content provider should offer the content such that actual media presentation time is as close as possible to the actual Period duration. It is recommended that the Period duration is the maximum of the presentation duration of all Representations contained in the Period.

A typical multi-Period offering is shown in table 11-4. This may for example represent a service offering where main content provided in Period 1 and Period 3 are interrupted by an inserted Period 2.

Table 6: Multi-Period service offering

MPD Information	Value
MPD@type	dynamic
MPD@availabilityStartTime	START
MPD@mediaPresentationDuration	PDURATION
MPD@suggestedPresentationDelay	SPD
MPD@minBufferTime	MBT
MPD@timeShiftBufferDepth	TSB
MPD.BaseURL	"http://example.com/"
Period@start	PSTART
SegmentTemplate@media	"1/$RepresentationID$/$Number$"
SegmentTemplate@startNumber	1
SegmentTemplate@duration	SDURATION1
Period@start	PSTART2
Representation@availabilityTimeOffset	ATO2
SegmentTemplate@media	"2/$RepresentationID$/$Number$"
SegmentTemplate@startNumber	1
SegmentTemplate@duration	SDURATION2
Period@start	PSTART3
SegmentTemplate@media	"1/$RepresentationID$/$Number$"
SegmentTemplate@startNumber	STARTNUMBER2
SegmentTemplate@duration	SDURATION1
SegmentTemplate@presentationTimeOffset	PTO

Based on the details in clause 11.2.2.2, the Segment Information is derived as:

- Period 1
  - PSwc[1] = START + PSTART
  - PEwc[1] = START + PSTART2
  - k1 = 1
  - k2 = ceil((PSTART2-PSTART1)/SDURATION)
  - for k = 1, ..., k2
    - SAST[*k*] = PSwc[1] + *k**SDURATION

- SAET[*k*] = SAST[*k*] + TSB + SDURATION
- SD[*k*] = SDURATION
- URL[*k*] = http://example.com/1/$RepresentationID$/k

- SAST[0] = PSwc[1]
- SAET[0] = SAET[k2]

- Period 2
  - PSwc[2] = START + PSTART2
  - PEwc[2] = START + PSTART3
  - k1 = 1
  - k2 = ceil((PSTART3-PSTART2)/SDURATION2)
  - for k = 1, ..., k2
    - SAST[*k*] = PSwc[2] + *k**SDURATION2
    - ASAST[*k*] = SAST[*k*] – ATO2
    - SAET[*k*] = SAST[*k*] + TSB + SDURATION2
    - SD[*k*] = SDURATION2
    - URL[*k*] = http://example.com/2/$RepresentationID$/k
  - SAST[0] = PSwc[2]
  - SAET[0] = SAET[k2]
- Period 3
  - PSwc[3] = START + PSTART3
  - PEwc[3] = START + PDURATION
  - k1 = 1
  - k2 = ceil((PSTART3-PDURATION)/SDURATION1)
  - for k = 1, ..., k2
    - SAST[*k*] = PSwc[3] + *k* × SDURATION1
    - SAET[*k*] = SAST[*k*] + TSB + SDURATION1
    - SD[*k*] = SDURATION1
    - URL[*k*] = "http://example.com/1/$RepresentationID$/(k+STARTNUMBER2-1)"
  - SAST[0] = PSwc[3]
  - SAET[0] = SAET[k2]

Note that the number *k* describes position in the Period. The actual number used in the segment template increased by the one less than the actual start number.

11.2.3.4 Joining recommendation

By default, an MPD with **`MPD`**`@type="dynamic"` suggests that the client would want to join the stream at the live edge, therefore to download the latest available segment (or close to, depending on the buffering model), and then start playing from that segment onwards.

However, there are circumstances where a dynamic MPD might be used with content intended for playback from the start, or from another position. For example, when a content provider offers 'start again' functionality for a live program, the intention is to make the content available as an on-demand program, but not all the segments will be available immediately.

This may be signalled to the DASH client by including an MPD Anchor, with either

- the `t` parameter, or
- both the `period` and `t` parameter, in the MPD URL provided to the DASH client, or

The format and behaviour of MPD Anchors is defined in clause C.4 of ISO/IEC 23009-1.

For example, to start from the beginning of the MPD the following would be added to the end of the MPD URL provided to the DASH client:

- #t=0

Or to start from somewhere other than the start, in this case 50 minutes from the beginning of the period with Period ID "`program_part_2`":

- #period=program_part_2&t=50:00

Where an MPD Anchor is used, it should refer to a time for which segments are currently available in the MPD.

11.2.4 Client operation, recommendations and guidelines (informative)

11.2.4.1 Basic operation

A DASH client is guided by the information provided in the MPD. A simple client model is shown in figure 11-1.

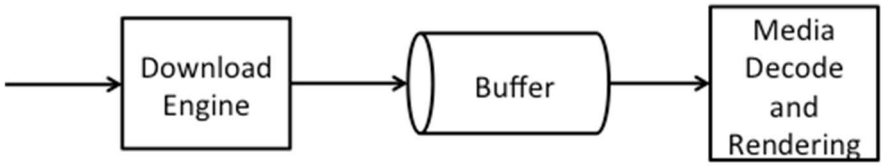


Figure 11-1 Simple client model

Assume that the client has access to an MPD and the MPD contains the parameters in table 11-1 and table 11-3, i.e. it consumes a dynamic service with fixed media presentation duration, but possibly multiple Periods.

The following example client behaviour may provide a continuous streaming experience to the user:

1) The client parses the MPD, selects a collection of Adaptation Sets suitable for its environment based on information provided in each of the **`AdaptationSet`** elements.

2) Within each Adaptation Set it selects one Representation, typically based on the value of the `@bandwidth` attribute, but also taking into account client decoding and rendering capabilities.

   The client creates a list of accessible Segments at least for each selected Representation taking into account the information in the MPD as documented in table 11-1 and table 11-3 and the current time *JOIN* in the client and in particular the segment closest to the live edge referred to the *live edge segment*. For details refer to clause 11.2.4.2.

   For this it needs to take into account the latest Period *i*[*NOW*]. The latest Period and the latest segment are obtained as follows with i* the index of the last Period:

- if *NOW* ≤ PSwc[1]
- no segment is yet available
- else if *NOW* > PSwc[i*]
- the last one and the latest segment is available is k2[i*]
- else if *NOW* > PSwc[i*] + TSB
- no segment is available any more
- else if PSwc[1] < *NOW* ≤ PEwc[i*]
  - *i*' the such that PSwc[i'] < NOW ≤ PEwc[*i*']
  - k[*NOW*] = min(`floor` ((*NOW* - PEwc[*i*'] - PSwc[*i*'])/`SDURATION[i']`), k2)

    Note again that if k[*NOW*] is *0*, then only the Initialization Segment is available. If the Period is not the first one, then the last available Media Segment is the last Media Segment of the previous Period.

3) The client downloads the initialization segment of the selected Representations and then accesses the content by requesting entire Segments or byte ranges of Segments. Typically, at any time the client downloads the next segment at the larger of the two: (i) completion of download of current segment or (ii) the Segment Availability Start Time of the next segment. Note that if the `@availabilityTimeOffset` is present, then the segments may be downloaded earlier, namely at the adjusted segment availability start time. Based on the buffer fullness and other criteria, rate adaptation is considered. Typically, the first media segment that is downloaded is the *live edge segment*, but other decisions may be taken in order to minimize start-up latency. For details on initial buffering, refer to clause 11.2.4.4.

4) According to figure 11-1 media is fed into buffer and at some point in time, the decoding and rendering of the media is kicked off. The downloading and presentation is done for the selected Representation of each selected Adaptation. The synchronization is done using the presentation time in the Period. For synchronized playout, the exact presentation times in the media is expected to be used.

5) Once presentation has started, the playout process is continuous. The playout process expects media to be present in the buffer continuously. If the **`MPD`**`@suggestedPresentationDelay` is present, then this value may be used as the presentation delay *PD*. If the **`MPD`**`@suggestedPresentationDelay` is not present, but the client is expected to consume the service at the live edge, then a suitable presentation delay should be selected, typically between the value of `@minBufferTime` and the value of `@timeShiftBufferDepth`. It is recommended that the client starts rendering the first sample of the downloaded media segment *k* with earliest presentation time EPT(*k*) at *PSwc*[*i*] + (EPT(*k*) − *o*[*r*,*i*]) + *PD*. For details on selecting and minimizing end-to-end latency as well as the start-up latency, see clause 11.2.4.4.

6) The client may request Media Segments of the selected Representations by using the generated Segment list during the availability time window.

7) Once the presentation has started, the client continues consuming the media content by continuously requesting Media Segments or parts of Media Segments and playing content that according to the media presentation timeline. The client may switch Representations taking into updated information from its environment, e.g. change of observed throughput. In a straight-forward implementation, with any request for a Media Segment starting with a stream access point, the client may switch to a different Representation. If switching at a stream access point, the client is expected to switch seamlessly at such a stream access point.

8) With the wall-clock time *NOW* advancing, the client consumes the available Segments. As *NOW* advances the client possibly expands the list of *available* Segments for each Representation in the Period according to the procedures specified in clause 11.2.4.2.

9) Once the client is consuming media contained in the Segments towards the end of the announced media in the Representation, then either the Media Presentation is terminated, a new Period is started (see subsection or the MPD needs to be refetched. Once the client is consuming media contained in the Segments towards the end of the announced media in the Representation, and the Representation is contained not in the last Period, then the DASH clients generally needs to reselect the Adaptation Sets and a Representation in same manner as described in bullet 1) and 2). Also steps 3), 4), 5) and 6) need to be carried out at the transition of a Period. Generally, audio/video switching across period boundaries may not be seamless. According to ISO/IEC 23009-1,

clause 7.2.1, at the start of a new Period, the playout procedure of the media content components may need to be adjusted at the end of the preceding Period to match the *PeriodStart* time of the new Period as there may be small overlaps or gaps with the Representation at the end of the preceding Period. Overlaps (respectively gaps) may result from Media Segments with actual presentation duration of the media stream longer (respectively shorter) than indicated by the Period duration. Also, at the beginning of a Period, if the earliest presentation time of any access unit of a Representation is not equal to the presentation time offset signalled in the `@presentationTimeOffset`, then the playout procedures need to be adjusted accordingly.

The client should play the content continuously across Periods, but there may be implications in terms of implementation to provide fully continuous and seamless playout. It may be the case that at Period boundaries, the presentation engine needs to be reinitialized, for example due to changes in formats, codecs or other properties. This may result in a re-initialization delay. Such a re-initialization delay should be minimized. If the Media Presentation is of type dynamic, the addition of the re-initialisation delay to the playout may result in drift between the encoder and the presentation engine. Therefore, the playout should be adjusted at the end of each Period to provide a continuous presentation without adding drift between the time documented in the MPD and the actual playout, i.e. the difference between the actual playout time and the Period start time should remain constant.

If the client presents media components of a certain Adaptation Set in one Period, and if the following Period has assigned an identical Asset Identifier, then the client should identify an associated Period and, in the absence of other information, continue playing the content in the associated Adaptation Set.

If furthermore the Adaptation Sets are period-continuous, i.e. the presentation times are continuous and this is signalled in the MPD, then the client is expected to seamlessly play the content across the Period boundary. Most suitably the client may continue playing the Representation in the Adaptation Set with the same `@id`, but there is no guarantee that this Representation is available. In this case the client is expected to seamlessly switch to any other Representation in the Adaptation Set.

For details on MPD updates and refetching, please refer to clause 11.3.

11.2.4.2 Joining, initial buffering and playout recommendations

11.2.4.2.1 General

A DASH client is expected to start playout from:

- The time indicated by the MPD Anchor, if one is present
- The live edge, if there is no MPD Anchor and **`MPD`**`@type="dynamic"`.

11.2.4.2.2 Joining at the live edge

For joining at the live edge there are basically two high-level strategies:

- Every client participating in the service commits to the same presentation delay (PD) relative to the announced segment availability start time at start-up and in continuous presentation, possible using one suggested by the Content Provider and then attempts to minimise start-up latency and maintain the buffer. The content provider may have provided the **`MPD`**`@suggestedPresentationDelay` (SPD) or may have provided this value by other means outside the DASH formats. The content author should be aware that the client may ignore the presence of **`MPD`**`@suggestedPresentationDelay` and may choose its own suitable playout scheduling.
- The client individually picks the presentation delay (PD) in order to maximize stable quality and does this dependent on its access, user preferences and other considerations.

In both cases the client needs to decide, which segment to download first and when to schedule the playout of the segment based on the committed PD.

A DASH client would download an available segment and typically render the earliest presentation time EPT(k) of the segment at $PSwc[i] + (EPT(k) - o[r,i]) + PD$. As PD may be quite large, for example in order to provision for downloading in varying bitrate conditions, and if a segment is downloaded that was just made available it may result in larger start up delay.

Therefore, a couple of strategies may be considered as a trade-off of for start-up delay, presentation delay and sufficient buffer at the beginning of the service, when joining at the live edge:

1) The client downloads the next available segment and schedules playout with delay PD. This maximizes the initial buffer prior to playout, but typically results in undesired long start-up delay.

2) The client downloads the latest available segment and schedules playout with delay PD. This provides large initial buffer prior to playout, but typically results in undesired long start-up delay.

3) The client downloads the earliest available segment that can be downloaded to schedules playout with delay PD. This provides a smaller initial prior to playout, but results in reasonable start-up delay. The buffer may be filled gradually by downloading later segments faster than their media playout rate, i.e. by initially choosing Representations that have lower bitrate than the access bandwidth.

11.2.5 Considerations on live edge

A DASH client should avoid being too aggressive in requesting segments exactly at the computed segment availability start time, especially if it is uncertain to be fully synchronized with the server. If the DASH client observes issues, such as 404 responses, it should back up slightly in the requests.

In addition, for a content authoring to avoid too aggressive requests and possible 404 responses, the content author may schedule the segment availability start time in the MPD with a small safety delay compared to the actual publish time. This also provides the content author a certain amount of flexibility in the publishing of Segments. However, note that such safety margins may lead to slightly increased end-to-end latencies, so it is a balance to be taken into account.

11.3 Live services with MPD updates

11.3.1 Background and assumptions

If many cases, the service provider cannot predict that an MPD that is once offered, may be used for the entire Media Presentations. Examples for such MPD changes are:

- The duration of the Media Presentation is unknown
- The Media Presentation may be interrupted for advertisements which requires proper splicing of data, for example by adding a Period
- Operational issues require changes, for example the addition of removal of Representations or Adaptation Sets.
- Operational problems in the backend.
- Changes of segment durations, etc.

In this case the MPD typically only can describe a limited time into the future. Once the MPD expires, the service provider expects the client to recheck and get an updated MPD in order to continue the Media Presentation.

The main tool in MPEG-DASH is Media Presentation Description update feature as described in clause 5.4 of ISO/IEC 23009-1. The MPD is updated at the server and the client is expected to obtain the new MPD information once the determined Segment List gets to an end.

If the MPD contains the attribute **MPD**@minimumUpdatePeriod, then the MPD in hand will be updated. The DASH client typically frequently polls the MPD update server whether an MPD update is available or the existing MPD can still be used. The update frequency is controlled by MPD based on the attribute **MPD**@minimumUpdatePeriod.

11.3.2 Preliminaries

11.3.2.1 MPD information

As the MPD is typically updated over time on the server, the MPD that is accessed when joining the service as well as the changes of the MPD are referred to as MPD instances in the following. This expresses that for the same service, different MPDs exist depending on the time when the service is consumed.

Assume that an MPD instance is present on the DASH server at a specific wall-clock time *NOW*. For an MPD-based Live Service Offering, the MPD instance may among others contain information as available in table 11-5. Information included there may be used to compute a list of announced Segments, Segment Availability Times and URLs.

Table 11-5: Information related to Live Service Offering with MPD-controlled MPD Updates

MPD Information	Status	Comment
MPD@type	mandatory, set to "dynamic"	The type of the Media Presentation is dynamic, i.e. Segments get available over time.
MPD@availabilityStartTime	mandatory	The start time is the anchor for the MPD in wall-clock time. The value is denoted as *AST*.
MPD@minimumUpdatePeriod	mandatory	Mandatory except for the case where the **MPD**@mediaPresentationDuration is present. However, such an MPD falls then in an instance as documented in clause 11.2.
Period@start	mandatory	Start time of the Period relative to the MPD availability start time. The value is denoted as *PS*.
SegmentTemplate@media	mandatory	Template for the Media Segment
SegmentTemplate@startNumber	optional default	Number of the first segment in the Period. The value is denoted as *SSN*.
SegmentTemplate@duration	mandatory	Duration of each Segment in units of a time. The value divided by the value of @timescale is denoted as *MD*[*k*] with k=1, 2, ... The segment timeline may contain some gaps.

11.3.2.2 Segment Information derivation

Based on an MPD instance including information as documented in table 11-5 and available at time *NOW* on the server, a DASH client may derive the information of the list of Segments for each Representation in each Period.

If the Period is the last one in the MPD and the **MPD**@minimumUpdatePeriod is present, then the time *PEwc*[*i*] is obtained as the sum of *NOW* and the value of **MPD**@minimumUpdatePeriod. Note that with the MPD present on the server and *NOW* progressing, the Period end time is extended. This issue is the only change compared to the segment information generation in clause 11.2.2.2.

If the **MPD**@minimumUpdatePeriod is set to 0, then the MPD documents all available segments on the server.

11.3.3 Service offering requirements and guidelines

11.3.3.1 General

The same service requirements as in clause 11.2.2.1 hold for any time *NOW* the MPD is present on the server with the interpretation that the Period end time *PEwc*[*i*] of the last Period is obtained as the sum of *NOW* and the value of **MPD**@minimumUpdatePeriod.

In order to offer a simple live service with unknown presentation end time, but only a single Period and the following details are known in advance,

- start at wall-clock time START,
- location of the segments for each Representation at "http://example.com/$RepresentationID$/$Number$",

a service provider may offer an MPD with values according to table 11-6.

Table 11-6: Basic service offering with MPD updates

MPD Information	Value
MPD@type	dynamic
MPD@availabilityStartTime	START
MPD@publishTime	PUBTIME1
MPD@minimumUpdatePeriod	MUP

MPD.BaseURL	"http://example.com/"
Period@start	PSTART
SegmentTemplate@media	"$RepresentationID$/$Number$"
SegmentTemplate@startNumber	1
SegmentTemplate@duration	SDURATION

Based on the details in clauses 11.2.2.2 and 11.4.2.2, the Segment Information can be derived at each time *NOW* by determining the end time of the Period *PEwc*[1] = *NOW* + MUP.

The service provider may leave the MPD unchanged on the server. If this is the case the Media Presentation may be terminated with an updated MPD that

- adds the attribute **MPD**@mediaPresentationDuration with value PDURATION
- removes the attribute **MPD**@minimumUpdatePeriod
- changes the **MPD**@publishTime attribute to PUBTIME2

The MPD must be published latest at the end of the Media Presentation minus the value of MUP, i.e. PUBTIME2 <= START + PSTART + PDURATION - MUP.

The minimum update period may also be changed during an ongoing Media Presentation. Note that as with any other change to the MPD, this will only be effective with a delay in media time of the value of the previous MUP.

The principles in this document also holds for multi-period content, for which an MPD update may add a new Period. In the same way as for signalling the end of the Media Presentation, the publish time of the updated MPD with the new period needs to be done latest at the start of the new Period minus the value of the **MPD**@minimumUpdatePeriod attribute of the previous MPD.

11.3.3.2 Setting the minimum update period value

Setting the value of the minimum update period primarily affects two main service provider aspects: A short minimum update period results in the ability to change and announce new content in the MPD on shorter notice. However, by offering the MPD with a small minimum update period, the client requests an update of the MPD more frequently, potentially resulting in increased uplink and downlink traffic.

A special value for the minimum update period is 0. In this case, the end time of the period is the current time *NOW*. This implies that all segments that are announced in the MPD are actually available at any point in time. This also allows changing the service provider to offer changes in the MPD that are instantaneous on the media timeline, as the client, prior for asking for a new segment, has to revalidate the MPD.

11.3.3.3 Permitted updates in an MPD

The permitted updates are provided in clause 5.4 of ISO/IEC 23009-1.

In addition, updates in the MPD only extend the timeline. This means that information provided in a previous version of the MPD shall not be invalidated in an updated MPD.

In order to make the MPD joining friendly and to remove data that is available in the past, any segments that have fallen out of the time shift buffer may no longer be announced in the MPD. In this case, the Period start may be moved by changing one or both, **MPD**@availabilityStartTime and **Period**@start. However, this requires that the @startNumber and @presentationTimeOffset need to be updated such that the Segment Information is correct.

If Representations and Adaptations Sets are added or removed or the location of the Segments is changed, it is recommended to update the MPD and provide Adaptation Sets in a period-continuous manner.

11.3.3.4 Last segment message

If the @segmentProfiles contains the 'lmsg' brand for a certain Representation, then the 'lmsg' brand for signalling the last segment shall be applied for any content with **MPD**@minimumUpdatePeriod present and the **MPD**@type="dynamic".

DASH clients operating based on such an MPD and consuming the service at the live edge typically need to request a new MPD prior to downloading a new segment. However, in order to minimise MPD requests and resulting traffic load, the client may use one or more of the following optimisations:

- If the client fetches the MPD using HTTP, the client should use conditional GET methods to reduce unnecessary network usage in the downlink.

- If the @segmentProfiles contains the `'lmsg'` brand clients may also rely on the `'lmsg'` message and request a new MPD only in case a segment is received with an 'lmsg' brand. Otherwise the client may use template constructions to continue deter-mining the URL and the segment availability start time of segments.

If the attribute **`MPD`**`@minimumUpdatePeriod` is set to a value greater than 0 then all Segments with availability start time less than the sum of the request time and the value of the **`MPD`**`@minimumUpdatePeriod` will eventually get available at the advertised position at their computed segment availability start time. Note that by providing a **`MPD`**`@minimumUpdatePeriod` is set to a value greater than 0, DASH servers reduce the polling frequency of clients, but at the same time cannot expect that clients will request an updated MPD to be informed on changes in the segment URL constructions, e.g. at the start of a new Period.

11.3.4 MPD-based live client operation based on MPD

In an extension to the description in clause 11.2.4, the client now has access to an MPD and the MPD contains the **`MPD`**`@minimumUpdatePeriod`, for example following the parameters in table 11-5. The start time of each Period is computed as period start time *PSwc*[*i*] and the MPD-URL does not include any fragment parameters.

The client fetches an MPD with parameters in table 11-5 at time `FetchTime`, at its initial location if no **`MPD.Location`** element is present, or at a location specified in any present **`MPD.Location`** element. `FetchTime` is defined as the time at which the server processes the request for the MPD from the client. The client typically should not use the time at which it actually successfully received the MPD, but is expected to take into account delay due to MPD delivery and processing. The fetch is considered successful either if the client obtains an updated MPD or the client verifies that the MPD has not been updated since the previous fetching.

If the client fetches the MPD using HTTP, the client is expected to use conditional GET methods as specified in RFC 9112 [9] to reduce unnecessary network usage in the downlink.

In an extension of bullet 3 in clause 11.2.4.1:

- The client creates a list of accessible Segments at least for each selected Representation taking into account the information in the MPD as documented in table 11-5 and the current time *NOW* by using the Period end time of the last Period as `FetchTime + MUP`.

In an extension of bullet 9 in clause 11.2.4.1:

- The client consumes media in last announced Period. Once the client is consuming media contained in the Segments towards the end of the announced Period, i.e. requesting segments with segment availability start time close to the validity time of the MPD defined as `FetchTime + MUP`, them, then the DASH client needs to fetch an MPD at its initial location if no **`MPD.Location`** element is present, or at a location specified in any present **`MPD.Location`** element.

- If the client fetches the updated MPD using HTTP, the client is expected to use conditional GET methods to reduce unnecessary network usage in the downlink.

- The client parses the MPD and generates a new segment list based on the new `FetchTime` and `MUP` of the updated MPD. The client searches for the currently consumed Adaptation Sets and Representations and continues the process of downloading segments based on the updated Segment List.

11.4 Provisioning of live content in On-Demand mode

11.4.1 Scenario

A common scenario for DASH distribution results that a live generated service is also made available for On-Demand offering after the live program is completed. The typical scenario is as follows:

- The Segments as generated for the live service are also used for the On-Demand case. This avoids reformatting and also permits to reuse the Segments that are already cached.
- The MPD is modified to reflect that the content is available as On-Demand now.
- Problems that results from live delivery may be solved, e.g. variable segment durations, or issues of segment unavailability.
- The content may be augmented with ads.
- The content may be trimmed from a longer, e.g. 24/7 stream, at the beginning and/or end.

11.4.2 Content Offering Requirements and Recommendations

In order to provide live content as On-Demand in the above scenario, the following is recommended:

- The same Segments as generated for the live distribution are reused also for static distribution.
- Typically, the Segments also will have the same URL in order to exploit caching advantages.
- An MPD should be generated latest at the end of the live session, but also may be created during an ongoing live session to document a certain window of the program that is offered for On-Demand.
- A new MPD is generated that should contain the following information
  - The **`MPD`**`@type` is set to `static`.
  - The **`MPD`**`@availabilityStartTime` may be set to any time in the past, for example the time of the original "live" MPD may reused.
  - The attributes `@timeShiftBufferDepth` and `@minimumUpdatePeriod` are not present (in contrast to the live MPD) and a `@mediaPresentationDuration` attribute is added.
  - The window offered by the MPD is expressed by appropriately setting the **`Period`**`@start` value (including the presentation time offset and the start number) and the `@mediaPresentationDuration` attribute. The wall-clock time should be maintained by offsetting the **`Period`**`@start` without changing the **`MPD`**`@availabilityStartTime`.
  - Content may be offered in the same Period structure as for live or in a different one.
  - If Periods are continuous, it is preferable to remove the Period structure.
  - If new Periods are added for Ad Insertion, the Periods preferably be added in a way that they are at Segment boundaries.
  - The same templating mode as used in the live service should also be used for static distribution.

11.4.3 Client behaviour

For a DASH client, there is basically no difference on whether the content was generated from a live service or the content is provided as On-Demand. However, there are some aspects that may be "left-overs" from a live service distribution that a DASH client is expected to be aware of:

- The Representations may show gaps in the media offering by using early terminated Periods. Such gaps are expected to be recognized and properly handled.

11.5 Availability time synchronization between client and server

11.5.1 Background

According to ISO/IEC 23009-1, in order to properly access MPDs and Segments that are available on origin servers or get available over time, DASH servers and clients should synchronize their clocks to a globally accurate time standard.

Specifically Segment Availability Times are expected to be wall-clock accurately announced in the MPD and the client needs to have access to the same time base as the MPD generation in order to enable a proper service. In order to ensure this, this clause provides server and client requirements to ensure proper operation of a live service.

11.5.2 Service Provider requirements and guidelines

If the Media Presentation is dynamic or if the **`MPD`**`@availabilityStartTime` is present then the service shall provide a Media Presentation as follows:

- The segment availability times announced in the MPD should be generated from a device that is synchronized to a globally accurate timing source, preferably using NTP.
- The MPD should contain at least one **`UTCTiming`** element with `@schemeIdURI` set to one of the following:
  - urn:mpeg:dash:utc:ntp:2014
  - urn:mpeg:dash:utc:http-head:2014
  - urn:mpeg:dash:utc:http-xsdate:2014
  - urn:mpeg:dash:utc:http-iso:2014
  - urn:mpeg:dash:utc:http-ntp:2014
- If the MPD does not contain any element **`UTCTiming`** then the segments shall be available latest at the announced segment availability time using a globally accurate timing source.
- If the MPD contains an element **`UTCTiming`** then
  - the announced timing information in the **`UTCTiming`** shall be accessible to the DASH client, and
  - the segments shall be available latest at the announced segment availability time in the MPD for any device that uses one of announced time synchronization methods at the same time.

If `urn:mpeg:dash:utc:http-head:2014` is used, then the server specified in the `@value` attribute of the **`UTCTiming`** element should be the server hosting the DASH segments such that with each request the Date general-header field in the HTTP header can be used by the client to maintain synchronization.

Note that in practical deployments segment availability may be an issue due to failures, losses, outages and so on. In this case the Server should use methods as defined in clause 11.7 to inform DASH clients about potential issues on making segments available.

A leap second is added to UTC every 18 months on average. A service provider should take into account the considerations in RFC 7164 [Y]. The MPD time does not track leap seconds. If these occur during a live service, they may advance or retard the media against the real time.

11.5.3 Client requirements and guidelines

If the Media Presentation is dynamic or if the **`MPD`**`@availabilityStartTime` is present then client is expected to do the following:

- If the MPD does not contain any element **`UTCTiming`** it should acquire an accurate wall-clock time from its system. The anticipated inaccuracy of the timing source should be taken into account when requesting segments close to their segment availability time boundaries.

- If the MPD contains one or several elements **`UTCTiming`** then the client is expected to at least use one of the announced timing information in the **`UTCTiming`** to synchronize its clock. The client must not request segments prior to the segment availability start time with reference to any of the chosen **`UTCTiming`** methods.
- The client may take into account the accuracy of the timing source as well as any transmission delays if it makes segment requests.
- Clients is expected to observe any difference between their time zone and the one identified in the MPD, as MPDs may indicate a time which is not in the same time zone as the client.
- If the client observes that segments are not available at their segment availability start time, the client is expected to use the recovery methods defined in clause 11.7.
- Clients is expected to not access the **`UTCTiming`** server more frequently than necessary.

11.6 Robust operation

11.6.1 General robustness

General Guidelines are provided ISO/IEC 23009-1 DASH spec in clause A.7 and in clause A.9 of the present document. DASH clients and servers should follow those guidelines.

11.6.2 Synchronization loss of segmenter

In order to address synchronization loss issues at the segmenter, the following options from the DASH standard should be considered with preference according to the order below:

1) The server is required to always offer a conforming media stream. In case the input stream or encoder is lost, the content author may always add dummy content. This may be done using a separate Period structure and is possible without any modifications of the standard.
2) Early Terminated Periods as included Cor.1 of the second edition of ISO/IEC 23009-1. Early Terminated Periods may be added that contain both **`Period`**`@start` and **`Period`**`@duration`. This expresses that for this Period no media is present at least for the time as expressed by the `@duration` attribute. Such Periods should only be used if Media Presentation author is experiencing issues in generating media, e.g. due to failures of a live feed. The MPD is updated using the `@minimumUpdatePeriod`, i.e. the timeline is progressing. This permits server to signal that there is an outage of media generation, but that the service is continuing. It is then up to the client to take appropriate actions.

11.6.3 Encoder clock drift

In order to support robust offering even under encoder drift circumstances, the segmenter should avoid being synced to the encoder clock. In order to improve robustness, in the case of an MPD-based offering Periods should be added in a period continuous manner. In the case of MPD and segment-based control, the producer reference box should be added to media streams in order for the media pipeline to be aware of such drifts. In this case the client should parse the segment to obtain this information.

11.6.4 Segment unavailability

To address signalling of segment unavailability between the client and server and to indicate the reason for this, it is recommended to use regular 404s. In addition, unless a UTC Timing has been defined prior in the MPD, the Date-Header specifying the time of the server should be used. In this case, the DASH client, when receiving a 404, knows that if its time is matching the Date Header, then the loss is due to a segment loss.

11.6.5 Swapping across redundant tools

To enable swapping across redundant tools doing hot and warm swaps, the following should be considered

1) the content author is offering the service redundant to the client (for example using multiple BaseURLs) and the client determines the availability of one or the other. This may be possible under certain circumstances

2) Periods may be inserted at a swap instance in order to provide the new information after swap. If possible, the offering may be continuous, but the offering may also be non-continuous from a media time perspective.

3) A completely new MPD is sent that removes all information that was available before any only maintains some time continuity. However, this tool is not fully supported yet in any DASH standard and not even considered.

There is a clear preference for the bullets above in their order 1, 2 and 3 as the service continuity is expected to be smoother with higher up in the bullet list. At the same time, it may be the case that the failure and outages are severe and only the third option may be used.

12 Ad insertion enablers

12.1 Background

This clause provides recommendations for implementing server-based ad insertion in DASH. In the server-based model, all ad-related information is expressed via MPD and segments, and ad decisions are triggered by client requests for MPDs and for resources described in them (Segments, remote periods).

The server-based model is inherently MPD-centric – all data needed to trigger ad decision is concentrated in the MPD. In case where ad break location (i.e., its start time) is unknown at the MPD generation time, it is necessary to rely on MPD update functionality.

12.2 Enablers

12.2.1 Period elements and Remote Periods

A single ad may be expressed as a single Period element.

Periods with content that is expected to be interrupted as a result of ad insertion should contain explicit start times (**`Period`**`@start`), rather than durations. This allows insertion of new periods without modifying the existing periods. If a period has media duration longer then the distance between the start of this period and the start of next period, use of start times implies that a client will start the playout of the next period at the time stated in the MPD, rather than after finishing the playout of the last segment.

An upcoming ad break is expressed as one or more Period elements. Remote periods may have default content that will be played in case dereferencing fails. After dereferencing MPD may contain zero-duration periods or/and remote Periods.

Remote Periods are resolved on demand into one or more than one Period elements. It is possible to embed parameters into the XLink URL of the corresponding remote period, in order to have them passed to the ad decision system via XLink resolver at resolution time.

The remote period may be resolved into a period with zero duration. Such a period element will not contain any Adaptation Sets.

In case of **`Period`**`@xlink:actuate="onRequest"`, XLink resolution should be done sufficiently early to ensure that there are no artefacts due to insufficient time given to download the inserted content. Care needs to be taken so that the client is given a sufficient amount of time to dereference the upcoming remote period.

In case of Period`@xlink:actuate="onRequest"`, MPD update and XLink resolution should be done sufficiently early to ensure that there are no artefacts due to insufficient time given to download the inserted content.

`Period@xlink:actuate="onRequest"` shall not be used if `MPD@type ="dynamic"`

12.2.2 Asset identifiers

AssetIdentifier descriptors identify the asset to which a Period belongs. This can be used for implementation of client functionality that depends on distinguishing between ads and main content (e.g. progress bar). Periods with the same AssetIdentifier should have identical Adaptation Sets, Initialization Segments and same DRM information (i.e., DRM systems, licenses). This allows reuse of at least some initialization data across periods of the same asset, and ensures seamless continuation of playback if inserted periods have zero duration. Period continuity may be signalled.

AssetIdentifier descriptor shall be used for distinguishing parts of the same asset within a multi-period MPD, hence it shall be used for main content.

In order to enable better tracking and reporting, unique IDs should be used for different assets.

In the absence of other asset identifier schemes, a 3GPP-defined scheme may be used with the value of `@schemeIdUri` set to `"urn:org:pss:dash:asset-id:2015"`. If used, the value of `@value` attribute descriptor is unspecified. It shall be the same for all parts of an asset.

If a Period has one-off semantics (i.e., an asset is completely contained in a single period, and its continuation is not expected in the future), the author shall not use asset identifier on these assets.

Periods that do not contain non-remote Adaptation Set elements, as well as zero-length periods shall not contain the AssetIdentifier descriptor.

12.2.3 MPD updates

MPD updates are used to implement dynamic behaviour. An updated MPD may have additional (possibly – remote) Periods. Hence, MPD update should occur as close as possible to the moment the start time of the upcoming ad break is known. Ad breaks can also be cancelled prior to their start, and such cancellation will also result in a newer MPD version (i.e., in a need for MPD update).

Frequent regular MPD updates are sufficient for implementing dynamic ad insertion. Unfortunately they create an overhead of unnecessary MPD traffic – ad breaks are rare events, while MPD updates need to be frequent enough if a cue message is expected to arrive only several seconds before the splice point. Use of HTTP conditional GET requests (i.e., allowing the server to respond with "304 Not Modified" if MPD is unchanged) is helpful in reducing this overhead, but asynchronous MPD updates avoid this overhead entirely.

13 Server And Network-assisted DASH (SAND) support

13.1 Introduction

Server And Network-assisted DASH (SAND) functionality is specified in ISO/IEC 23009-5 [54]. SAND support in PSS is described in TS 26.233 [2]. An overview of SAND functionality in ISO/IEC 23009-5 [54], 3GPP-targeted architectural considerations for SAND support, and initial use cases for SAND in context of 3GPP are described in TR 26.957 [55].

This clause describes the following:

- SAND modes to use for 3GP-DASH described in clause 13.2.
- DANE discovery procedures for SAND described in clause 13.3.
- SAND messages and protocols to use for 3GP-DASH described in clause 13.4.
- Normative behaviours on SAND message handling for DANE and 3GP-DASH client described in clause 13.5.
- Use of SAND functionality for enabling network assistance, proxy caching and consistent QoE/QoS described in clauses 13.6, 13.7 and 13.8, respectively.
- XML schema for SAND extension messages, described in clause 13.9.
- Use of SAND functionality for enabling SAND for Multi-Network support (SAND4M) in clause 13.10.

13.2 SAND modes for 3GP-DASH

3GP-DASH clients supporting SAND functionality shall support at least one of the following modes:

- 'Network Assistance', as specified in clauses 13.4, 13.5 and 13.6
- 'Proxy Caching', as specified in clauses 13.4, 13.5 and 13.7
- 'Consistent QoE/QoS', as specified in clauses 13.4, 13.5 and 13.8
- 'SAND for Multi-Network Access' as specified in clause 13.10.

13.3 DANE discovery

The SAND specification [54] provides the *sand:Channel* element in the MPD to inform the client about the location and method to communicate with the DANE. That method of DANE discovery may be used for DANEs that are in-band with respect to the media delivery path, i.e. when the MPD server may be aware of SAND functionality in the network.

An in-band DANE may also be discovered by a SAND header field in the HTTP header of an HTTP response to a request for DASH resources as defined in ISO/IEC 23009-5 [54]. In this case, no session is created between the DANE and the DASH client.

When the DANE is out-of-band with respect to the media delivery path, as is the case with the Network Assistance DANE and the Consistent QoE/QoS DANE, a more generic method for DANE discovery may be used, namely using the DNS protocol as described in TS 23.003 [57]. Toward this purpose, the UE needs a DANE Fully Qualified Domain Name (FQDN) for the DANE. The procedures for addressing and identification for Bootstrapping MBMS Service Announcement as described in clause 15.5 of TS 23.003 [57] shall be used for DANE discovery. Accordingly, the Fully Qualified Domain Name (FQDN) for the DANE shall be "dane.mnc<MNC>.mcc<MCC>.pub.3gppnetwork.org". The URL to establish the out-of-band connection with the DANE shall be:

http://dane.mnc<MNC>.mcc<MCC>.pub.3gppnetwork.org/

The DANE FQDN is composed of six labels. The last three labels shall be "pub.3gppnetwork.org". The second and third labels together shall uniquely identify the PLMN, as described in clause 15.5 of TS 23.003 [57]. The first label shall be "dane".

When receiving a DNS query on the DANE FQDN, the DNS server shall respond with the information, including IP address, of the DANE or DANEs that are available to the UE for SAND functionality, according to any of the defined SAND modes.

Specific modes of DANE may be identified with targeted DANE FQDNs, as follows:

- A Proxy-Caching DANE, if provided, shall be located at the FQDN "pcdane.mnc<MNC>.mcc<MCC>.pub.3gppnetwork.org" (first label is "pcdane").
- A Network Assistance DANE, if provided, shall be located at the FQDN "nadane.mnc<MNC>.mcc<MCC>.pub.3gppnetwork.org" (first label is "nadane");
- A Consistent QoE/QoS DANE, if provided, shall be located at the FQDN "qoedane.mnc<MNC>.mcc<MCC>.pub.3gppnetwork.org" (first label is "qoedane");

If a specific mode of DANE is queried, using the specific DANE FQDN, then the response informs of the IP address of that mode of DANE only.

13.4 SAND messages and protocols for 3GP-DASH

3GP-DASH clients supporting SAND functionality in the 'Proxy Caching' mode shall support the following SAND messages (as described further in clause 13.7):

- `ClientCapabilities`, as defined in clause 6.4.7 of ISO/IEC 23009-5 [54]
- `DaneCapabilities`, as defined in clause 6.5.9 of ISO/IEC 23009-5 [54]

- `AnticipatedRequests`, as defined in clause 6.4.1 of ISO/IEC 23009-5 [54]
- `AcceptedAlternatives`, as defined in clause 6.4.3 of ISO/IEC 23009-5 [54]
- `DeliveredAlternative`, as defined in clause 6.5.8 of ISO/IEC 23009-5 [54]
- `ResourceStatus`, as defined in clause 6.5.1 of ISO/IEC 23009-5 [54]
- `MPDValidityEndTime`, as defined in clause 6.5.4 of ISO/IEC 23009-5 [54]

In a PSS service environment with SAND support, DANEs in the 'Proxy Caching' mode shall also support these messages.

3GP-DASH clients supporting SAND functionality in the 'Network Assistance' mode shall support the following SAND messages (as described further in clause 13.6):

- `ClientCapabilities`, as defined in clause 6.4.7 of ISO/IEC 23009-5 [54]
- `DaneCapabilities`, as defined in clause 6.5.9 of ISO/IEC 23009-5 [54]
- `SharedResourceAssignment`, as defined in clause 6.5.3 of ISO/IEC 23009-5 [54]
- `SharedResourceAllocation`, as defined in clause 6.4.2 of ISO/IEC 23009-5 [54]
- `NetworkAssistanceInitiationRequest`, as defined in clause 13.6
- `NetworkAssistanceInitiationResponse`, as defined in clause 13.6
- `NetworkAssistanceTermination`, as defined in clause 13.6
- `SegmentDuration`, as defined in clause 13.6
- `DeliveryBoostRequest`, as defined in clause 13.6
- `DeliveryBoostResponse`, as defined in clause 13.6

In a PSS service environment with SAND support, DANEs in the 'Network Assistance' modes shall also support these messages.

3GP-DASH clients supporting SAND functionality in the 'Consistent QoE/QoS' mode shall support the following SAND messages (as described further in clause 13.8):

- `ClientCapabilities`, as defined in clause 6.4.7 of ISO/IEC 23009-5 [54]
- `DaneCapabilities`, as defined in clause 6.5.9 of ISO/IEC 23009-5 [54]
- `SharedResourceAssignment`, as defined in clause 6.5.3 of ISO/IEC 23009-5 [54]
- `SharedResourceAllocation`, as defined in clause 6.4.2 of ISO/IEC 23009-5 [54]
- `QoSInformation`, as defined in clause 6.5.7 of ISO/IEC 23009-5 [54]

In a PSS service environment with SAND support, DANEs in the 'Consistent QoE/QoS' modes shall also support these messages.

In case the DASH client connects to a DANE, the 3GP-DASH client shall send the status message `ClientCapabilities` in order to inform the DANE about the SAND mode(s) it supports. The 3GP-DASH client shall use the messageSetUri parameter to indicate which SAND mode(s) it supports based on the following URNs:

- `urn:3gpp:dash:sand:messageset:pc:2016` to indicate support for the 'Proxy Caching' mode
- `urn:3gpp:dash:sand:messageset:na:2016` to indicate support for the 'Network Assistance' mode
- `urn:3gpp:dash:sand:messageset:qoe:2016` to indicate support for the 'Consistent QoE/QoS' mode

- `urn:3gpp:dash:sand:messageset:sand4m:2018` to indicate support for the 'SAND for Multi-Network Access' mode

Depending on the SAND mode(s) supported by the 3GP-DASH client, one or more of these URNs may be included in the `ClientCapabilities` message.

In case the DANE connects to a 3GP-DASH client, the DANE shall send the PER message `DaneCapabilities` in order to inform the 3GP-DASH client about the SAND mode(s) it supports. The DANE shall use the messageSetUri parameter to indicate which SAND mode(s) it supports based on the following URNs:

- `urn:3gpp:dash:sand:messageset:pc:2016` to indicate support for the 'Proxy Caching' mode
- `urn:3gpp:dash:sand:messageset:na:2016` to indicate support for the 'Network Assistance' mode
- `urn:3gpp:dash:sand:messageset:qoe:2016` to indicate support for the 'Consistent QoE/QoS' mode
- `urn:3gpp:dash:sand:messageset:sand4m:2018` to indicate support for the 'SAND for Multi-Network Access' mode

Depending on the SAND mode(s) supported by the DANE, one or more of these URNs may be included in the `DaneCapabilities` message.

If the 3GP-DASH client has already discovered the DANE via the use of mode-specific FQDNs provided in clause 13.3 (i.e., with first labels 'nadane', 'qoedane' and 'pcdane' as defined in clause 13.3), the exchange of `ClientCapabilities` and `DaneCapabilities` messages shall not be performed on connection to a DANE.

If SAND is supported, HTTP shall be supported as the minimum transport protocol for carrying SAND messages. This does not preclude that other additional transport protocols could also be implemented. The mandatory use of HTTP as a minimum transport protocol for SAND messages shall be in accordance with table 13-1 (bold font represents mandatory):

Table 13-1: Mandatory usages of HTTP for carrying SAND messages

Metrics messages	**HTTP POST** HTTP headers may be used for small metrics messages.
Status messages	**HTTP headers** **HTTP POST**
PER messages	**HTTP GET**

For SAND status messages, carriage in HTTP headers shall be used for communicating with in-band DANEs, while HTTP POST shall be used for communicating with out-of-band DANEs. This distinction between two kinds of DANE is introduced in the high-level architecture in clause 4.2.8 of TS 26.233 [2].

3GP-DASH clients supporting SAND functionality as well as DANEs in the 'Network Assistance' or 'Consistent QoE/QoS' modes shall further support the WebSocket protocol specified in IETF RFC 6455 [56], provided that HTTP over TLS (HTTPS) is supported by the respective 3GP-DASH client or DANE. If HTTP over TLS (HTTPS) is not supported by a 3GP-DASH client, then that 3GP-DASH client should support the WebSocket protocol for the 'Consistent QoE/QoS' and 'Network Assistance' modes. Similarly, if HTTP over TLS (HTTPS) is not supported by a DANE, then that DANE should support the WebSocket protocol for the 'Consistent QoE/QoS' and 'Network Assistance' modes. When WebSockets is supported for the 'Consistent QoE/QoS' mode, as specified in ISO/IEC 23009-5 [54], the MPD shall contain a `sand:Channel` element whose `@schemeIdUri` is `"urn:mpeg:dash:sand:channel:websocket:2016"` and WebSocket URI in the `@endpoint` attribute.

13.5 SAND message handling behaviours for DANEs and DASH clients

13.5.1 DASH client behaviour

3GP-DASH clients supporting one or more of the following SAND modes: (i) 'Network Assistance', (ii) 'Proxy Caching', (iii) 'Consistent QoE/QoS', shall comply with the actions in table 13-2 in handling of the corresponding SAND messages.

Table 13-2: Normative behaviours on SAND message handling for 3GP-DASH clients

SAND Message	Actions	Nature
On DaneCapabilities and SharedResourceAllocation supported	Send SharedResourceAllocation	Mandatory
On SharedResourceAssignment	Select Representations to fit in `@bandwidth`	Optional
On DaneCapabilities and AnticipatedRequests supported	Send `AnticipatedRequests` with ALL future segment requests.	Mandatory
On DaneCapabilities and AcceptedAlternatives supported	Send `AcceptedAlternatives` with ANY future segment requests.	Mandatory
On MPDValidityEndTime	If `@mpdUrl` present, fetch MPD before `@validityEndTime`.	Mandatory
	Else (`@mpd` is present by SAND specification), use `@mpd` as new MPD version when `@validityEndTime` has passed.	Mandatory

13.5.2 DANE behaviour

In a PSS service environment with one or more of the following SAND modes: (i) 'Network Assistance', (ii) 'Proxy Caching', (iii) 'Consistent QoE/QoS', DANEs shall comply with the actions in table 13-3 in handling of the corresponding SAND messages.

Table 13-3: Normative behaviours on SAND message handling for DANEs in a PSS environment

SAND message	Actions	Nature
On `SharedResourceAllocation`	1. Add client to the sharing strategy	Optional
	2. Update allocation strategy	Optional
	3. Send `SharedResourceAssignment` to clients in the sharing strategy	Mandatory
On `AnticipatedRequests`	Cache resources indicated by `AnticipatedRequests` and send `ResourceStatus` to signal available resources	Mandatory
On `AcceptedAlternatives`	Send `DeliveredAlternatives` in case DANE delivers an alternative segment rather than the requested segment	Mandatory

13.6 Use of SAND for Network Assistance

13.6.1 General description

The Network Assistance function enables a 3GP-DASH client to improve the QoE of content streaming sessions, and is provided by the DANE. The DANE for this mode is out-of-band, i.e. it is not in the media delivery path. The Network Assistance communication is independent from the media server communication, hence the Network Assistance communication occurs in a separate path to the transfer of the MPD and the content segments. The media server does not need to be aware of the Network Assistance function.

Network Assistance may be made available to certain clients only, for example subject to subscription options. Client authentication may also be applied before granting access to Network Assistance. Clients are able to discover the availability and information about the Network Assistance DANE, and to establish a Network Assistance session with the DANE.

Network Assistance is based on the model of the client requesting network assistance and the DANE responding to the request. The Network Assistance functionality may be granted to a client supporting the delivery of 3GP-DASH content with either only the first or with both of the two functions below, in both cases based on the 3GP-DASH client having made a request to the DANE for Network Assistance:

- The DANE indicates to the 3GP-DASH client the highest suitable media rate for the next segment download, based on the available Representations for the content item;
- The DANE indicates to the 3GP-DASH client a temporary delivery boost for occasions when the content playback input buffer on the client risks suffering from under-run.

Once a Network Assistance session is active, the client may issue a Network Assistance call prior to fetching the next media segment from the server. The Network Assistance call consists of a single SAND signalling exchange. This exchange with the DANE activates either the first of the above functions or a sequence of both functions; the second only if the 3GP-DASH client was granted access to the function. If the client does not need a delivery boost, then the DANE omits the second function in the response to the 3GP-DASH client.

The Network Assistance functions depend on only a small part of the set of SAND capabilities.

Clause 13.6.2 specifies common functions.

Clauses 13.6.3 and 13.6.4 describe the two constituent functions of Network Assistance, namely rate recommendation and temporary delivery boost, respectively.

Clause 13.6.5 specifies the Network Assistance functions and their mapping to SAND messages.

Clause 13.6.6 specified how the Network Assistance request and response calls are constructed using the SAND message container that contains the appropriate Network Assistance messages.

Clause 13.6.7 describes an example workflow for Network Assistance.

13.6.2 Common functions

13.6.2.1 Introduction

Unless the DANE location is already known to the 3GP-DASH client, for example by pre-installing a commonly used DANE location for the operator to which the client is subscribed, the client needs to discover the DANE for Network Assistance, before being able to use Network Assistance functionality.

The DANE manages the population of clients that are eligible for Network Assistance by accepting Network Assistance session initiation calls from clients. In this way the DANE is able to provide a more reliable Network Assistance function by being aware of which clients might need relevant network resources allocated at any time.

13.6.2.2 DANE discovery

DANE discovery procedures relevant for the Network Assistance mode are described in clause 13.3.

13.6.2.3 Network Assistance session initiation

The facility of Network Assistance requires that the network is aware of its possible intended usage by every client in advance of first usage of the facility. Hence the client needs to register with the DANE, providing the location of the media server delivering the content item and the IP port at which it will be delivered, in advance of the playback starting. The network thus has the possibility to apply any authentication or policy procedure on that connection, as well as be prepared for Network Assistance usage by all clients that may register for it.

The 3GP-DASH client shall initiate a Network Assistance session with the Network Assistance DANE at a convenient stage in the process of preparing to receive PSS content. When this takes place may be dependent on the nature of the application that streams media content items, but in any case the scope of the Network Assistance session is intended

such that it starts with the start of playback of an individual content item, and ends when the playback of that content item is stopped.

13.6.2.4 Network Assistance session termination

When the 3GP-DASH client no longer requires Network Assistance facilities, it shall terminate the Network Assistance session. This could be the case for example when the playback of a streamed media content item is stopped, or the converse operation to that which occurred when the session was initiated.

13.6.3 Rate recommendation function

The 3GP-DASH client uses this function of the DANE to obtain a recommendation of the highest suitable media rate for an upcoming media segment download.

The 3GP-DASH client provides the necessary information, such as available media versions with the required bit rates, and buffer level, to the DANE. The DANE provides the response with the recommendation of the highest suitable media rate. The recommended rate is based on network estimations or predictions of available link bandwidth for the ensuing period of time. The recommended rate is neither enforced by the network, nor does the network make any commitment that the recommended rate will be honoured. How the information is gathered and relayed to the Network Assistance function of the DANE is out of scope of the present specification.

At the Network Assistance logical level this function includes the option for the client to indicate that it would benefit from a delivery boost during the following media segment download. This is specified in clause 13.6.5.2.

13.6.4 Temporary delivery boost function

The 3GP-DASH client uses this function to indicate to the network that a temporary boost, i.e. a temporary increase of network throughput for this client, would be needed in order to avoid the risk of media playback stalling due to buffer under-run, which could otherwise occur during the next media segment or soon after.

Throughput boosting may be used also at the start of a playback session to shorten the time to playout, giving a better experience for the user.

The network informs the client when the network applies the delivery boost, in order to ensure that the client is not misled as to the available link throughput, since this could lead to the client making an erroneous media rate selection when the throughput is back to normal again, without boost, and select a higher media rate than suitable for the next segment download. During a delivery boost period the client shall not select a higher media rate than indicated with the rate recommendation function. The client may return to its own normal media rate selection method only when the delivery boost period has ended. After the delivery of a segment with network boost, the network reverts to normal delivery, i.e. without boost.

13.6.5 SAND messages usage and extensions

13.6.5.1 Introduction

This clause contains the specification of the Network Assistance functions firstly as generic function calls, and how they could be mapped to SAND messages.

The Network Assistance (NA) DANE is out-of-band, i.e. not located in the media path. The 3GP-DASH client shall send the NA SAND messages as the body of HTTP requests directly to the NA DANE, using the HTTP POST method to send a Network Assistance message to the DANE.

The Network Assistance transactions between the 3GP-DASH client and the DANE at the logical level consist of the Network Assistance initiation and termination, the Network Assistance request and the Network Assistance response.

A combination of existing SAND messages as defined in the SAND specification, with usage as defined in clause 13.6.5.2, and SAND extension messages as defined in clause 13.6.5.3, are used.

The XML schema for the SAND extension messages is provided in clause 13.9.

13.6.5.2 Use of existing SAND messages

13.6.5.2.1 Shared resource allocation

The SAND status message *SharedResourceAllocation* is used in the NA request from the 3GP-DASH client to the DANE, in order to provide information about the available media bitrates for the content item to be accessed.

The parameters *operationPoints* and *bandwidth* in the SAND message *SharedResourceAllocation* shall represent each of the available media bitrates, indicated as the sum of all media components in each case.

13.6.5.2.2 Buffer level

In the case that the 3GP-DASH client is requesting a network delivery boost during the following segment, the SAND metrics message *BufferLevel* is used to inform the DANE of the current buffer level for the content item being accessed.

13.6.5.2.3 Shared resource assignment

The SAND PER message *SharedResourceAssignment* is used in the NA response from the DANE to the 3GP-DASH client, in order to provide the recommended choice of bitrate version for the next segment of the content item being accessed.

13.6.5.3 SAND message extensions

13.6.5.3.1 Network Assistance session initiation

The 3GP-DASH client initiates a Network Assistance session by sending the *NetworkAssistanceInitiation* SAND extension message to the DANE within a SAND message envelope. The generic procedure for the Network Assistance session initiation request is shown in table 13-4.

Table 13-4: Network Assistance session initiation generic procedure

Function call	Originator → destination	Parameters
Network Assistance session initiation request	UE → DANE	Media server IP address, Media delivery port number
Network Assistance session initiation response	DANE → UE	session id, port number, WebSocket requirement

In the Network Assistance session initiation request message, the following parameters shall be set by the UE:

- *MediaServerIPAddress* – the IP address of the media server;
- *MediaDeliveryPortNumber* – the port number used for the delivery of the media segments.

The Network Assistance session initiation request message syntax is shown in table 13-5.

Table 13-5: Network Assistance session initiation request message syntax

Parameter	Type	Cardinality	Description
SandMessage = NetworkAssistanceInitiationRequest	Object	1	
MediaServerIPAddress	String	1	IP address of the media server delivering content segments in the present Network Assistance session.
MediaDeliveryPortNumber	Integer	1	Port number for media server delivering content segments in the present Network Assistance session.

The *NetworkAssistanceInitiationRequest* message shall be carried in a SAND message envelope, as specified in ISO/IEC 23009-5 [54], with the following constraints:

- The *senderID* element shall be included, in order to provide a reference by which the session initiation request can be authenticated or otherwise authorised at the DANE or elsewhere in the network, relayed by the DANE receiving the message;
- The *generationTime* element may be omitted;
- The *messageId* element may be omitted.

The *NetworkAssistanceInitiationRequest* message is defined in the schema whose @schemeIdUri is:

"urn:3gpp:dash: schema:sandmessageextension:2017"

In the Network Assistance session initiation response message, the following parameters are set by the DANE:

- *SessionID* – this is used to identify the context of any future messages relating to this Network Assistance session;
- *PortNumber* – the DANE sets the port number at which all further NA communications in this session shall occur;
- *WebSocketRequirement* – the DANE indicates whether the UE shall set up a WebSocket connection to the DANE for all further NA communications in this session.

The Network Assistance session initiation response SAND extension message syntax is shown in table 13-6.

Table 13-6: Network Assistance session initiation response message syntax

Parameter	Type	Cardinality	Description
SandMessage = NetworkAssistanceInitiationResponse	Object	1	
SessionID	Integer	1	Reference to the network assistance session, allocated by the DANE. A value of 0 is reserved to indicate failure, or refusal of the NA session initiation. In case of failure, no further parameters shall be added to this message.
PortNumber	Integer	0..1	The DANE informs the 3GP-DASH client of the port number that shall be used for NA communications in the present session.
WebSocketRequirement	Object	0..1	The DANE informs the 3GP-DASH client whether a WebSocket shall be set up for the present session.

The *NetworkAssistanceInitiationResponse* message shall be carried in a SAND message envelope, as specified in ISO/IEC 23009-5 [54], with the following constraints:

- The *senderID* element shall be included, and shall be the same as that used in the corresponding session initiation request;

- The *generationTime* element may be omitted;

The *messageId* element may be omitted. The *NetworkAssistanceInitiationResponse* message is defined in the schema whose @schemeIdUri is:

"urn:3gpp:dash: schema:sandmessageextension:2017"

13.6.5.3.2 Network Assistance session termination

The 3GP-DASH client terminates a Network Assistance session by sending the *NetworkAssistanceTermination* SAND extension message to the DANE within a SAND message envelope, with parameter S*essionID* as allocated by the DANE when the session was initiated. Table 13-7 shows the generic procedure of Network Assistance session termination.

Table 13-7: Network Assistance session termination generic procedure

Function call	Originator → destination	Parameters
Network Assistance session termination request	UE → DANE	Session id
Network Assistance session termination response	DANE → UE	Session id

The Network Assistance session termination message syntax is shown in table 13-8. The same SAND message is used for both request and response, only with different semantics for the *SessionID* parameter.

Table 13-8: Network Assistance session termination message syntax

Parameter	Type	Cardinality	Description
SandMessage = NetworkAssistanceTermination	Object	1	
SessionID	Integer	1	Reference to the network assistance session that was allocated by the DANE after successful initiation. The request message shall not use value 0 for SessionID, since it is an invalid setting. A response message indicating SessionID = 0 means that an unknown SessionID was provided in the request, or some other failure.

The *NetworkAssistanceTermination* SAND extension message is defined in the schema whose @schemeIdUri is:

"urn:3gpp:dash: schema:sandmessageextension:2017"

The *NetworkAssistanceTermination* message shall be carried in a SAND message envelope, as specified in ISO/IEC 23009-5 [54], with the following constraints:

- The senderID element shall be included, in order to provide the same reference by which the Network Assistance session had previously been initiated;

- The *generationTime* element may be omitted;

- The *messageId* element may be omitted.

13.6.5.3.3 Segment duration

The segment duration SAND extension status message is used to provide the nominal segment duration for the upcoming segment, as obtained from the MPD of the content item. The generic procedure is shown in table 13-9.

Table 13-9: Segment duration provision generic procedure

Function call	Originator → destination	Parameters
Segment duration	UE → DANE	Segment duration

The *segmentDuration* message syntax is shown in table 13-10.

Table 13-10: Segment duration message syntax

Parameter	Type	Cardinality	Description
SandMessage = SegmentDuration	Object	1	
segmentDuration	Integer	1	Segment duration in milliseconds

The *SegmentDuration* message is used within the compound *NetworkAssistanceRequest* SAND transaction specified in clause 13.6.6.2.

The *SegmentDuration* message is defined in the schema whose @schemeIdUri is:

"urn:3gpp:dash: schema:sandmessageextension:2017"

13.6.5.3.4 Delivery boost request

The 3GP-DASH client may include a request for delivery boost in the NA request, when there is a risk that buffer underflow is imminent. The request is valid for the time period of the upcoming media segment. The client may request delivery boost for the fetch of the first media segment, in order to accelerate the start of playback. If buffer under-run remains to be an imminent risk then the client may repeat the delivery boost request for the following segment. The client may request delivery boosts until it attains a sufficient media buffer fullness level. The decision to grant or decline any boost request lies entirely with the network or DANE policy.

The network informs the client when the network applies the delivery boost, in order to ensure that the client is not misled as to the available link throughput, since this could lead to the client making an erroneous media rate selection when the throughput is back to normal again, without boost, and select a higher media rate than suitable for the next segment download. During a delivery boost period the client shall not select a higher media rate than indicated with the rate recommendation function. The client may return to its own normal media rate selection method only when the delivery boost period has ended. After the delivery of a segment with network boost, the network reverts to normal delivery, i.e. without boost.

The delivery boost request is actuated by including the *DeliveryBoostRequest* message in the Network Assistance request. It is classed as a SAND PED message extension. Table 13-11 shows the generic procedure.

Table 13-11: Delivery boost request generic procedure

Function call	Originator → destination	Parameters
Delivery boost request	UE → DANE	None

The syntax of the *DeliveryBoostRequest* message is shown in table 13-12.

Table 13-12: Delivery boost request message syntax

Parameter	Type	Cardinality
SandMessage = DeliveryBoostRequest	Object	1

The *DeliveryBoostRequest* message is used within the compound *NetworkAssistanceRequest* SAND transaction specified in clause 13.6.6.2.

The *DeliveryBoostRequest* message is defined in the schema whose @schemeIdUri is:

"urn:3gpp:dash: schema:sandmessageextension:2017"

13.6.5.3.6 Delivery boost response

The delivery boost response is actuated by including the *DeliveryBoostResponse* message in the Network Assistance response. It is classed as a SAND PED message extension. It indicates whether the DANE grants or declines the corresponding boost request. Table 13-13 shows the generic procedure.

Table 13-13: Delivery boost response generic procedure

Function call	Originator → destination	Parameters
Delivery boost response	DANE →UE	Request granted or not

The *DeliveryBoostResponse* message syntax is shown in table 13-14.

Table 13-14: Delivery boost response message syntax

Parameter	Type	Cardinality	Description
SandMessage = DeliveryBoostResponse	Object	1	
Status	Enum	1	Status of requested delivery boost

The status semantics are shown in table 13-15.

Table 13-15: Delivery boost response status values

Status	Semantics
boostGranted	Delivery boost granted
boostDeclined	Delivery boost declined

The *DeliveryBoostResponse* message is used within the compound *NetworkAssistanceResponse* SAND transaction specified in clause 13.6.6.3.

The *DeliveryBoostResponse* message is defined in the schema whose @schemeIdUri is:

"urn:3gpp:dash: schema:sandmessageextension:2017"

13.6.6 Network Assistance transactions

13.6.6.1 General

The Network Assistance transactions consist of the Network Assistance request and Network Assistance response compound messages. Each of these compound messages consists of several SAND messages and/or several SAND extension messages as defined in the present specification, contained in a single SAND envelope message.

13.6.6.2 Network Assistance request

The generic procedure for the Network Assistance request is shown in table 13-16.

Table 13-16: Network Assistance request generic procedure

Function call	Originator → destination	Parameters
Network Assistance request	Client → DANE	Segment duration Available bit rates Delivery boost request Buffer level

The Network Assistance request is realised by using a single SAND message envelope containing the following SAND messages:

- *SegmentDuration* SAND extension message;

- *SharedResourceAllocation* SAND status message*;*
- Optionally the *DeliveryBoostRequest* SAND extension message, if the client is requesting delivery boost during the upcoming media segment;
- BufferLevel SAND metrics message, which shall be included if the *DeliveryBoostRequest* message is included in the Network Assistance request.

The SAND message envelope is used to carry the Network Assistance request compound message, with the following constraints:

- The *senderId* element shall be used by the UE as a reference for the Network Assistance transaction;
- The *generationTime* element is not required for Network Assistance, hence it shall be omitted;
- The *messageId* element is not used and shall be omitted in all Network Assistance messages.

The segment duration and available media bit rates are derived by the client from the information contained in the MPD.

The media segment duration shall be provided using the *MediaSegmentDuration* message, as defined in clause 13.6.5.4. The *validityTime* element in the *SharedResourceAllocation* SAND message is not required to be included.

The parameters operationPoints and bandwidth in the SAND message *SharedResourceAllocation* shall represent each of the available media bitrates, indicated as the sum of all media components.

The delivery boost request is activated by the inclusion of the *DeliveryBoostRequest* message in the SAND message envelope. It has no additional parameters. If the message is present in the Network Assistance SAND message envelope then the delivery boost request is actuated. If it is not present, then the 3GP-DASH client is not making a delivery boost request for the respective segment.

The buffer level parameter may be omitted, but if the client is requesting a boost for this segment, then the buffer level shall be communicated.

An example of a complete Network Assistance request compound message structure is shown in table 13-17. Depicted is a specific example with six operation points available to the client, and the client is asserting the delivery boost request. The syntax of each of the component messages is specified normatively in clause 13.6.5.

Table 13-17: Network assistance request compound message structure example

Parameter
CommonEnvelope
senderId
SandMessage = SegmentDuration
segmentDuration
SandMessage = SharedResourceAllocation
operationPoints
Bandwidth1
Bandwidth2
Bandwidth3
Bandwidth4
Bandwidth5
Bandwidth6
SandMessage = DeliveryBoostRequest
SandMessage = BufferLevel
T
Level

13.6.6.3 Network Assistance response

The generic procedure for the Network Assistance request is shown in table 13-18:

Table 13-18: Network Assistance response generic procedure

Function call	Originator → destination	Parameters
Network Assistance response	DANE → Client	Recommended bitrate Delivery boost response

The Network Assistance response is realised by using a single SAND message envelope containing the following SAND messages:

- One *SharedResourceAssignment* SAND PER message;
- Optionally, one *DeliveryBoostResponse* SAND extension message.
- The DANE provides the recommended bitrate from those listed in the Network Assistance request using the *Bandwidth* element in the *SharedResourceAssignment* message.

The *validityTime* element is used to express the end time of validity of the recommendation. If it is omitted then the client shall assume the validity is for the whole media segment duration, i.e. for the duration indicated in the Network Assistance request.

The *DeliveryBoostResponse* message indicates whether the requested boost, if requested for the upcoming media segment, is granted or declined. If no boost request was made then the Network Assistance response shall not contain a *DeliveryBoostResponse* message.

The syntax of a complete Network Assistance response compound message example is shown in table 13-19. In this example the DANE communicates the recommended bandwidth and includes the response to the delivery boost request that was received from the client in the preceding Network Assistance request compound message.

Table 13-19: Network assistance response compound message structure example

Parameter
CommonEnvelope
SandMessage = SharedResourceAssignment
validityTime
Bandwidth
SandMessage = DeliveryBoostResponse
Status

13.6.7 Example workflow for Network Assistance

Figure 13.1 shows an example workflow for the Network Assistance use case. Here it is assumed that the Network Assistance DANE location is already known by the 3GP-DASH client, hence the discovery procedure of the out-of-band DANE is not shown. It is also assumed that the client is already aware of the location of the PSS server from which it will access the content item, about which the client informs the DANE. The individual workflow steps are described in figure 13.1.

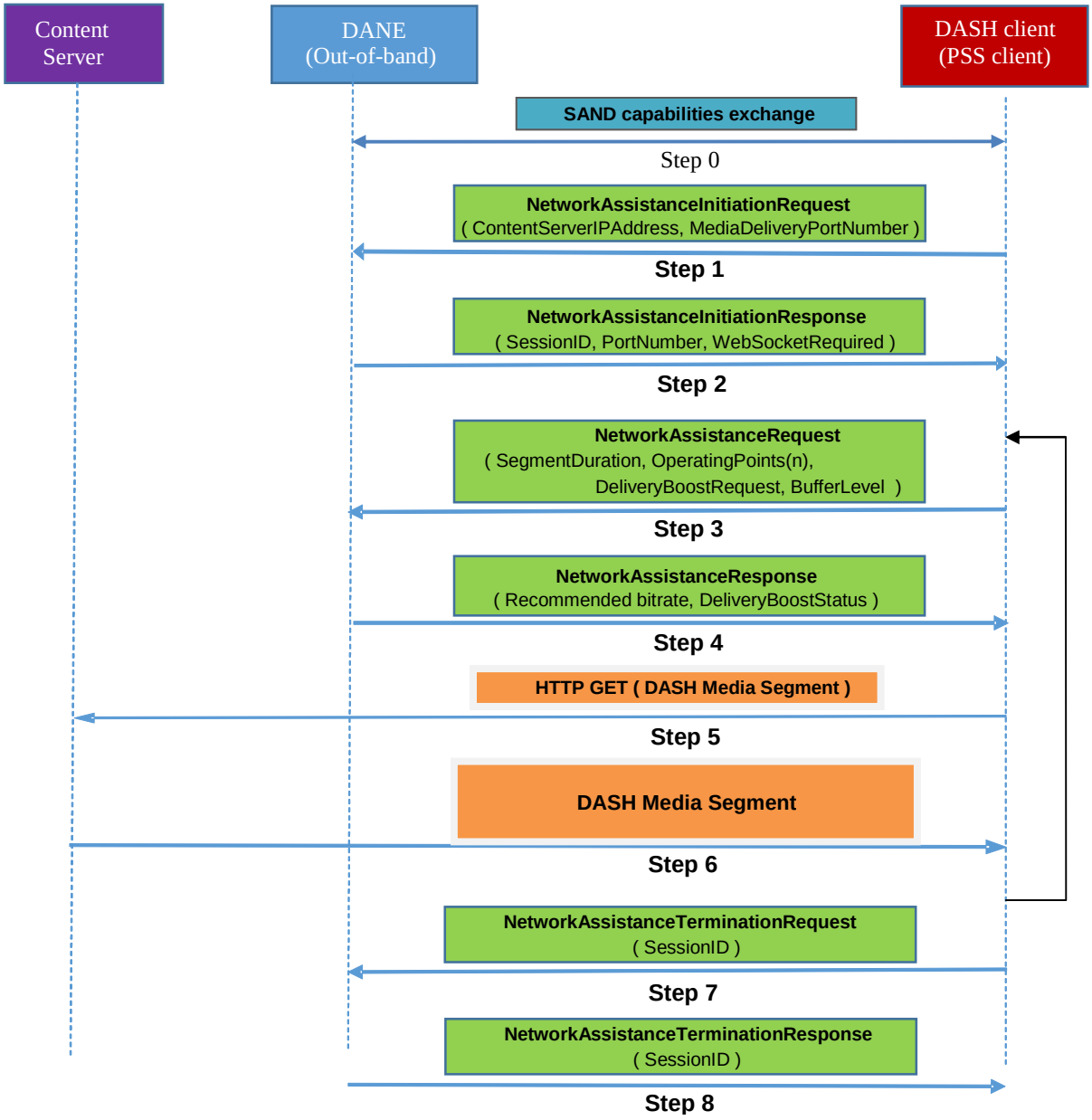


Figure 13.1: Example SAND workflow for Network Assistance

Step 0: The client and DANE exchange SAND capabilities exchange messages, as described in clause 13.4.

Step 1: The client registers its intention to make use of the Network Assistance functionality in the DANE by initiating the Network Assistance session, using the *NetworkAssistanceInitiationRequest* message, providing the IP address of the media server and port number used for delivery of media segments.

Step 2: The DANE responds to the request in step 1 with the *NetworkAssistanceInitiationResponse* message. If the request is accepted by the DANE, the response contains the session id, the port number for all further Network Assistance messages, and the indication of whether a WebSocket connection shall be set up by the client for further Network Assistance communications in the present Network Assistance session.

Step 3: The client sends the *NetworkAssistanceRequest* message to the DANE, indicating the media content item segment duration and the available versions in terms of total bitrate. If a network delivery boost is requested for the upcoming segment then the *DeliveryBoostRequest* and the *BufferLevel* messages are included in addition.

Step 4: The DANE responds with the *NetworkAssistanceResponse* message, indicating the recommended version of the content item in terms of the recommended bitrate, and whether a delivery boost is foreseen during the upcoming media segment.

Step 5: The client issues an HTTP GET to the media server in order to initiate delivery of the media segment.

Step 6: The media server delivers the requested segment to the client. If a delivery boost was granted for the present segment, the DANE or network may deliver a part or parts of the segment at a significantly higher bitrate than the media playback bitrate, in order to facilitate filling of the media buffer in the client.

Steps 3-6 are repeated for as long as the content item playback continues.

Step 7: The client has ended playback of the content item, so it sends the *NetworkAssistanceTerminationRequest* message to the DANE, indicating the session id of the Network Assistance session.

Step 8: The DANE confirms the termination of the Network Assistance session, also confirming the corresponding session id.

13.7 Use of SAND for proxy caching

13.7.1 Introduction

To realize partial representation caching, SAND can be used to inform DASH clients about partially cached representations, e.g., via use of the PER messages *ResourceStatus*, *DeliveredAlternative* and *MPDValidityEndTime*, described in clause 13.7.3. Moreover, toward realizing next segment caching, SAND can be used by DASH clients to inform the network (i.e., DANE) anticipated DASH segments, acceptable alternative content, etc. leading to next segment caching, e.g., via use of the status messages *AnticipatedRequests* and *AcceptedAlternatives*, described in clause 13.7.2. An example workflow realizing next segment caching is presented in clause 13.7.4. Further details of the proxy caching use case are in TR 26.957 [55].

DANE discovery procedures relevant for the Proxy Caching mode are described in clause 13.3. SAND messages and protocols for the Proxy Caching mode are described in clause 13.4. SAND message handling behaviours for DANEs and 3GP-DASH clients in the Proxy Caching mode are described in clause 13.5.

13.7.2 Status messages for proxy caching

13.7.2.1 AnticipatedRequests

This message allows a 3GP-DASH client to announce to a DANE its interest to a specific set of segments. The intent is to signal the set of segments in representations that the DASH client is likely to select and request soon.

3GP-DASH clients sending an *AnticipatedRequests* message shall follow the syntax and semantics in table 3 of ISO/IEC 23009-5 [54], and shall include the *sourceUrl* parameter.

13.7.2.2 AcceptedAlternatives

This message allows 3GP-DASH clients to inform DANEs on the media delivery path (typically caching DANEs), that when they request a given DASH segment, they are willing to accept other DASH segment(s) as an alternative. A 3GP-DASH client shall not include alternative segments unless it is ready to receive them and be able to play them.

3GP-DASH clients sending an *AcceptedAlternatives* message shall follow the syntax and semantics in table 5 of ISO/IEC 23009-5 [54], and shall the sourceUrl parameter.

13.7.3 PER messages for proxy caching

13.7.3.1 ResourceStatus

This message allows for a DANE to inform a 3GP-DASH client – typically in advance – about knowledge of segment availability including the caching status of the segment(s) in the DANE.

3GP-DASH clients receiving a *ResourceStatus* message shall follow the syntax and semantics in tables 10-12 of ISO/IEC 23009-5 [54].

13.7.3.2 MPDValidityEndTime

This message provides the ability to signal to the client that a given MPD, whose @type is set to 'dynamic' and @minimumUpdatePeriod is present, can only be used up to at a certain wall-clock time.

3GP-DASH clients receiving an *MPDValidityEndTime* message shall follow the syntax and semantics in table 17 of ISO/IEC 23009-5 [54].

13.7.3.3 DeliveredAlternative

As a response to an AcceptedAlternatives message sent by a 3GP-DASH client, a DANE may deliver an alternative segment rather than the requested segment. If so, the DANE shall send a DeliveredAlternative message to the 3GP-DASH client to inform that the response contains a segment alternative and not the requested segment.

3GP-DASH clients receiving a *DeliveredAlternative* message shall follow the syntax and semantics in table 23 of ISO/IEC 23009-5 [54].

13.7.4 Example workflow on SAND use for proxy caching

An example workflow realizing next segment caching is depicted in figure 13.2, where DANE (PSS Server) caches content based on SAND-based status messages received from the DASH client (PSS client). For the SAND messages depicted in figure 13.2, the *messageType* codes in table 2 of ISO/IEC 23009-5 [54] are used.

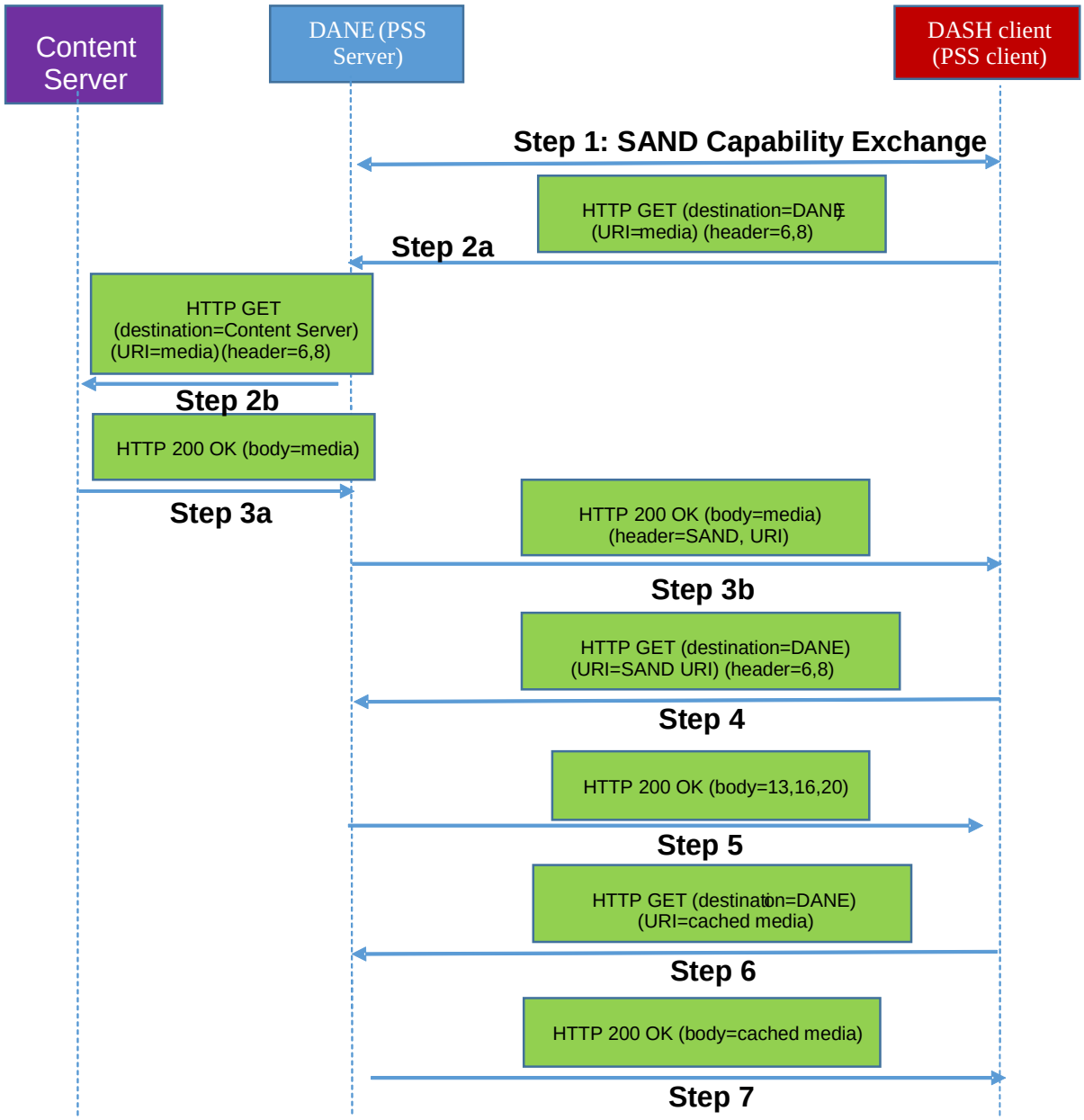


Figure 13.2: Example SAND workflow for proxy caching

Step 1: The SAND capability exchange between the DANE and client will negotiate the use of the related SAND messages for proxy caching (using the SAND messages ClientCapabilities and DaneCapabilities as described in clause 13.4). More specifically, the DANE and DASH client negotiate the use of the following SAND messages:

- PER: ResourceStatus, DeliveredAlternative, MPDValidityEndTime, DaneCapabilities
- Status Messages: AnticipatedRequests, AcceptedAlternatives, ClientCapabilities

Step 2a: Client issues an HTTP GET and sends request for media to the DANE. In the header of the HTTP request (per the standardized formats in ISO/IEC 23009-5 [54], as described in clause 13.4), client includes the SAND header that contains the status messages on proxy caching, namely on anticipated requests, accepted alternatives and/or next alternatives. DANE receives these status messages, processes them and then forwards the SAND header that contains the status messages.

Step 2b: The DANE forwards the HTTP request for the desired media to the content server, since the DANE does not have a cached version of the media. DANE forwards the HTTP headers carrying SAND messages to without any modification.

Step 3a: Content server responds with HTTP 200 OK with body containing media.

Step 3b: In the HTTP response, DANE includes SAND header to advertise availability of PER messages on proxy caching with the URI hosted at the DANE for the corresponding PER messages, namely on resource status, DANE resource status and/or delivered alternatives.

Step 4: Client issues an HTTP GET request targeting the URI hosted at the DANE to fetch the PER messages on proxy caching, namely on resource status, DANE resource status and/or delivered alternatives. In the header of the HTTP request, client may include the SAND header that contains further status messages on proxy caching, namely on anticipated requests, accepted alternatives and/or next alternatives.

Step 5: DANE responds with the HTTP OK with body containing the PER message on proxy caching, namely on resource status, DANE resource status and/or delivered alternatives.

Steps 6,7: Client requests and downloads cached media from DANE.

13.8 Use of SAND for consistent QoE/QoS

SAND can be an enabler for video-aware network resource management to provide consistent QoE/QoS for DASH clients. As such, DANE placed at the network core can allow the service provider to perform video-aware resource management. The DANE is also considered to contain a QoE reporting server to receive periodic feedback of media buffer levels from the DASH clients. This is done by establishing an application-level feedback connection between the DASH client and the DANE for each streaming flow, e.g., via an HTTP POST connection. Hence the DANE has a synthetic view of the buffer levels of all the connected streaming clients. It computes a maximum bit rate (MBR) MBR_j for each flow j and communicates the MBR_j of each flow j to the respective DASH client using SAND messages. Further details of the consistent QoE/QoS use case are in TR 26.957 [55].

Thus the DANE indirectly manages the network resources at various parts of the network by setting the QoS parameters that influence the DASH client adaptation behaviour. The DANE controls the rate adaptation of the video clients by communicating the respective MBR parameters to each client.

The video-aware resource management can be realized at the DANE via the use of SAND, where DASH clients can indicate their desired bandwidth levels through the use of the SAND status message *SharedResourceAllocation* and also can report QoE metrics. In addition, when the DANE determines the resource allocation across the DASH clients, it can inform the DASH clients about their resource assignments and throughput / QoS, which can be achieved by the SAND PER messages *SharedResourceAssignment*, *Throughput* and *QoSInformation*, as specified in clauses 6.5.3, 6.5.5 and 6.5.7 of ISO/IEC 23009-5 [54], respectively.

DANE discovery procedures relevant for the Consistent QoE/QoS mode are described in clause 13.3. SAND messages and protocols for the Consistent QoE/QoS mode are described in clause 13.4. SAND message handling behaviours for DANEs and 3GP-DASH clients in the Consistent QoE/QoS mode are described in clause 13.5.

The SAND status message *SharedResourceAllocation* shall follow the syntax and semantics in table 4 of ISO/IEC 23009-5 [54]. 3GP-DASH clients sending the *SharedResourceAllocation* message shall include the bandwidth parameter. In addition, the SAND message common envelope shall contain the senderId parameter.

The SAND PER message *SharedResourceAssignment* shall follow the syntax and semantics in table 16 of ISO/IEC 23009-5 [54]. 3GP-DASH clients receiving the *SharedResourceAssignment* message shall recognize the *clientId* and bandwidth parameters.

The SAND PER message *QoSInformation* shall follow the syntax and semantics in table 22 of ISO/IEC 23009-5 [54].

An example workflow realizing the consistent QoE/QoS is depicted in figure 13.3, where the use of the WebSocket protocol [56] is considered and the DANE functionality is hosted at the PSS server and SAND-capable DASH client capabilities are hosted in the PSS client (consistent with the SAND support in PSS as described in [2]). For the SAND messages depicted in figure 13.3, the messageType codes in table 2 of ISO/IEC 23009-5 [54] are used.

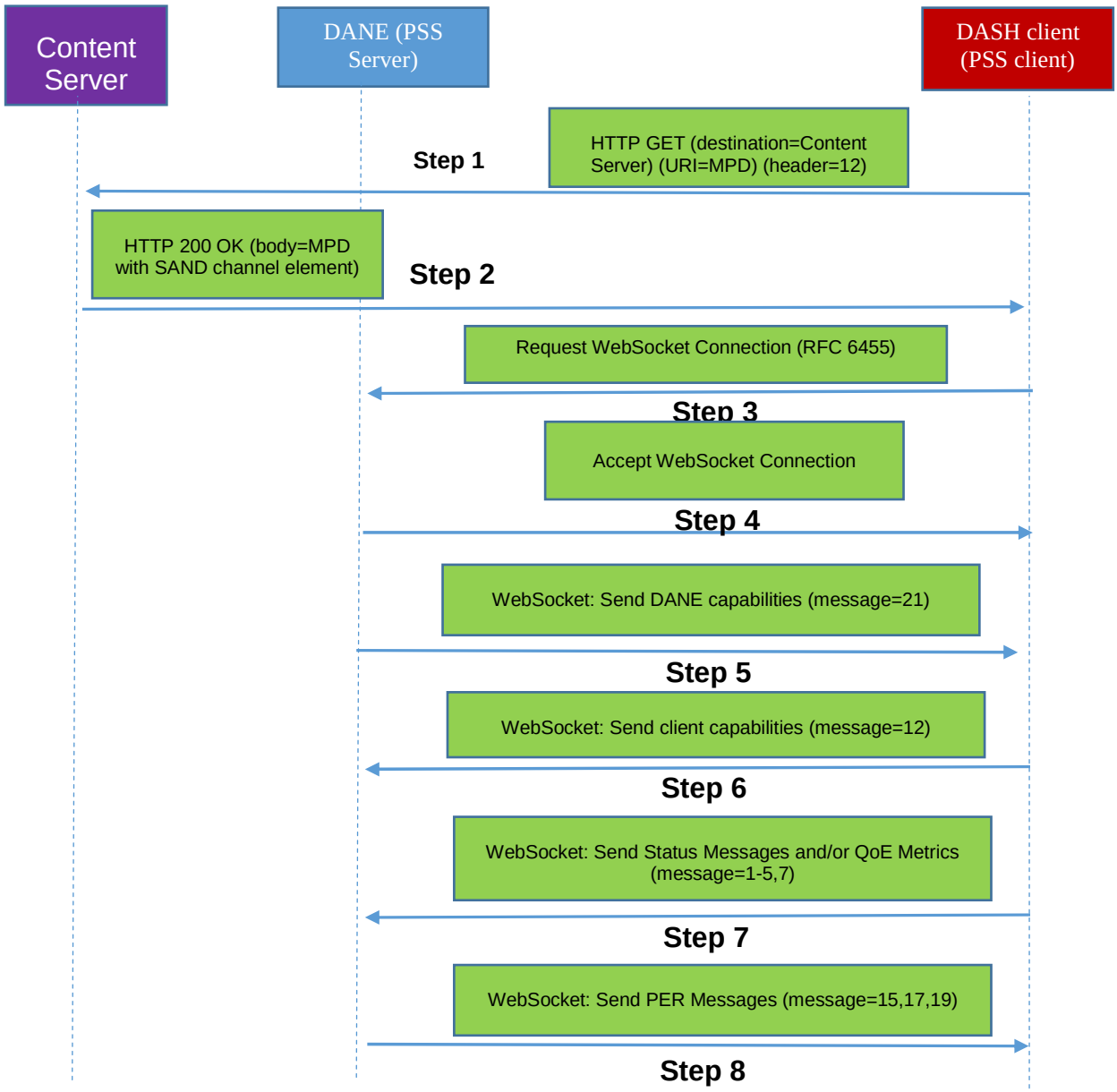


Figure 13.3: Example SAND workflow for consistent QoE/QoS

Step 1: Client issues an HTTP GET and sends request for MPD to the content server. In the header of the HTTP request, client includes the SAND header that contains the status messages on client capabilities.

Step 2: Content server responds with HTTP 200 OK with body containing the MPD. As specified in ISO/IEC 23009-5 [54], the MPD contains a sand:Channel element whose @schemeIdUri is "urn:mpeg:dash:sand:channel:websocket:2016" and WebSocket URI in the @endpoint attribute.

Steps 3, 4: The 3GP-DASH Client parses the MPD starts downloading the segments. In addition, the sand:Channel element is located in the MPD element. Using this information, the 3GP-DASH client initiates the WebSocket connection with the out-of-band DANE (e.g., located in the PSS server) as specified in RFC 6455 [56].

It is noted here that WebSocket-based SAND channel announcement may also be accomplished by the use of OMA DM [22].

Steps 5, 6, 7, 8: Upon successful establishment of the WebSocket connection between a 3GP-DASH Client and DANE, the 3GP-DASH client starts listening for incoming PER messages and may send metrics and status messages when needed. Since the WebSocket Protocol establishes a full-duplex connection, the DANE and the 3GP-DASH client may exchange SAND messages travelling simultaneously in opposite directions over the channel.

The DANE sends the DANE capabilities message to the 3GP-DASH Client (Step 5). When received, the 3GP-DASH Client replies with a client capabilities message (Step 6). As specified by ISO/IEC 23009-5 [54], the messages are WebSocket messages sent in the text format and formatted in XML. More specifically, the DANE and 3GP-DASH client negotiate the use of the following SAND messages for consistent QoE/QoS:

- PER: SharedResourceAssignment, DaneCapabilities
- Status Messages: SharedResourceAllocation, ClientCapabilities
- QoE Metrics: BufferLevel, PlayList

When the capabilities messages have been exchanged, the WebSocket connection stays open and further SAND messages relevant for consistent QoE/QoS may be exchanged, namely PER messages on shared resource assignment, QoS information and throughput, QoE metrics and status messages shared resource allocation.

13.9 SAND extension messages XML schema

The XML schema for the SAND extension messages is defined in table 13-20.

NOTE: Support for the schema is only required when required by the specific SAND mode.

Table 13-20: XML schema for SAND extension messages

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
  targetNamespace="urn:3gpp:dash:schema:sandmessageextension:2017"
  attributeFormDefault="unqualified"
  elementFormDefault="qualified"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  xmlns="urn:3gpp:dash:schema:sandmessageextension:2017">

  <xs:annotation>
    <xs:appinfo>SAND Messages Extension</xs:appinfo>
    <xs:documentation xml:lang="en">
      This Schema defines the Server And Network Assisted DASH (SAND) messages extension for 3GPP.
    </xs:documentation>
  </xs:annotation>

  <!-- SAND message: main element -->
  <xs:element name="SANDMessage" type="SANDEnvelopeType"/>

  <!-- SAND common envelope Type -->
  <xs:complexType name="SANDEnvelopeType">
    <xs:choice maxOccurs="unbounded">
      <xs:element name="NetworkAssistanceInitiationRequest"
type="NetworkAssistanceInitiationRequestType"/>
      <xs:element name="NetworkAssistanceInitiationResponse"
type="NetworkAssistanceInitiationResponseType"/>
      <xs:element name="NetworkAssistanceTermination" type="NetworkAssistanceTerminationType"/>
      <xs:element name="SegmentDuration" type="SegmentDurationType"/>
      <xs:element name="DeliveryBoostRequest" type="DeliveryBoostRequestType"/>
      <xs:element name="DeliveryBoostResponse" type="DeliveryBoostResponseType"/>
      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
    </xs:choice>
    <xs:attribute name="senderId" type="xs:token"/>
    <xs:attribute name="generationTime" type="xs:dateTime"/>
    <xs:anyAttribute namespace="##other" processContents="lax"/>
  </xs:complexType>

  <!-- SAND message base Type -->
  <xs:complexType name="SANDMessageType">
    <xs:attribute name="messageId" type="xs:unsignedInt"/>
    <xs:attribute name="validityTime" type="xs:dateTime"/>
  </xs:complexType>

  <!-- NetworkAssistanceInitiationRequest Type -->
  <xs:complexType name="NetworkAssistanceInitiationRequestType">
    <xs:attribute name="MediaServerIPAddress" type="xs:string" use="required"/>
    <xs:attribute name="PortNumber" type="xs:unsignedInt" use="required"/>
  </xs:complexType>

  <!-- NetworkAssistanceInitiationResponse Type -->
```

```
  <xs:complexType name="NetworkAssistanceInitiationResponseType">
    <xs:attribute name="sessionId" type="xs:unsignedInt" minOccurs="1" maxOccurs="1"/>
    <xs:attribute name="PortNumber" type="xs:unsignedInt" minOccurs="0" maxOccurs="1"/>
    <xs:attribute name="WebSocketRequired" type="WebSocketRequiredType" minOccurs="0"
maxOccurs="1"/>
  </xs:complexType>

  <!-- NetworkAssistanceTermination Type -->
  <xs:complexType name="NetworkAssistanceTerminationType">
    <xs:attribute name="sessionId" type="xs:unsignedInt" use="required"/>
  </xs:complexType>

  <!-- SegmentDuration Type -->
  <xs:complexType name="SegmentDurationType">
    <xs:attribute name="duration" type="xs:unsignedInt" use="required"/>
  </xs:complexType>

  <!-- DeliveryBoostRequest Type -->
  <xs:complexType name="DeliveryBoostRequestType">
    <xs:attribute name="DeliveryBoostRequest" type="xs:string" fixed="Affirmed"/>
  </xs:complexType>

  <!-- DeliveryBoostResponse Type -->
  <xs:complexType name="DeliveryBoostResponseType">
    <xs:attribute name="DeliveryBoostStatus" type="DeliveryBoostStatusType" use="required"/>
  </xs:complexType>

  <!—- DeliveryBoostStatus Type -->
  <xs:simpleType name="DeliveryBoostStatusType">
    <xs:restriction base="xs:string">
      <xs:enumeration value="granted"/>
      <xs:enumeration value="declined"/>
    </xs:restriction>
  </xs:simpleType>

  <!-- WebSocketRequiredType Type -->
  <xs:complexType name="WebSocketRequiredType">
    <xs:attribute name="WebSocketRequired" type="xs:string" fixed="Affirmed"/>
  </xs:complexType>

</xs:schema>
```

13.10 SAND for multi-network access mode

13.10.1 Introduction

The primary use case for SAND for Multi-Network Access results from the distribution of the DASH content over MBMS, for which the MBMS client acts as a DASH server or DANE in order to provide DASH formats in a manner compatible with the present document to the DASH client. The client architecture follows TS 26.347 [60], figure 5.2. In this case, DASH Server functionalities are resident on an MBMS client, that provides selected DASH functionalities. In a simplified version, the MBMS client may not provide full DASH Server functionalities but acts as a DASH Aware Network Element (DANE). This function is also particularly relevant, if the MBMS service is setup to support MBMS-operation-on-Demand (MooD) and or MBMS Service continuity. The DANE functionality in the MBMS client permits dynamic steering of the DASH client between different resources, for example broadcast distributed resources and those being provided over unicast. clause 7.4 in TS 26.347 [60] provides different options for DASH-specific interfaces between the MBMS client and the DASH client. The use of SAND in this context is also documented in clause 7.4.3 of TS 26.347 [60].

In addition to this, the second major use case is that the user may join the DASH-over-MBMS service while the service is already happening. This requires that the DASH server/DANE is able to express what Segments are cached and which ones are not cached as the user was not in broadcast coverage at earlier time. This also includes signalling of Segments that may have been lost or any of those that have been removed from the local cache.

This clause provides required and recommended functions for DANE and DASH client. Despite the requirements of this mode have been designed to fulfil the SAND for MBMS functionalities, it is not restricted for this use case and this mode may also be used in other context, in particular when using multiple networks for distribution and a dynamic steering across the network. Specifically, the following cases are considered:

- Networks go down dynamically and may re-appear

- Not all resources as announced in MPD are always accessible on all networks, e.g. broadcast resource is unavailable when device is outside broadcast coverage
- Networks may have different availability times
- Not all resources are available an all networks all the time
- The DANE may issue preferences for one network
- The information may be established by in-band channels and out of band.

13.10.2 DANE functionalities for SAND4M

To address the use cases, a DANE supporting SAND4M mode shall support the following SAND messages:

- `ResourceStatus`, as defined in clause 6.5.1 of ISO/IEC 23009-5 [54]
- `DaneResourceStatus`, as defined in clause 6.5.2 of ISO/IEC 23009-5 [54]

The DANE shall support offering the SAND messages using the assistance message channels defined in clause 13.10.4.2. The DANE may support offering the SAND messages using the enforcement and/or error message channels defined in clause 13.10.4.3 and clause 13.10.4.4, respectively.

When a SAND message is offered on a DANE, this message is expected to document the current status.

13.10.3 DASH client functionalities for SAND4M

3GP-DASH clients supporting SAND4M mode shall support the following SAND messages:

- `ResourceStatus`, as defined in clause 6.5.1 of ISO/IEC 23009-5 [54].
- `DaneResourceStatus`, as defined in clause 6.5.2 of ISO/IEC 23009-5 [54].

The DASH Client shall support the regular SAND in-band message channels defined in clause 13.10.4. The DASH client should support receiving the SAND messages using the enforcement and/or error message channels defined in clause 13.10.5.3 and clause 13.10.5.4, respectively.

The DASH Client when receiving a SAND message from the DANE is expected to use the information in the message such that it expresses the current status of the announced networks.

As long the DASH client consumes the media, it should regularly re-request the SAND message from the same URL as provided in the message channel defined in clause 13.10.4.2. If doing so, it should use a conditional HTTP GET in order to be informed if the SAND message has been updated.

13.10.4 Message Channel

13.10.4.1 General

The following scenarios are considered for exchange between the DANE and the DASH client in the SAND4M mode.

- *Client assistance:* A scenario for which the message is provided as auxiliary information for the client, but the service will be continued even if the client ignores the message. This is for example the case when the service provider provides information on the availability of additional networks that may be accessed by the DASH client to request the content. For protocols and methods, see clause 13.10.4.2.
- *Client enforcement:* A scenario for which the client requires to act, the network provides suitable alternatives for future requests. The DANE cannot or is not willing to respond to the request with a valid resource, but provides suitable alternatives. For protocols and methods, see clause 13.10.4.3.
- *Error cases:* A scenario for which the client is informed that the request is not valid and the network provides the reason and possible resolutions for the problem. The DANE cannot respond to the request with a valid resource. For protocols and methods, see clause 13.10.4.4.

13.10.4.2 Assistance

For assistance, a dedicated HTTP header field into the response for a segment request that indicates a notification that the DANE has SAND messages to send to the DASH client shall be used. Upon receiving an HTTP entity that contains the SAND header field in its entity head, the DASH client should issue an HTTP GET request to the indicated element to receive the SAND message.

The following ABNF syntax applies for the header field:

- SAND-header-field = "MPEG-DASH-SAND" ":" element-address
- element-address = absolute-URI

The field `absolute-URI` takes the syntax from RFC3986 [17]. The SAND header field provides the URI to the SAND message that is to be fetched by the DASH client using an HTTP GET method.

The message channel may be offered with segment requests or MPD requests and MPD update requests.

13.10.4.3 Enforcement

For enforcement, the DANE shall use a 300 Multiple Choices response with the following details:

- the response includes an entity containing a SAND message. The entity format is specified by the media type given in the Content-Type and shall be set to `sand+xml`, as defined in annex C of ISO/IEC 23009-5 [54].
- The response should not include the Location field to avoid the use of the `Location` field value by the user agent for automatic redirection.

This response is cacheable unless indicated otherwise.

13.10.4.4 Error case

For error cases, the DANE shall use of a suitable 4xx error code. The response may include a SAND message from which the client can deduce the reason for the error code and potential resolution of the problem.

14 Auxiliary metrics and reporting

14.1 General

Auxiliary metrics pertain to parameters known to the 3GP-DASH client other than QoE metrics as defined in clause 10, In this version of the specification, those metrics are specific to the measurement of service interactivity related usage/consumption, and their corresponding definition, reporting scheme and report protocol.

Interactivity usage reporting by the 3GP-DASH client is optional. However, a client which supports reporting of interactivity usage metrics shall do so in accordance to the metrics configuration and reporting mechanism as described in the sub-clauses of clause 14.2.

14.2 Interactivity Usage metrics and reporting

14.2.1 Introduction

This clause specifies the interactivity usage metrics that the DASH client, if capable and instructed to do so, shall report to a network-based report server. These metrics are conveyed by the MPD (via the **`MPD.Metrics.Reporting`** element) and the DASH client will use the DASH API as described in clause 14.3 to pass the metrics to the appropriate interactivity-capable application. Consequently, interactivity usage measurements performed by the application, in accordance to the metrics, are returned to the DASH client via the same API. The interactivity usage metric syntax is described in clause 14.2.2.1, and the detailed metric definitions are provided in clauses 14.2.2.2 and 14.2.2.3.

The reporting scheme is defined in clause 14.2.3, the metrics to be supported by 3GP-DASH clients supporting the interactivity usage metrics reporting feature are specified in clause 14.2.4, and the interactivity usage reporting protocol is defined in clause 14.2.5.

14.2.2 Interactivity Usage metric definitions

14.2.2.1 Introduction

This clause provides the interactivity usage metric definitions and measurement framework.

The semantics are defined using an abstract syntax. clause 14.2.5 provides a mapping to an XML schema. Items in this abstract syntax have one of the following primitive types (`Integer`, `Real`, `Boolean`, `Enum`, `String`) or one of the following compound types:

- `Objects`: an unordered sequence of (`key`, `value`) pairs, where the key always has string type and is unique within the sequence.
- `List`: an ordered list of items.
- `Set`: an unordered set of items.

Additionally, there are two kinds of timestamp defined, i.e., *real time* (wall-clock time) and *media time.*

14.2.2.2 Interactivity summary

The **`IntySummary`** element as shown in table 14.2.2.2.1 summarizes the measurements of interactivity usage according to different metrics such as user consumption of rendered interactivity content or engagement with user interface (UI) functionality, such as viewing, clicking on or selection of hyperlinks, radio buttons, check boxes and other forms of UI displays or controls. Occurrences of opportunities for such user input may be pre-scheduled or unplanned during the playout of a main program. Examples of interactivity content include advertisements and auxiliary content accessible under user control during the occurrence of an interactivity event during a main program.

Table 14.2.2.2.1: Interactivity summary

Key	Type	Description
`IntySummary`	`Object`	Summarized information on interactivity usage associated with the set of interactivity events summarized by an interactivity usage report
`Parameters`	`List`	Collection of one or more parameters pertaining to interactivity usage
`consumptionDuration`	`Integer`	Total measured time duration in milliseconds of user consumption of rendered interactivity content (e.g., video or audio)
`engagementInterval`	`Integer`	Total measured time duration in milliseconds of user engagement with interactivity content via UI controls
`clickthroughStart`	`List`	List of start times, by media presentation time, of click-through actions associated with user access of interactivity-related resources offered by UI controls
`PrivateExt`	`List`	Container for proprietary or application-specific extensions on parameters to be included in the interactivity usage report

14.2.2.3 Interactivity event list

The **IntyEventList** element as shown in table 14.2.2.3.1 contains a time-ordered list of interactivity usage events and the details of the measured interactivity usage information during each event.

Table 14.2.2.3.1: Interactivity event list

IntyEventList			List	A time-ordered list of interactivity events occurring during the playout of the main program, each containing detailed information on the incidences of interactivity usage during that event, as covered by an instance of the interactivity usage report.
	Entry		Object	A record of a single interactivity event.
		mStart	Media Time	The media presentation time at the start of the interactivity event.
		mStop	Media Time	The media presentation time at the end of the interactivity event.
		Rendering	List	A list of time intervals within the duration of an interactivity event during which the user consumes the rendered interactivity content (e.g., video or audio).
		rStart	Media Time	The media presentation time at the start of interactivity content rendering resulting from, for example, the user's click of an embedded hyperlink, or engagement with UI controls, present in an advertisement that is initially displayed during an interactivity event.
		rStop	Media Time	The media presentation time at the end of consumption of interactivity content as measured by the time of conclusion of that content, or when the user selects another interactivity content item for rendering, whichever occurs earlier.
		Engagement	List	A list of time occurrences within the duration of an interactivity event at which the user engages with the interactivity content via UI controls.
		eStart	Media Time	The media presentation time at the start of user engagement with interactivity content via UI controls via UI controls.
		Clickthrough	List	A list of click-through actions performed by the user to access interactivity-related resources, as denoted by embedded hyperlinks in the interactivity content.
		cStart	Media Time	The media presentation time at which user performs a click/selection.
		PrivateExt	List	Container for proprietary or application-specific extensions on parameters to be included in this interactivity event entry.

14.2.3 Interactivity usage reporting scheme

This clause specifies a 3GP-DASH interactivity usage reporting scheme.

The interactivity usage reporting scheme is signalled using the **Reporting** element in the **Metrics** element. The URN to be used for the **Reporting**@schemeIdUri shall be "urn:3GPP:ns:PSS:DASH:IU15".

The reporting scheme shall use the interactivity usage reporting protocol as defined in clause 14.2.5.

The semantics and XML syntax of the scheme information for the 3GP-DASH interactivity usage reporting scheme are specified in table 14.2.3.1 and table 14.2.3.2, respectively.

Table 14.2.3.1: Semantics of Interactivity Usage reporting scheme information

Element or attribute name	Use	Description
@metrics	M	This attribute lists all interactivity usage metrics (as a list of interactivity metric keys as defined in clauses 14.2.2.2 and 14.2.2.3).
@apn	O	This attribute gives the access point that should be used for sending the interactivity usage reports.
@format	O	This field gives the requested format for the reports. Possible formats are: "uncompressed" and "gzip".
@samplepercentage	O	Percentage of the clients that should report interactivity usage. The client should use a random number generator and compare the generated output with the given percentage in determining whether to perform reporting.
@reportingserver	M	The reporting server URL to which the interactivity usage reports will be sent.
@reportTime	O	If present, is indication to the DASH client that each interactivity usage report by the DASH client should be accompanied by the wall-clock time at which the report is sent.
@reportinginterval	O	Indicates the time interval(s) between successive transmission of interactivity usage reports. If present, @reportingInterval=n indicates that the client should send a report every n-th second provided that new metrics information has become available since the previous report.
GroupID	0..1	A set of one or more identifiers, representing aliases associated with and known to the client devices, used to identify that/those device(s) which should perform interactivity usage reporting. Presence of this attribute and @samplePercentage should be mutually exclusive.
Legend: For attributes: M=Mandatory, O=Optional, OD=Optional with Default Value, CM=Conditionally Mandatory. For elements: <minOccurs>…<maxOccurs> (N=unbounded) Elements are **bold**; attributes are non-bold and preceded with an @		

Table 14.2.3.2: Syntax of Interactivity Usage reporting scheme information

```
<?xml version="1.0"?>
<xs:schema targetNamespace="urn:3GPP:ns:PSS:AdaptiveHTTPStreaming:2009:qm"
    attributeFormDefault="unqualified"
    elementFormDefault="qualified"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:xlink="http://www.w3.org/1999/xlink"
    xmlns="urn:3GPP:ns:PSS:AdaptiveHTTPStreaming:2018:iu">

    <xs:annotation>
        <xs:appinfo>3GPP DASH Interactivity Usage Reporting</xs:appinfo>
        <xs:documentation xml:lang="en">
            This Schema defines the interactivity reporting scheme information for 3GPP DASH.
        </xs:documentation>
    </xs:annotation>

    <xs:element name="ThreeGPIntyUsageReporting" type="SimpleIntyReportingType"/>

    <xs:complexType name="SimpleIntyReportingType">
        <xs:sequence>
            <xs:element name="GroupID" type="xs:string" minOccurs="0"/>
        </xs:sequence>
        <xs:attribute name="metrics" type="xs:string" use="required"/>
        <xs:attribute name="apn" type="xs:string" use="optional"/>
        <xs:attribute name="format" type="FormatType" use="optional"/>
        <xs:attribute name="samplePercentage" type="xs:double" use="optional"/>
        <xs:attribute name="reportingServer" type="xs:anyURI" use="required"/>
        <xs:attribute name="reportingTime" type="xs:boolean" use="optional"/>
        <xs:attribute name="reportingInterval" type="xs:unsignedInt" use="optional"/>
        <xs:anyAttribute processContents="skip"/>
    </xs:complexType>

    <xs:simpleType name="FormatType">
        <xs:restriction base="xs:string">
            <xs:enumeration value="uncompressed" />
            <xs:enumeration value="gzip" />
        </xs:restriction>
    </xs:simpleType>

</xs:schema>
```

14.2.4 Interactivity metrics

The following metrics shall be supported by 3GP-DASH clients supporting the interactivity usage metrics reporting feature.

- Interactivity Usage Summary information (clause 14.2.2.2); and
- Interactivity Event List information (clause 14.2.2.3).

The `@metrics` attribute of the interactivity usage reporting scheme information in clause 14.2.3 contains a list of interactivity usage metric keys comprising the entirety of such metrics that the DASH client shall collect and report.

14.2.5 Interactivity Usage reporting protocol

14.2.5.1 General

The interactivity usage reporting protocol consists of:

- the XML-based report format defined in clause 14.2.5.2; and
- the reporting protocol defined in clause 14.2.5.3.

The MIME type of an XML-formatted interactivity usage report shall be "`application/3gpdash-iu-report+xml`" as defined in annex J.

14.2.5.2 Report format

The interactivity usage report is formatted as an XML document that complies with the following XML schema:

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema"
xmlns="urn:3gpp:metadata:2018:HSD:intyusagereport"
targetNamespace="urn:3gpp:metadata:2018:HSD:intyusagereport" elementFormDefault="qualified">
   <xs:element name="IntyUsageReport" type="IntyUsageReportType"/>
   <xs:complexType name="IntyUsageReportType">
      <xs:choice>
         <xs:element name="IntySummary">
            <xs:complexType>
               <xs:sequence>
                  <xs:element name="ClickThrough" minOccurs="0" maxOccurs="unbounded">
                     <xs:complexType>
                        <xs:attribute name="cStart" type="xs:dateTime"/>
                        <xs:anyAttribute processContents="skip"/>
                     </xs:complexType>
                  </xs:element>
                  <xs:element name="PrivateExtension" minOccurs="0"/>
                  <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
               </xs:sequence>
               <xs:attribute name="consumptionDuration" type="xs:duration"/>
               <xs:attribute name="engagementInterval" type="xs:duration"/>
               <xs:anyAttribute processContents="skip"/>
            </xs:complexType>
         </xs:element>
         <xs:element name="IntyEventList">
            <xs:complexType>
               <xs:sequence>
                  <xs:element name="Entry" maxOccurs="unbounded">
                     <xs:complexType>
                        <xs:sequence>
                           <xs:element name="Rendering" minOccurs="0" maxOccurs="unbounded">
                              <xs:complexType>
                                 <xs:attribute name="rStart" type="xs:unsignedLong"
use="required"/>
                                 <xs:attribute name="rStop" type="xs:unsignedLong"/>
                              </xs:complexType>
                           </xs:element>
                           <xs:element name="Engagement" minOccurs="0" maxOccurs="unbounded">
```

```
                              <xs:complexType>
                                 <xs:attribute name="eStart" type="xs:unsignedLong"
use="required"/>
                              </xs:complexType>
                           </xs:element>
                           <xs:element name="ClickThrough" minOccurs="0" maxOccurs="unbounded">
                              <xs:complexType>
                                 <xs:attribute name="cStart" type="xs:dateTime"/>
                                 <xs:anyAttribute processContents="skip"/>
                              </xs:complexType>
                           </xs:element>
                           <xs:element name="PrivateExtension" minOccurs="0"/>
                           <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
                        </xs:sequence>
                        <xs:attribute name="mStart" type="xs:unsignedLong" use="required"/>
                        <xs:attribute name="mStop" type="xs:unsignedLong" use="required"/>
                        <xs:anyAttribute processContents="skip"/>
                     </xs:complexType>
                  </xs:element>
               </xs:sequence>
            </xs:complexType>
         </xs:element>
      </xs:choice>
      <xs:attribute name="mediaPresentationId" type="xs:string" use="required"/>
      <xs:attribute name="periodId" type="xs:string" use="required"/>
      <xs:attribute name="reportTime" type="xs:dateTime" use="required"/>
      <xs:anyAttribute processContents="skip"/>
   </xs:complexType>
</xs:schema>
```

14.2.5.3 Reporting protocol

In the MPD configuration, if a specific metrics server has been configured, the client shall send interactivity usage reports using the HTTP (RFC 2616) [9] POST request carrying XML formatted metadata in its body. An example interactivity usage reporting based on HTTP POST request signalling is shown below:

```
POST http://www.exampleserver.com HTTP/1.1
Host: 192.68.1.1
User-Agent: Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 6.1; Trident/4.0)
Content-Type: text/xml; charset=utf-8
Content-Length: 4408
<?xml version="1.0" encoding="UTF-8"?>
<IntyUsageReport xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns="urn:3gpp:metadata:2018:HSD:intyusagereport"
xsi:schemaLocation="urn:3gpp:metadata:2018:HSD:intyusagereport IntyUsageReport.xsd">
mediaPresentationId="NFLGame-xyz_123" periodId="8" reportTime="2019-02-17T09:30:47Z"
    <IntyEventList>
        <Entry mStart="12345678294967295" mStop="12345678305767295">
            <Rendering rStart="12345678295417295" cStop="12345678300817295"/>
            <Engagement eStart="12345678301267295"/>
        </Entry>
    </IntyEventList>
</IntyUsageReport>
```

15 Service Interactivity support via event signalling and DASH APIs

The notification of upcoming occurrences of events pertaining to service interactivity, in the context of a DASH streaming service, and regardless of whether that service is delivered via unicast or broadcast, is provided by the signalling of interactivity-specific event information. Such interactivity event signalling may be instantiated as a DASH Event Stream, in the form of MPD Events or in-band event messages (i.e., "`emsg`") as defined in ISO/IEC 23009-1 [43], or as samples in an ISOBMFF timed metadata track as defined in ISO/IEC 14496-12 [11].

As described in clause 8.8, interactivity-related events are identified by the appropriate scheme identifier such that the DASH client can forward the event metadata to the subscribing interactivity application.

The 3GP-DASH client should support the event and timed metadata processing model and the WebIDL API as specified in [62]. If supported, then:

1. The processing of interactivity event signalling, and subsequent delivery of event metadata to the interactivity application shall comply with the DASH-IF's "DASH Player's Application Events and Timed Metadata Processing Model and APIs" specification [62].

2. The API between the 3GP-DASH client and consuming interactivity application, pertaining to the subscription and notification delivery of such interactivity event signalling information, shall comply with the WebIDL API as specified in [62].

The service interactivity application could be implemented as either a native application or a Web application.

Annex A (informative): Example DASH client behaviour

A.1 Introduction

The information on client behaviour is purely informative and does not imply any normative procedures on DASH client implementations. The annex primarily refers to annex A of ISO/IEC 23009-1 [43].

A.2 Overview

See ISO/IEC 23009-1 [43], clause A.2.

A.3 Segment List generation

See ISO/IEC 23009-1 [43], clause A.3.

Table A.1: Void

A.4 Seeking

See ISO/IEC 23009-1 [43], clause A.4.

A.5 Support for trick modes

See ISO/IEC 23009-1 [43], clause A.5.

A.6 Switching Representations

See ISO/IEC 23009-1 [43], clause A.6.

A.7 Reaction to error codes

See ISO/IEC 23009-1 [43], clause A.7.

A.8 Encoder clock drift control

See ISO/IEC 23009-1 [43], clause A.8.

A.9 Handling partial file responses

If the DASH client issued a regular HTTP GET request including an accept header with `application/3gpp-partial`, i.e. a partial-file-accept request. it may receive one of the following responses.

- 200 OK with the Content-Type set to `application/3gpp-partial` and the message format according to the definition in clause 7.9.2.2 of TS 26.346 [42].
- `416 Requested Range Not Satisfiable` with the additional information according to the definition in clause 7.9.2.2 of TS 26.346 [42].

If the **`416`** code is received, the 3GP-DASH client should conceal the lost media that is contained in this lost segment and should continue its regular operation, by downloading the next Segment in the Representation that contains a random access point, i.e. a SAP type 1 or 2.

If the 200 OK code (or other 2xx success status codes) with the Content-Type set to `application/3gpp-partial` is received, the 3GP-DASH client should continue its regular operation, by downloading the next Segment in the Representation that contains a random access point, i.e. a SAP type 1 or 2. In addition, the 3GP-DASH client may use the information in the body of the response in order to advantageously conceal the lost media as follows:

- If the content range starting from the first byte is included in the response,
  - the 3GP-DASH client may decode and present all complete access units that are included in the initial byte range.
  - The 3GP-DASH client may parse the Segment Index and/or movie fragment header in order to identify SAPs or sync samples that may be used to decode and render additional information.
- If the content range starting from the first position is not included in the response and the extension header `3gpp-access-position` is not present, the 3GP DASH client may discard all information and treat the media segment as lost.
- If the content range starting from the first position is not included in the response and the extension header `3gpp-access-position` is present and has a value that is part of the content range,
  - then the 3GP-DASH client may access and decode from the indicated position using regular box parsing mechanisms in order to identify SAPs or sync samples that may be used to decode and render additional information contained in this range or in other ranges.

A.10 Utilization of QoS information

A DASH client may take into consideration available QoS information when requesting representations such that the consumed content bandwidth remains within the limits established by the signalled QoS information, see clause 6.8.3 of 3GPP TR 26.938 [46].

A.11 Utilization of Quality Metadata

Quality metadata refers to video quality metrics based on associated measurements of the media data, as described in clause 16 of TS 26.244 [4]. A DASH client may take into consideration Quality Metadata when requesting representations, see clause 6.17 of TR 26.938 [46].

Annex B (normative): Media Presentation Description Schema

B.1 Introduction

The main schema is provided in clause B.2 in table B-1. The main schema refers to the extension schema in clause B.3 and in clause 8.3.

B.2 Main schema

The XML schema for the MPD is defined in annex B of ISO/IEC 23009-1 [43]. Table B-1 provides the MPD type and the inclusion of the 3GPP extensions in the main schema.

Table B-1: 3GPP extensions in main schema

```
<!-- MPD Type -->
<xs:complexType name="MPDtype">
  <xs:sequence>
    <xs:element name="ProgramInformation" type="ProgramInformationType" minOccurs="0"
maxOccurs="unbounded"/>
    <xs:element name="UTCTiming" type="DescriptorType" minOccurs="0" maxOccurs="unbounded"/>
    <xs:element name="BaseURL" type="BaseURLType" minOccurs="0" maxOccurs="unbounded"/>
    <xs:element name="Location" type="xs:anyURI" minOccurs="0" maxOccurs="unbounded"/>
    <xs:element name="Period" type="PeriodType" maxOccurs="unbounded"/>
    <xs:element name="Metrics" type="MetricsType" minOccurs="0" maxOccurs="unbounded"/>
    <xs:element ref="x3gpp:DeltaSupport" minOccurs="0"/>
  </xs:sequence>
  <xs:attribute name="id" type="xs:string"/>
  <xs:attribute name="profiles" type="xs:string" use="required"/>
  <xs:attribute name="type" type="PresentationType" default="static"/>
  <xs:attribute name="availabilityStartTime" type="xs:dateTime"/>
  <xs:attribute name="availabilityEndTime" type="xs:dateTime"/>
  <xs:attribute name="mediaPresentationDuration" type="xs:duration"/>
  <xs:attribute name="minimumUpdatePeriod" type="xs:duration"/>
  <xs:attribute name="minBufferTime" type="xs:duration" use="required"/>
  <xs:attribute name="timeShiftBufferDepth" type="xs:duration"/>
  <xs:attribute name="suggestedPresentationDelay" type="xs:duration"/>
  <xs:attribute name="maxSegmentDuration" type="xs:duration"/>
  <xs:attribute name="maxSubsegmentDuration" type="xs:duration"/>
  <xs:anyAttribute namespace="##other" processContents="lax"/>
</xs:complexType>
```

B.3 3GPP extension schema

Table B-2: XML schema of the 3GPP extensions for MPD

```
<?xml version="1.0"?>
<xs:schema targetNamespace="urn:3GPP:ns:DASH:MPD-ext:2011"
    attributeFormDefault="unqualified" elementFormDefault="qualified"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns="urn:3GPP:ns:DASH:MPD-ext:2011">
    <xs:annotation>
        <xs:appinfo>Extensions to Media Presentation Description for 3GPP</xs:appinfo>
    </xs:annotation>

    <xs:element name="DeltaSupport" type="DeltaSupportType"/>

    <!--DeltaSupport for the MPD -->
    <xs:complexType name="DeltaSupportType">
        <xs:sequence>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
```

```
        <xs:attribute name="sourceURL" type="xs:anyURI" use="required"/>
        <xs:attribute name="availabilityDuration" type="xs:duration"/>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>


</xs:schema>
```

Annex C (normative): Descriptor scheme definitions

C.1 Introduction

This annex defines descriptors that are defined in the present document. In particular the following descriptors are defined

- 3D Video disparity range descriptor scheme in clause C.4.
- 3D Video display information descriptor scheme in clause C.5.

NOTE: Other descriptors are defined in ISO/IEC 23009-1 [43], clause 5.9.

C.2 Void

Table C.1: Void

C.3 Void

C.4 3D Video Disparity Range description scheme

The 3D Video Disparity Range description scheme may only be signalled in **`SupplementalProperty`** element with SupplementalProperty@schemeIdUri set to urn:3GPP:dash:3dVideoDisparityRange:2013. When present the 3D Video Disparity Range description scheme shall be on the Adaptation Set level. The SupplementalProperty@value of 3D Video Disparity Range description scheme shall be present and is defined below using ABNF [40]:

@value="disparityRange"

disparityRange = disparityMinValue disparityMaxValue

disparityMinValue = [%x2D]1*3DIGIT

disparityMaxValue = [%x2D]1*3DIGIT

disparityMinValue and *disparityMaxValue* are expressed as the minimum and the maximum disparity, in units of pixels, divided by the width of the 3D video content, in units of pixels, multiplied by one hundred. *disparityMinValue* and *disparityMaxValue* shall be the same for all Representations included in an Adaptation Set containing this descriptor scheme.

NOTE 1: The disparity is the difference between the horizontal positions of a pixel representing the same point in space in the right and left views of a stereoscopic 3D video frame. Positive disparity values move the object away from the viewer whilst negative values move it towards the viewer. A value of zero places the object in the plane of the display screen.

NOTE 2: The DASH client can calculate the perceived depth of the stereoscopic 3D content based on the provided disparity range and the prevailing contextual information, such as a screen size used to display the content, and the calculated or predefined distance of a viewer from the display. Based on the calculated depth range perceived by a user, the DASH client can determine applicability of the stereoscopic 3D video content provided for a given Adaptation Set [41].

C.5 3D Video Display Information description scheme

The 3D Video Display Information description scheme may only be signalled in **`SupplementalProperty`** element with SupplementalProperty@schemeIdUri urn:3GPP:dash:3dVideoDisplayInformation:2013. When present the 3D Video Display Information description scheme shall be present on the Adaptation Set level. The SupplementalProperty@value of 3D Video Display Information description scheme shall be present and is defined below using ABNF [40]:

@value="displayInformation"

displayInformation = displayWidth [viewingDistance]

displayWidth = 1*5DIGIT

viewingDistance = 1*5DIGIT

displayWidth and *viewingDistance* provide actual values for which the stereoscopic 3D video content is optimized and are expressed in millimetres. The *displayWidth* defines the width of the part of the screen actually used for displaying the 3D video content.

NOTE: The DASH client can choose the Adaptation Set which contains the 3D Video Display Information Scheme with *displayWidth* closest to the actual display size used by the DASH client to display the content.

Annex D (informative): MPD examples

D.1 On-Demand service

Table D.1 provides an example MPD for an On-Demand service.

Table D.1: Example MPD for an On-Demand service

```
<?xml version="1.0"?>
<MPD
    profiles="urn:3GPP:PSS:profile:DASH10"
    type="static"
    minBufferTime="PT10S"
    mediaPresentationDuration="PT2H"
    availabilityStartTime="2010-04-01T09:30:47Z"
    availabilityEndTime="2010-04-07T09:30:47Z"
    xsi:schemaLocation="urn:mpeg:dash:schema:mpd:2011 3GPP-Rel10-MPD.xsd"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns="urn:mpeg:dash:schema:mpd:2011">
    <ProgramInformation moreInformationURL="http://www.example.com">
        <Title>Example</Title>
    </ProgramInformation>
    <BaseURL>http://www.example.com</BaseURL>
    <Period start="PT0S">
        <AdaptationSet mimeType="video/3gpp">
            <ContentComponent contentType="video"/>
            <ContentComponent contentType="audio" lang="en"/>
            <Representation codecs="s263, samr" bandwidth="256000" id="256">
                <BaseURL>"rep1"</BaseURL>
                <SegmentList duration="1000" timescale="100">
                    <Initialization sourceURL="seg-init.3gp"/>
                    <SegmentURL media="seg-1.3gp"/>
                    <SegmentURL media="seg-2.3gp"/>
                    <SegmentURL media="seg-3.3gp"/>
                </SegmentList>
            </Representation>
            <Representation codecs="mp4v.20.9, mp4a.E1" bandwidth="128000" id="128">
                <BaseURL>"rep2"</BaseURL>
                <SegmentList duration="10">
                    <Initialization sourceURL="seg-init.3gp"/>
                    <SegmentURL media="seg-1.3gp"/>
                    <SegmentURL media="seg-2.3gp"/>
                    <SegmentURL media="seg-3.3gp"/>
                </SegmentList>
            </Representation>
        </AdaptationSet>
    </Period>
    <Period start="PT30S">
        <SegmentTemplate
            duration="10"
            initialization="seg-init-$RepresentationId$.3gp"
            media="http://example.com/$RepresentationId$/$Number$.3gp"/>
        <AdaptationSet mimeType="video/3gpp" codecs="mp4v.20.9, mp4a.E1">
            <ContentComponent contentType="video"/>
            <ContentComponent contentType="audio" lang="en"/>
            <Representation bandwidth="256000" id="1"/>
            <Representation bandwidth="128000" id="2"/>
        </AdaptationSet>
    </Period>
</MPD>
```

D.2 Live service

Table D.2 provides an example MPD for a live service.

Table D.2: Example MPD for a live service

```
<?xml version="1.0"?>
<MPD
    profiles="urn:3GPP:PSS:profile:DASH10"
    type="dynamic"
    minBufferTime="PT3S"
    availabilityStartTime="2010-04-26T08:45:00-08:00"
    minimumUpdatePeriod="PT5M0S"
    timeShiftBufferDepth="PT1H30M0S"
    xsi:schemaLocation="urn:mpeg:dash:schema:mpd:2011 3GPP-Rel10-MPD.xsd"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns="urn:mpeg:dash:schema:mpd:2011">
    <ProgramInformation moreInformationURL="http://www.example.com">
        <Title>Example 3: 3GPP SA4 Meeting in Vancouver as Live Broadcast</Title>
        <Source>3GPP</Source>
    </ProgramInformation>
    <Period start="PT0S" id="0">
        <AdaptationSet mimeType='video/3gpp' codecs="avc1.42E00B" width="320" height="240"
contentType="video">
            <SegmentTemplate
                duration="60"
                initialization="http://www.ad-server.com/1-day-black/QVGA/0.3gp"
                media="http://www.ad-server.com/1-day-black/QVGA/$Number$.3gp">
            </SegmentTemplate>
            <Representation id="Ad-QVGA" bandwidth="10000">
        </Representation>
        </AdaptationSet>
    </Period>
    <Period start="PT15M0S" id="1">
        <SegmentTemplate
            duration="10"
            initialization="http://www.example.com/Period-2010-04-26T08-45-00/rep-
$RepresentationID$/seg-0.3gp"
            media="http://www.example.com/Period-2010-04-26T08-45-00/rep-$RepresentationID$/seg-
$Number$.3gp"/>
        <AdaptationSet mimeType='video/3gpp'>
            <ContentComponent contentType="video"/>
            <ContentComponent contentType="audio" lang="en"/>
            <Representation
id="QVGA-LQ" codecs="avc1.42E00C, mp4a.40.2" bandwidth="192000" width="320" height="240"/>
            <Representation id="QVGA-HQ" codecs="avc1.42E00C, mp4a.40.2" bandwidth="384000"
width="320" height="240"/>        <Representation id="VGA-LQ" mimeType='video/3gpp'
codecs="avc1.64001E, mp4a.40.2" bandwidth="512000" width="640" height="480"/>
            <Representation id="VGA-HQ" codecs="avc1.64001E, mp4a.40.2" bandwidth="1024000"
width="640" height="480"/>
        </AdaptationSet>
    </Period>
    <Period start="PT2H01M22.12S" id="2">
        <SegmentTemplate duration="10"
            media="http://www.ad-server.com/15min-Ads/$RepresentationID$/$Number$.3gp"
            initialization="http://www.ad-server.com/15min-Ads/$RepresentationID$/0.3gp"/>
        <AdaptationSet mimeType='video/3gpp'>
            <ContentComponent contentType="video"/>
            <ContentComponent contentType="audio" lang="en"/>
            <Representation id="QVGA" codecs="avc1.42E00C, mp4a.40.2" bandwidth="256000"
width="320" height="240"/>
            <Representation id="VGA"  codecs="avc1.64001E, mp4a.40.2" bandwidth="512000"
width="640" height="480"/>
        </AdaptationSet>
    </Period>
    <Period start="PT2H16M22.12S" id="3">
        <SegmentTemplate
            duration="10"
            media="http://www.example.com/Period-2010-04-26T11-01-22/rep-$RepresentationID$/seg-
$Number$.3gp"
            initialization="http://www.example.com/Period-2010-04-26T11-01-22/rep-
$RepresentationID$/seg-0.3gp"
        />
        <AdaptationSet mimeType='video/3gpp' contentType="video">
            <Representation id="QVGA-LQ" codecs="avc1.42E00C" bandwidth="192000" width="320"
height="240"/>
```

```
            <Representation id="QVGA-HQ" codecs="avc1.42E00C" bandwidth="384000" width="320"
height="240"/>
            <Representation id="VGA-LQ" codecs="avc1.64001E" bandwidth="512000" width="640"
height="480"/>
            <Representation id="VGA-HQ" codecs="avc1.64001E" bandwidth="1024000" width="640"
height="480"/>
        </AdaptationSet>
        <AdaptationSet mimeType='audio/3gpp' contentType="audio" lang="en">
            <Representation id="audio" codecs="mp4a.40.2" bandwidth="32000"/>
            <Representation id="audio" codecs="mp4a.40.2" bandwidth="64000"/>
        </AdaptationSet>
    </Period>
</MPD>
```

D.3 MPD assembly

Table D.3 provides an example MPD with reference to external Period element as provided in table D.4. An equivalent MPD to the one in table D.3 after dereferencing with the **`Period`** element in table D.4 is shown in table D.2.

Table D.3: Example MPD with reference to external Period element

```
<?xml version="1.0"?>
<MPD
    profiles="urn:3GPP:PSS:profile:DASH10"
    type="dynamic"
    minBufferTime="PT3S"
    availabilityStartTime="2010-04-26T08:45:00-08:00"
    minimumUpdatePeriod="PT5M0S"
    timeShiftBufferDepth="PT1H30M0S"
    xsi:schemaLocation="urn:mpeg:Ddash:schema:mpd:2011 3GPP-Rel10-MPD.xsd"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns:xlink="http://www.w3.org/1999/xlink"
    xmlns="urn:mpeg:dash:schema:mpd:2011">
    <ProgramInformation moreInformationURL="http://www.example.com">
        <Title>Example 3: 3GPP SA4 Meeting in Vancouver as Live Broadcast</Title>
        <Source>3GPP</Source>
    </ProgramInformation>
    <Period start="PT0S" id="0">
        <AdaptationSet mimeType='video/3gpp' codecs="avc1.42E00B" width="320" height="240"
contentType="video">
            <SegmentTemplate
                duration="60"
                initialization="http://www.ad-server.com/1-day-black/QVGA/0.3gp"
                media="http://www.ad-server.com/1-day-black/QVGA/$Number$.3gp">
            </SegmentTemplate>
            <Representation id="Ad-QVGA" bandwidth="10000">
            </Representation>
        </AdaptationSet>
    </Period>
    <Period start="PT15M0S" id="1">
        <SegmentTemplate
            duration="10"
            initialization="http://www.example.com/Period-2010-04-26T08-45-00/rep-
$RepresentationID$/seg-0.3gp"
            media="http://www.example.com/Period-2010-04-26T08-45-00/rep-$RepresentationID$/seg-
$Number$.3gp"/>
        <AdaptationSet mimeType='video/3gpp'>
            <ContentComponent contentType="video"/>
            <ContentComponent contentType="audio" lang="en"/>
            <Representation id="QVGA-LQ" codecs="avc1.42E00C, mp4a.40.2" bandwidth="192000"
width="320" height="240"/>
            <Representation id="QVGA-HQ" codecs="avc1.42E00C, mp4a.40.2" bandwidth="384000"
width="320" height="240"/>        <Representation id="VGA-LQ" mimeType='video/3gpp'
codecs="avc1.64001E, mp4a.40.2" bandwidth="512000" width="640" height="480"/>
            <Representation id="VGA-HQ" codecs="avc1.64001E, mp4a.40.2" bandwidth="1024000"
width="640" height="480"/>
        </AdaptationSet>
    </Period>
    <Period xlink:ref="http://www.example.com/Period.xml" id="2"/>
    <Period start="PT2H16M22.12S" id="3">
        <SegmentTemplate
            duration="10"
            media="http://www.example.com/Period-2010-04-26T11-01-22/rep-$RepresentationID$/seg-
$Number$.3gp"
```

```
            initialization="http://www.example.com/Period-2010-04-26T11-01-22/rep-
$RepresentationID$/seg-0.3gp"
        />
        <AdaptationSet mimeType='video/3gpp' contentType="video">
            <Representation id="QVGA-LQ" codecs="avc1.42E00C" bandwidth="192000" width="320"
height="240"/>
            <Representation id="QVGA-HQ" codecs="avc1.42E00C" bandwidth="384000" width="320"
height="240"/>
            <Representation id="VGA-LQ" codecs="avc1.64001E" bandwidth="512000" width="640"
height="480"/>
            <Representation id="VGA-HQ" codecs="avc1.64001E" bandwidth="1024000" width="640"
height="480"/>
        </AdaptationSet>
        <AdaptationSet mimeType='audio/3gpp' contentType="audio" lang="en">
            <Representation id="audio" codecs="mp4a.40.2" bandwidth="32000"/>
            <Representation id="audio" codecs="mp4a.40.2" bandwidth="64000"/>
        </AdaptationSet>
    </Period>
</MPD>
```

Table D.4: External Period

```
<?xml version="1.0"?>
<Period start="PT2H01M22.12S">
    <SegmentTemplate duration="10"
        media="http://www.ad-server.com/15min-Ads/$RepresentationID$/$Number$.3gp"
        initialization="http://www.ad-server.com/15min-Ads/$RepresentationID$/0.3gp"/>
    <AdaptationSet mimeType='video/3gpp'>
        <ContentComponent contentType="video"/>
        <ContentComponent contentType="audio" lang="en"/>
        <Representation id="QVGA" codecs="avc1.42E00C, mp4a.40.2" bandwidth="256000" width="320"
height="240"/>
        <Representation id="VGA"  codecs="avc1.64001E, mp4a.40.2" bandwidth="512000" width="640"
height="480"/>
    </AdaptationSet>
</Period>
```

D.4 MPD deltas

In the following MPD example, the content is 30 minutes in duration. There are 3 Periods, each of 10 minutes duration. Each Period has 3 Representations and each Representation is contained within one 3gp file. Each Representation has audio encoded with Low Complexity-AAC. One Representation of each Period (p1rep1.3gp, p2rep1.3gp, and p3rep1.3gp) has video resolution 320x240 encoded with H.264 baseline profile level 1.1. Another Representation of each Period (p1rep2.3gp, p2rep2.3gp, and p3rep2.3gp) has resolution 320x240 encoded with H.264 baseline profile level 1.3. Finally, a third representation in each period (p1rep3.3gp, p2rep3.3gp, and p3rep3.3gp) has resolution 480×240 encoded with H.264 baseline profile level 2.1. One Representation of each Period has bandwidth of 239 kbps, a second representation has bandwidth of 478 kbps, and a third representation has bandwidth of 892 kbps.

Since each representation is contained in one file, the Initialization Segments and the Media Segments for a representation are accessed with byte ranges. Each **`SegmentURL`** element in the MPD contains a **`mediaRange`** attribute and the corresponding byte range for the Initialization Segment or Media Segment. For the example each Segment of all representations is 10 seconds in duration.

Line numbers of the MPD in the example are shown for clarity, although these would not be present in the MPD.

EXAMPLE 1 (add)

The change of adding the SegmentURL element for the next Segment to the Representation of the third Period of the MPD in the example with 239K bandwidth can be described as follows.

```
492a
                <SegmentURL mediaRange="17339554-17642841"/>
.
```

The SegmentURL element is added on a new line after line number 492.

EXAMPLE 2 (replace)

Replacing the line containing the **`DeltaSupport`** element to correspond to the MPD after the next update can be described as follows.

```
625c
      <DeltaSupport sourceURL="delta2.mpdd" availabilityDuration="PT120S"/>
.
```

EXAMPLE 3 (delete)

If lines 8 through 10 of the original MPD are deleted and not present in the updated MPD, the delta to express this is:

```
8,10d
.
```

Below is what the MPD looks like after approximately 29 minutes and 40 seconds. In this case, the MPD is updated approximately every 10 seconds.

```
1<?xml version="1.0"?>
2<MPD   profiles="urn:3GPP:PSS:profile:DASH10"
3       type="dynamic"
4       availabilityStartTime="2010-07-01T05:00:00Z"
5      availabilityEndTime="2010-07-08T05:00:00Z"
6       mediaPresentationDuration="PT2H"
7       minimumUpdatePeriod="PT10S"
8       minBufferTime="PT10S"
9       timeShiftBufferDepth="PT30M"
10      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
11      xmlns:x3gpp="urn:3GPP:ns:DASH:MPD-ext:2011"
12      xsi:schemaLocation="urn:mpeg:dash:schema:mpd:2011 3GPP-Rel10-MPD.xsd"
13      xmlns="urn:mpeg:dash:schema:mpd:2011">
14
15 <ProgramInformation moreInformationURL="http://www.example.com">
16 <Title>Example</Title>
17  <Source>Example</Source>
18  <Copyright>Example</Copyright>
19 </ProgramInformation>
20 <BaseURL>http://www.example.com</BaseURL>
21 <Period start="PT0S" bitstreamSwitching="true" id="0">
22  <AdaptationSet mimeType="video/3gpp">
23    <ContentComponent contentType="video"/>
24    <ContentComponent contentType="audio" lang="en"/>
25    <Representation id="0" bandwidth="239000" width="320" height="240" codecs="avc1.42E00b,
mp4a.40.2">
26       <BaseURL>"p1rep1.3gp"</BaseURL>
27       <SegmentList duration="10" >
28          <Initialization range="0-985" />
29                              <SegmentURL  mediaRange="986-293761" />
30                              <SegmentURL  mediaRange="293762-592501" />
                                                .
                                                .
                                                .

88                              <SegmentURL  mediaRange="17600065-17894640" />
89       </SegmentList>
90    </Representation>
91    <Representation id="1" bandwidth="478000" width="320" height="240" codecs="avc1.42E00d,
mp4a.40.2" >
92       <BaseURL>"p1rep2.3gp"</BaseURL>
93       <SegmentList duration="10">
94          <Initialization range="0-985" />
95                              <SegmentURL  mediaRange="986-586538" />
96                              <SegmentURL  mediaRange="586539-1184019" />
                                                .
                                                .
                                                .
154                             <SegmentURL  mediaRange="35199171-35788323" />
155      </SegmentList>
156    </Representation>
157    <Representation id="2" bandwidth="892000" width="480" height="240" codecs="avc1.42E015,
mp4a.40.2">
158      <BaseURL>"p1rep3.3gp"</BaseURL>
159      <SegmentList duration="10">
160           <Initialization range="0-985" />
161                              <SegmentURL  mediaRange="986-1093691" />
```

```
162                             <SegmentURL  mediaRange="1093692-2208656" />
                                                .
                                                .
                                                .
220                             <SegmentURL  mediaRange="65684646-66784068" />
221      </SegmentList>
222   </Representation>
223 </AdaptationSet>
224</Period>
225<Period start="PT10M0S" bitstreamSwitching="true" id="1">
226 <AdaptationSet mimeType="video/3gpp">
227   <ContentComponent contentType="video"/>
228   <ContentComponent contentType="audio" lang="en"/>
229   <Representation  id="0" bandwidth="239000" width="320" height="240" codecs="avc1.42E00b,
mp4a.40.2">
230     <BaseURL>"p2rep0.3gp"</BaseURL>
231    <SegmentList duration="10">
232         <Initialization range="0-985" />
233                             <SegmentURL  mediaRange="986-296011" />
234                             <SegmentURL  mediaRange="296012-595787" />
                                                .
                                                .
                                                .

292                             <SegmentURL  mediaRange="17647666-17946154" />
293     </SegmentList>
294    </Representation>
295    <Representation id="1" bandwidth="478000" width="320" height="240" codecs="avc1.42E00d,
mp4a.40.2">
296      <BaseURL>"p2rep1.3gp"</BaseURL>
297      <SegmentList duration="10">
298          <Initialization range="0-985" />
299                             <SegmentURL  mediaRange="986-591037" />
300                             <SegmentURL  mediaRange="591038-1190590" />
                                                .
                                                .
                                                .
358                             <SegmentURL  mediaRange="35294377-35891354" />
359      </SegmentList>
360   </Representation>
361   <Representation id="2" bandwidth="892000" width="480" height="240" codecs="avc1.42E015,
mp4a.40.2">
362     <BaseURL>"p2rep2.3gp"</BaseURL>
363    <SegmentList duration="10">
364         <Initialization range="0-985" />
365                             <SegmentURL mediaRange="986-1102088" />
366                             <SegmentURL mediaRange="1102089-2220920" />
                                                .
                                                .
                                                .
424                             <SegmentURL mediaRange="65862331-66976355" />
425     </SegmentList>
426   </Representation>
427 </AdaptationSet>
428</Period>
429<Period start="PT20M0S" bitstreamSwitching="true" id="2">
430 <AdaptationSet mimeType="video/3gpp">
431   <ContentComponent contentType="video"/>
432   <ContentComponent contentType="audio" lang="en"/>
433   <Representation  id="0" bandwidth="239000" width="320" height="240" codecs="avc1.42E00b,
mp4a.40.2">
432     <BaseURL>"p3rep0.3gp"</BaseURL>
433    <SegmentList duration="10">
434         <Initialization range="0-985" />
435                             <SegmentURL  mediaRange="986-302469" />
436                             <SegmentURL  mediaRange="302470-597839" />
                                                .
                                                .
                                                .
492                             <SegmentURL  mediaRange="17040002-17339553" />
493    </SegmentList>
494   </Representation>
495   <Representation id="1" bandwidth="478000" width="320" height="240" codecs="avc1.42E00d,
mp4a.40.2" >
496     <BaseURL>"p3rep1.3gp"</BaseURL>
497    <SegmentList duration="10">
498         <Initialization range="0-985" />
499                             <SegmentURL  mediaRange="986-603953" />
```

```
500                              <SegmentURL  mediaRange="603954-1194693" />
                                                .
                                                .
                                                .
556                              <SegmentURL  mediaRange="34079046-34678149" />
557    </SegmentList>
558   </Representation>
559   <Representation id="2" bandwidth="892000" width="480" height="240" codecs="avc1.42E015,
mp4a.40.2" >
560     <BaseURL>"p3rep2.3gp"</BaseURL>
561    <SegmentList duration="10">
562          <Initialization range="0-985" />
563                              <SegmentURL  mediaRange="986-1126190" />
564                              <SegmentURL  mediaRange="1126191-2228575" />
                                                .
                                                .
                                                .
620                              <SegmentURL  mediaRange="63594383-64712374" />
621    </SegmentList>
622   </Representation>
623  </AdaptationSet>
624 </Period>
625 <x3gpp:DeltaSupport sourceURL="delta1.mpdd" availabilityDuration="PT120S"/>
626</MPD>
```

Since the value of `@sourceURL` in the above MPD is "`delta1.mpdd`", `delta1.mpdd` is an empty file at the time of publication of the above MPD.

The following file is `delta1.mpdd` after the next MPD update. Notice that clients have access to the new value of `@sourceURL` referenced by the latest MPD via the delta.

```
625c
      <DeltaSupport sourceURL="delta2.mpdd" availabilityDuration="PT120S"/>
.
620a
                                 <SegmentURL mediaRange="64712375-65844316"/>
.
556a
                                 <SegmentURL mediaRange="34678150-35284727"/>
.
492a
                                 <SegmentURL mediaRange="17339554-17642841"/>
.
```

At the next MPD update, "`delta1.mpdd`" would contain the cumulative update for 2 MPD updates.

```
625c
      <DeltaSupport sourceURL="delta3.mpdd" availabilityDuration="PT120S"/>
.
620a
                                 <SegmentURL mediaRange="64712375-65844316"/>
                                 <SegmentURL mediaRange="65844317-66966044"/>
.
556a
                                 <SegmentURL mediaRange="34678150-35284727"/>
                                 <SegmentURL mediaRange="35284728-35885833"/>
.
492a
                                 <SegmentURL mediaRange="17339554-17642841"/>
                                 <SegmentURL mediaRange="17642842-17943394"/>
.
```

Annex E (normative):
Void

Annex F (normative): OMA DM QoE Management Object

As an alternative to configuring the QoE reporting for each session via MPD, OMA-DM can be used to specify the QoE configuration. If such an OMA-DM QoE configuration has been specified, it shall be evaluated by the client for all subsequent sessions.

For the OMA-DM QoE configuration the parameters are specified according to the following Managed Object (MO), and represents the same information as specified in clauses 10.4 and 10.5. Version numbering is included for possible extension of the MO.

The Management Object Identifier shall be: `urn:oma:mo:ext-3gpp-pss-dash-qoe:1.0`.

Protocol compatibility: The MO is compatible with OMA Device Management protocol specifications, version 1.2 and upwards, and is defined using the OMA DM Device Description Framework as described in the Enabler Release Definition OMA-ERELD _DM-V1_2 [22].

The nodes and leaf objects as provided in figure F.1 shall be contained under the 3GPP_PSS_DASH_QOE node if a client supports the feature described in this clause.

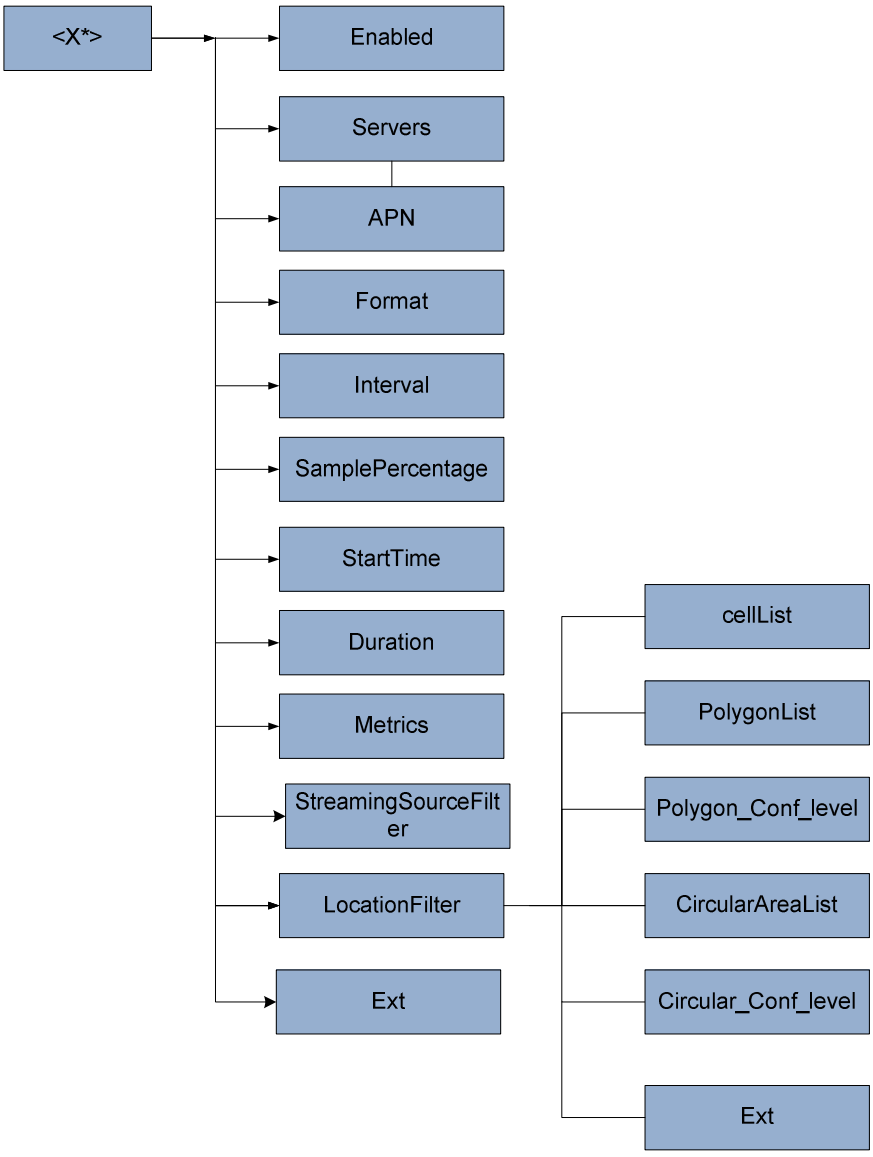


Figure F.1: Nodes and leaf objects

Node: /<X>

This interior node specifies the unique object id of a QoE metrics management object. The purpose of this interior node is to group together the parameters of a single object.

- Occurrence: ZeroOrOne
- Format: node
- Minimum Access Types: Get

The following interior nodes shall be contained if the client supports the QoE Management Object.

/<X>/Enabled

This leaf indicates if QoE reporting is requested by the provider.

- Occurrence: One
- Format: bool
- Minimum Access Types: Get

/<X>/Servers

This leaf contains a space-separated list of servers to which the QoE reports are transmitted. It is URI addresses, e.g. http://qoeserver.operator.com. In case of multiple servers, the client randomly selects one of the servers from the list, with uniform distribution.

- Occurrence: One
- Format: chr
- Minimum Access Types: Get
- Values: URI of the servers to receive the QoE report.

/<X>/APN

This leaf contains the Access Point Name that should be used for establishing the PDP context on which the QoE metric reports will be transmitted. This may be used to ensure that no costs are charged for QoE metrics reporting. If this leaf is not defined then any QoE reporting is done over the default access point.

- Occurrence: ZeroOrOne
- Format: chr
- Minimum Access Types: Get
- Values: The Access Point Name

/<X>/Format

This leaf specifies the format of the report. If this leaf is not defined the QoE reports shall be sent uncompressed.

- Occurrence: ZeroOrOne
- Format: chr
- Minimum Access Types: Get
- Values: "uncompressed", "gzip"

/<X>/Interval

This leaf specifies how often QoE reports shall be sent. If this leaf is not defined only one QoE report shall be sent after the complete session.

- Occurrence: ZeroOrOne
- Format: int
- Minimum Access Types: Get
- Values: seconds

/<X>/SamplePercentage

This leaf specifies the percentage of sessions for which QoE metrics shall be reported. The client evaluates a random number at start of each session to determine if reporting shall be done for the specific session. If this leaf is not defined QoE reports are sent for every session.

- Occurrence: ZeroOrOne
- Format: float
- Minimum Access Types: Get
- Values: 0.0-100.0.

/<X>/StartTime

This leaf specifies when collection of QoE metrics shall start. It is specified in seconds and is relative to the start of the session. If this leaf is not defined, the QoE collection shall be done from the start of the session.

- Occurrence: ZeroOrOne
- Format: int
- Minimum Access Types: Get
- Values: seconds

/<X>/Duration

This leaf specifies for how long QoE collection shall be done. It is specified in seconds and is relative to the start time of QoE collection. If this leaf is not defined QoE collection shall be done until the end of the session.

- Occurrence: ZeroOrOne
- Format: int
- Minimum Access Types: Get
- Values: seconds.

/<X>/Metrics

This leaf specifies a list of white-space separated metrics which shall be reported, and follows the same syntax as specified for the "@metrics" attribute in table 32. If this leaf is not defined no QoE reporting shall be done.

- Occurrence: ZeroOrOne
- Format: chr
- Minimum Access Types: Get
- Values: Metrics as specified in clause 10.4.

/<X>/StreamingSourceFilter

This leaf controls the Media Presentations for which QoE metrics are collected and reported. The filter is defined as a URL pattern. Each URL pattern is represented by a POSIX regular expression as specified by IEEE Std 1003.1-2008 [52].

If the MPD URL of a Media Presentation does not match any of the regular expressions as provided by the URL patterns in the StreamingSourceFilter values, then reporting is disabled and the DASH client shall not report the requested QoE metrics for that Media Presentation. If no StreamingSourceFilter leaf is present, then reporting is enabled for any Media Presentation.

- Occurrence: ZeroOrMore
- Format: chr
- Minimum Access Types: Get
- Values: a POSIX regular expression providing the URL pattern.

/<X> /<LocationFilter>

This interior node specifies any location filters.

- Occurrence: ZeroOrOne
- Format: node
- Minimum Access Types: Get

/<X>/<LocationFilter>/CellList

This leaf specifies a list of cellIDs.

- Occurrence: ZeroOrOne
- Format: chr
- Minimum Access Types: Get
- Values: a list of ECGI or CGI.

/<X>/<LocationFilter>/PolygonList

This leaf specifies a list of shapes defined as ‘Polygon’ by OMA MLP[51].

- Occurrence: ZeroOrOne
- Format: chr
- Minimum Access Types: Get
- Values: a list of ‘Polygon’ defined by OMA MLP[51].

/<X>/<LocationFilter>/Polygon_Conf_Level

This leaf indicates the probability in percent that the DASH client is located in the corresponding polygon area specified by leaf ‘PolygonList’. It is defined as ‘lev_conf’ by OMA MLP. If not present, it has default value of 60.

- Occurrence: ZeroOrOne
- Format: int
- Minimum Access Types: Get
- Values: ‘lev_conf’ defined by OMA MLP[51].

/<X>/<LocationFilter>/CircularAreaList

This leaf specifies a list of shapes defined as 'CircularArea' by OMA MLP[51].

- Occurrence: ZeroOrOne
- Format: chr
- Minimum Access Types: Get
- Values: a list of 'CircularArea' defined by OMA MLP[51].

/<X>/<LocationFilter>/Circular_Conf_Level

This leaf indicates the probability in percent that the DASH client is located in the corresponding circular area specified by leaf 'CircularAreaList'. It is defined as 'lev_conf' by OMA MLP. If not present, it has default value of 60.

- Occurrence: ZeroOrOne
- Format: int
- Minimum Access Types: Get
- Values: 'lev_conf' defined by OMA MLP[51].

/<X>/Ext

The Ext node is an interior node where the vendor specific information can be placed (vendor includes application vendor, device vendor etc.). Usually the vendor extension is identified by vendor specific name under the ext node. The tree structure under the vendor identified is not defined and can therefore include one or more un-standardized sub-trees.

- Occurrence: ZeroOrOne
- Format: node
- Minimum Access Types: Get

Annex G (normative): File format extensions for 3GP-DASH support

G.1 Introduction

This clause documents extensions to the ISO Base Media File Format [11] for the support of 3GPP DASH.

All extensions are documented in [11].

G.2 Void

G.3 Void

G.4 Void

G.5 Void

G.6 Void

Annex H (normative): MIME type registration for MPD

H.1 MPD MIME type

H.1.1 Introduction

The MIME type of the MPD is registered and available at the registry at http://www.iana.org/assignments/media-types/application. For formal registration, refer to annex C of ISO/IEC 23009-1 [43].

H.1.2 Void

H.1.3 Void

H.2 MPD delta MIME type

H.2.1 Introduction

This clause provides the formal MIME type registration for the MPD delta. The MIME type is registered with IANA and available at http://www.iana.org/assignments/media-types/application/dashdelta.

H.2.2 MIME type and subtype

The MIME type and subtype are defined as follows:

- Media Type Name: application
- Subtype name: Standards Tree - dashdelta
- Required parameters: none
- Optional parameters: none
- Encoding considerations: 8-bit text
- Security considerations:

A Media Presentation Description (MPD) Delta contains text changes to an MPD. An MPD Delta is used together with a first MPD to construct a second MPD. As such, any security considerations for an MPD (media type application/dash+xml) may also be applicable to an MPD Delta. A MIME type handler would not launch a service with only an MPD Delta.

Further to this, as an MPD Delta performs editing operations on an MPD there are risks that deliberately malformed editing operations could cause security issues.

- Interoperability considerations:
- Published specification: 3GPP TS 26.247
- Applications which use this media type:

various including but not limited to On-Demand Streaming over the Internet, Live Streaming over the Internet, Internet Video, Internet Radio

- Additional information:
  1) 1. Magic number(s) : none
  2) 2. File extension(s) : mpdd
  3) 3. Macintosh file type code : none
  4) 4. Object Identifiers: none

- Person to contact for further information:
  1) 1. Name : David Furbeck
  2) 2. Email : dfurbeck@blackberry.com

- Intended usage : Common

- Author/Change controller : 3GPP TSG SA WG4

Annex I (informative): Signalling of DASH AVP values for QoS handling in the PCC

The PCC architecture is defined in TS 23.203 [31] and provides the Rx reference point, which enables the application layer to authorize a specific usage. In this architecture the DASH HTTP streaming server or any other function in the HTTP streaming path (e.g. an HTTP proxy) can act as Application Function and interact with the PCRF via the Rx reference point for QoS control. It is assumed here that the AF has knowledge of the application type and of the MPD.

The relevant AVPs are the ones enabling the PCRF to establish bearers with correct characteristics for DASH users. The AVPs are defined in TS 29.214 [33]. The further PCRF mapping from AVP to IP QoS parameter mapping is defined in TS 29.213 [32].

Table I.1: Example mapping of MPD parameters to Rx AVPs for 3GP-DASH (PSS)

AVP	Value	Comment
AF-Application-Identifier	"DASH"	Allows to signal the DASH based application hence giving the opportunity to enforce application specific policies.
Max-Requested-Bandwidth-DL (NOTE 1)	B1	`B1 = sum of all` **`MPD`**`@maxBandwidth` (see clause 8.4.3.3) of all media components simultaneously (not mutually exclusive) selectable by the DASH client plus HTTP/TCP/IP overhead and TCP messages for flow control. If this attribute is not present then `B1 = sum of` **`MPD`**`@bandwidth` attributes of all media components of the available media presentation corresponding to representations or subrepresentations with highest bandwidth simultaneously selectable (not mutually exclusive) by the DASH client plus HTTP/TCP/IP overhead and TCP messages for flow control. Note: the mapping rules to derive the TCP message flow control bandwidth are FFS.
Max-Requested-Bandwidth-UL (NOTE 1)	FFS	For Further Study. If included, should be greater than or equal to Min-Requested-Bandwidth-UL.
Min-Requested-Bandwidth-DL (NOTE 1)	B2	`B2 = sum of all` **`MPD`**`@minBandwidth` (see clause 8.4.3.3) of all media components simultaneously (not mutually exclusive) selectable by the DASH client plus HTTP/TCP/IP overhead and TCP messages for flow control. If this attribute is not present then `B2 = sum of` **`MPD`**`@bandwidth` attributes of all media components of the available media presentation corresponding to representations or subrepresentations with lowest bandwidth simultaneously (not mutually exclusive) selectable by the DASH client plus HTTP/TCP/IP overhead and TCP messages for flow control. Note: the mapping rules to derive the TCP message flow control bandwidth are FFS.
Min-Requested-Bandwidth-UL (NOTE 1)	FFS	For Further Study. Enough bitrate to cover TCP and HTTP GET requests.
Flow-Description AVP (NOTE 1)	IP addresses and ports	
NOTE 1: AVPs provided within the Media-Component-Description AVP, except Flow-Description AVP that is included within the Media-Sub-Component AVP. Omitted AVPs are not relevant for this functionality.		

Annex J (normative): MIME type registration for QoE reports

J.1 Introduction

This annex provides the formal MIME-type registration for "application/3gpdash-qoe-report+xml" that identifies XML documents following the schema "urn:3gpp:metadata:2011:HSD:receptionreport" defined in clause 10.6.2. It is referenced from the registry at http://www.iana.org/.

J.2 MIME type and subtype

The MIME Type and Subtype are defined as follows:

- Media Type name: application
- Subtype name: Standards Tree - 3gpdash-qoe-report+xml
- Required parameters:
  - None
- Optional parameters:
  - charset: If a charset parameter is provided its value must be "utf-8".Encoding considerations: 8bit
  - The utf-8 charset is always used for this type.
- Security considerations:
  - The general XML security issues are addressed in clause 10 of RFC 3023 [83].
  - The QoE report XML document instances contain information about what content, services and files a particular UE has received. The information also has privacy relevance as it reveals what a specific receiver, which usually can be connected to a specific user, has received.
  - This media type only contains quality of experience data; there is no defined executable content.
  - XML from other vocabularies is allowed in this media type. Such material will have its own security considerations.
  - If operating in an insecure environment and required by the content/service provider, elements and attributes of the report may be encrypted to protect their confidentiality.
  - If operating in an insecure environment and required by the content/service provider, digital signing and verification procedures may be used to protect data origin authenticity and integrity of the report.
  - Thus to prevent manipulation of that information it would need to be integrity protected.
- Interoperability considerations:
  - None
- Published specification:
  - 3GPP TS 26.247
- Applications which use this media type:
  - 3GPP DASH based applications
  - 3GPP progressive download applications
- Additional information:

- None

- Person & email address to contact for further information:
  - Ozgur Oyman (ozgur.oyman@intel.com)
    3GPP TSG SA WG4
- Intended usage: COMMON
- Restrictions on usage:
  - None
- Author:
  - 3GPP TSG SA WG4
- Change controller:
  - 3GPP TSG SA WG4

Annex K (normative): ITU-T P.1203 audio/video MOS estimation

K.1 Introduction

ITU-T P.1203 [49] defines an objective assessment model for Progressive Download and DASH streaming. This model predicts audio only, video only and audio-video MOS on a 5-point Absolute Category Rating scale (see ITU-T P.910 [50]) for monitoring of quality for operation and maintenance purposes. The model can execute in several modes, where Mode 0 is suitable for large-scale network supervision. For more details on this model, please refer to ITU-T P.1203 [49].

K.2 Calculation

The PSS server should support MOS estimation according to ITU-T P.1203 [49] Mode 0. To enable the MOS estimation, the QoE reporting should include the following metrics in either MPD*@metrics* or OMA DM QoE Management Object '*metrics*':

- Initial Playout Delay (clause 10.2.5)
- Play List (clause 10.2.7)
- MPD Information, unless already available by other means (clause 10.2.8)
- Device Information (clause 10.2.10)

The attribute **`MPD.Metrics.Range`**`@duration` or the OMA DM QoE Management Object '*duration*' shall be set to a value between 1 and 5 minutes.

The value of **`MPD.Metrics.Reporting`**`@reportinginterval` or OMA DM QoE Management Object '*Interval*' shall be larger or equal than the value of **`MPD.Metrics.Range`**`@duration` or OMA DM QoE Management Object '*duration*'.

The DASH client reports collected metrics using quality metric reporting protocol defined in clause 10.6.

The mapping between QoE metrics and input to ITU-T P.1203 [49] Mode 0 is provided in table K-1 to K-4 below.

Table K-1: Mapping between QoE metrics and input in ITU-T P.1203 mode 0 model for video stream

Video Metrics needed for ITU-T P.1203[49] model		QoE metrics defined in clause 10.2		Remark
Description	**Value**	**Metric**	**Description**	
Target Video bit rate	Bit rate in kbps.	Mpdinfo	Provides the MPD information for the representation or subrepresentation identified by representationid and subreplevel, if present. Related attributes: @bandwidth, @width, @height, and @codecs. Note: codec attribute includes video codec profile information and video frame rate information.	Target Video bit rate has been supported by @bandwidth attribute. Video frame rate and Video codec and profile have been supported by @ codec attribute. Video encoding resolution has been supported by @width and @height attributes.
Video frame rate	Frame rate in frames per second.			
Segment duration	Duration in seconds			
Video encoding resolution	Number of pixels (W×H) in transmitted video			
Video codec and profile	One of: H264-baseline, H264-high, H264-main			
		Playlist	Provides information on when and how long different representations or subrepresentations have been used during the session.	Note that the actual timing of the required input for the P.1203 quality model may be influenced by actions such as play/pause.

Table K-2: Mapping between QoE metrics and input in ITU-T P.1203 mode 0 model for audio stream

Audio Metrics needed for ITU-T P.1203 model		QoE metrics defined in clause 10.2		Remark
Description	**Value**	**Metric**	**Description**	
Target Audio bit rate	Bit rate in kbps.	Mpdinfo	Provides the MPD information for the representation or subrepresentation identified by representationid and subreplevel, if present. Related attributes: @bandwidth, and @codecs. Note: codec attribute includes audio codec profile information, audio sampling frequency and Number of audio channels information.	Target Audio bit rate has been supported by @bandwidth attribute. Audio codec, Audio sampling frequency and Number of audio channels have been supported by @ codecs attribute.
Segment duration	Duration in seconds			
Audio codec	One of: AAC-LC, AAC-HEv1, AAC-HEv2, AC3			
Audio sampling frequency	In Hz			
Number of audio channels	2			
		Playlist	Provides information on when and how long different representations or subrepresentations have been used during the session.	Note that the actual timing of the required input for the P.1203 quality model may be influenced by actions such as play/pause.

Table K-3: Mapping between QoE metrics and input in ITU-T P.1203 mode 0 model for stalling

Metrics needed for ITU-T P.1203 model		QoE metrics defined in clause 10.2		Remark
Description	**Value**	**Metric**	**Description**	
Buffering event start	The start time of the buffering/stalling event in seconds relative to the start of the original video clip, expressed in media time (not wall clock time) Note: This is 0 for initial buffering.	Playlist	The start time of a buffering/stalling event is equal to the end time of a trace entry with the stop reason equal to "rebuffering".	-
Event duration	The duration of the buffering/stalling event in seconds.	Playlist	The duration of each buffer event is calculated from the PlayList according to the example described in the Playlist metric clause.	-

Table K-4: Mapping between QoE metrics and input in ITU-T P.1203 mode 0 model for device information

Metrics needed for ITU-T P.1203 model		QoE metrics defined in clause 10.2		Remark
Description	**Value**	**Metric**	**Description**	
The resolution of the image displayed to the user	Number of pixels (W×H) in displayed video	Device Information	videoWidth and videoHeight	.
The device type on which the media is played	PC or mobile	Device Information	screenWidth × pixelWidth, screenHeight × pixelHeight	P.1203 formally classifies devices up to 10" (diagonal) as mobiles, but to cover most handheld devices (phones and tablets) devices up to 12" could be classified as mobiles.
The size of the video shown	W×H in mm	Device Information	videoWidth × pixelWidth, videoHeight × pixelHeight	

Annex L (normative): QoE measurement collection functionalities

L.1 Configuration and reporting

As an alternative to configuration via MPD or OMA-DM, the QoE configuration can optionally be specified by the QoE Measurement Collection (QMC) functionality. In this case the QoE configuration is received via specific RRC [53] messages for UMTS, RRC [59] messages for LTE, and RRC messages for NR [70] over the control plane, and the QoE reporting is also sent back via RRC messages over the control plane.

If QMC is supported, the UE shall support the following QMC functionalities:

- QoE Configuration: The QoE configuration is delivered via RRC to the UE as a container according to "Application Layer Measurement Configuration" (see [53]) for UMTS, "measConfigAppLayer" (see [59]) for LTE and “AppLayerMeasConfig” (see [70]) for NR. The container is an octet string with gzip-encoded data (see [18]) stored in network byte order. The maximum size of the container is 1000 bytes for UMTS (see [53]) and LTE (see [59]), and 8000 bytes for NR (see [70]). The container shall be uncompressed, and is then expected to conform to XML-formatted QoE configuration data according to clause L.2 in the present document. This QoE Configuration shall be forwarded to the DASH client. The interface towards the RRC signalling is handled by the AT command +CAPPLEVMC for UMTS and LTE, and AT command +CAPPLEVMCNR for NR [61]. In the case where QMC is enabled for streaming services delivered via the MBS communication service, the `@communicationServiceType` attribute in the QoE configuration (see clause 10.5) shall indicate whether the DASH client is requested to collect and report QoE metrics about content received via unicast, via MBS broadcast, via MBS multicast, or via all of these communication service types.

- QoE Metrics: QoE Metrics from the DASH client shall be XML-formatted according to clause 10.6 in the present document. The XML data shall be compressed with gzip (see [18]) and stored in network byte order into an octet string container. The maximum size of the container is 8000 bytes for UMTS (see [53]) and LTE (see [59]). For NR [70], the maximum size is 8000 bytes if RRC segmentation is not enabled, and 144000 bytes if enabled. The container shall be delivered via RRC to the RNC according to "Application Layer Measurement Reporting" (see [53]) for UMTS, to the eNB according to "measReportAppLayer" (see [59]) for LTE, and to the gNB according to “MeasurementReportAppLayer” (see [70]) for NR. The behaviour if the compressed data is larger than the maximum container size is unspecified in this version of the specification. The interface towards the RRC signalling is handled by the AT command +CAPPLEVMR for UMTS and LTE, and AT command +CAPPLEVMRNR for NR [61].

- The UE shall also set the QMC capability "QoE Measurement Collection for streaming services" (see [53]) to TRUE for UMTS, include the QMC capability "qoe-MeasReport" (see [59]) for LTE and include the QMC capability “qoe-Streaming-MeasReport” (see [70]) for NR.

- When a new session is started, the QoE reporting AT command +CAPPLEVMRNR [61] shall be used to send a Recording Session Indication. Such an indication does not contain any QoE report, but indicates that QoE recording has started for a session.

- When the QoE configuration is to be released, an unsolicited result code, associated with the AT command +CAPPLEVMC or AT command +CAPPLEVMCNR [61] and containing the parameter <start-stop_reporting> or <start-stop_measurement> set to "1" shall be sent to the DASH client as notification of a discard request. Then the DASH client shall stop collecting quality metrics and discard any already collected information [63].

For NR, the RAN visible QoE may be supported. The gNB can use RAN visible QoE configurations to instruct the UE to collect application layer measurements for network optimization.

- The RAN visible QoE configuration generated by the gNB shall be forwarded by the UE AS layer to the DASH client via AT command +CAPPLEVMCNR, including the required RAN visible QoE metrics, service type, the RRC identifier and optionally reporting periodicity. The set of RAN visible QoE metrics is a subset of the QoE metrics defined in clause 10.4. In this release of the specification, the set of RAN visible QoE metrics include "Buffer Level" and "Playout Delay for Media Startup". If the reporting periodicity for RAN visible QoE metrics is not specified, the reporting periodicity follows the baseline NR QoE configuration. The measurement interval for "Buffer Level" metric collection is given by reporting periodicity divided by "numberOfBufferLevelEntries" as specified in [70].

- Based on the RAN visible QoE configuration, the RAN visible QoE report shall be delivered to the UE AS layer via AT command +CAPPLEVMRNR and the collected metrics shall be sent to the gNB via the "MeasurementReportAppLayer" message. The PDU session ID(s) corresponding to the service that is subject to RAN visible QoE measurement can also be reported by the DASH client along with the RAN visible QoE report.

- When the RAN visible QoE measurement is deactivated by the gNB, the DASH client shall be notified to terminate and release the RAN visible QoE measurement.

NOTE: The RAN visible QoE metrics collection can be configured only if baseline NR QoE measurements are configured for the same service type. When the baseline NR QoE measurements are released, the RAN visible QoE configuration shall also be released.

The exact implementation is not specified here, but example signalling diagrams for UMTS, LTE and NR below show the QMC functionality with a hypothetical "QMC Handler" entity.

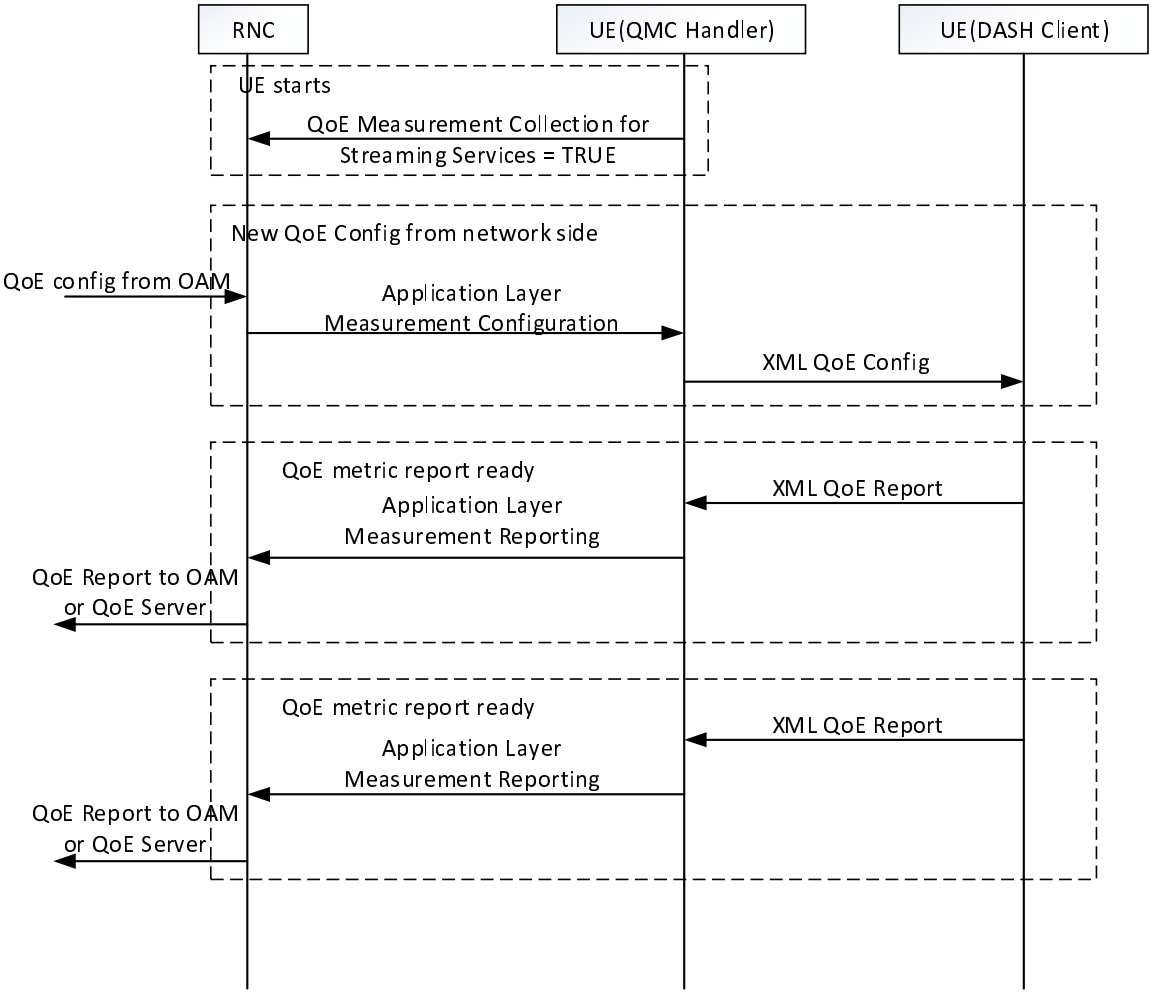


Figure L-1: Example signalling diagram for UMTS

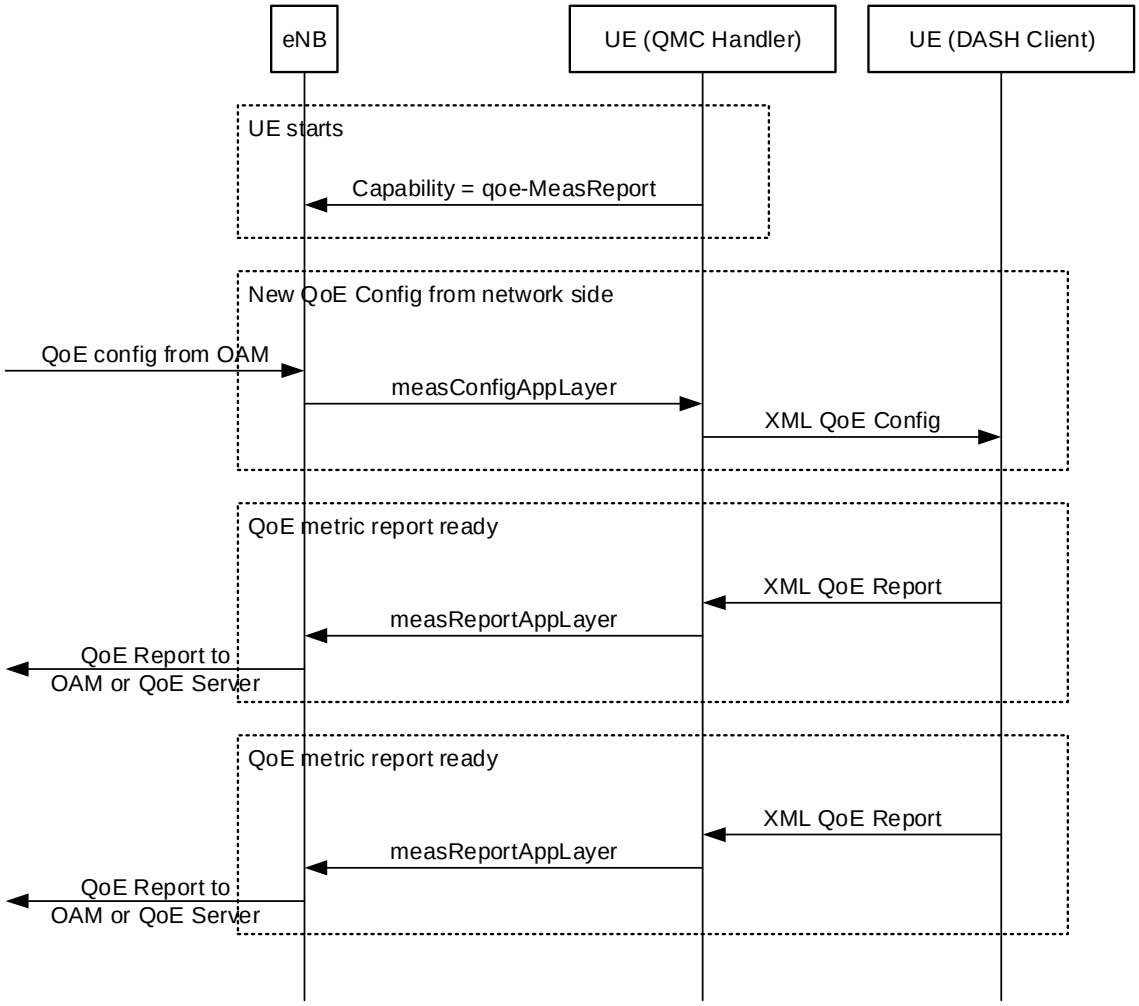


Figure L-2: Example signalling diagram for LTE

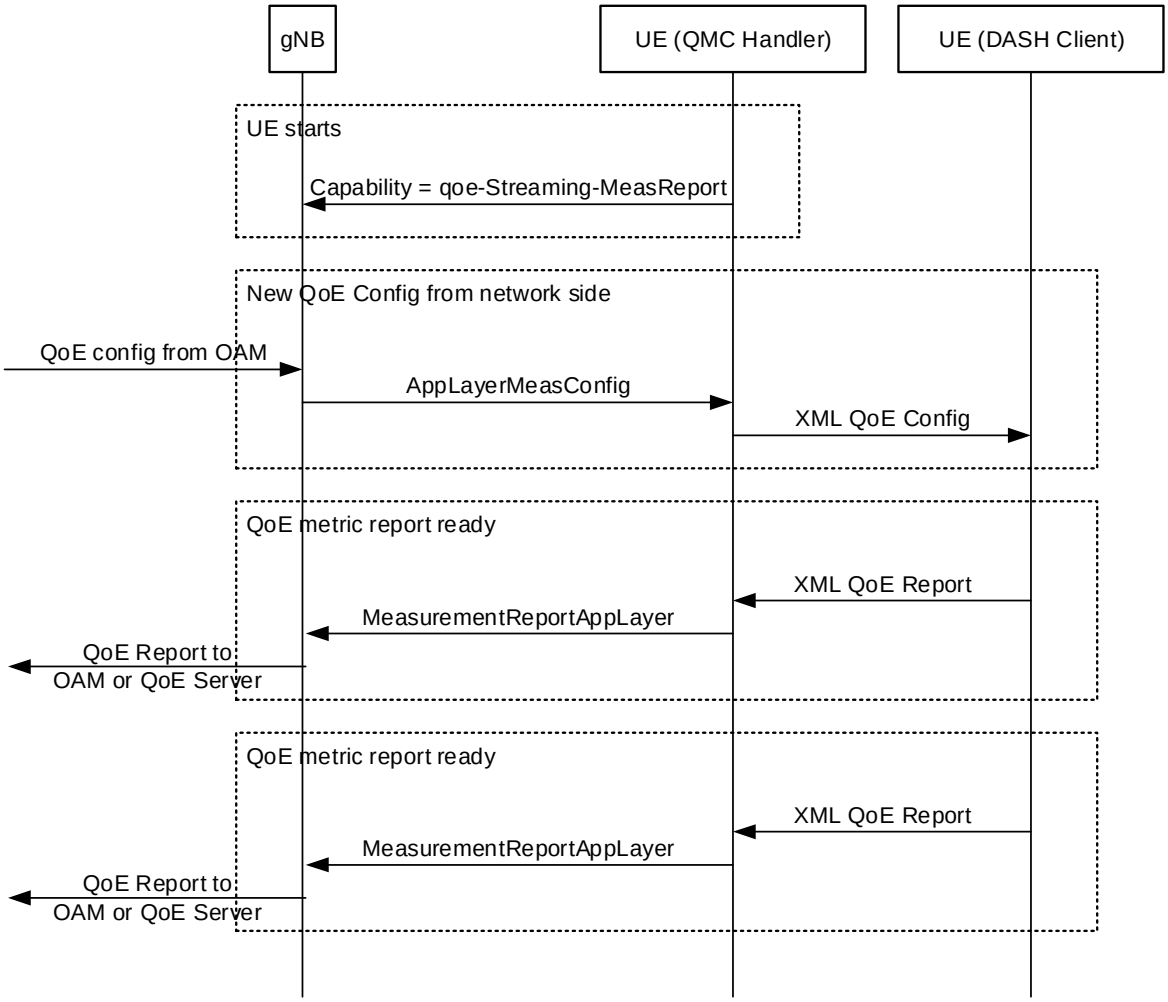


Figure L-3: Example signalling diagram for NR

Note that the QMC Handler is only shown here as one possible implementation, and it need not be implemented as such. The corresponding QMC functionality could be built into the DASH client or into other UE entities. In this version of the specification the detailed implementation of the above functionalities is left to the UE vendor.

L.2 XML configuration

When QoE reporting is configured via the QMC functionality, the configuration is done according to the XML schema below. The interpretation of the different elements and attributes are the same as described in clauses 10.4, 10.5 and annex F in the present document.

Note that if geographical filtering is handled on the network side (i.e. QoE reporting is turned on/off by the network depending on the UE location), no *LocationFilter* should be specified in the QoE Configuration, as this would mean two consecutive filterings.

Also note that the optional attribute *qoeReferenceId* is a reference set by the network side (see [63]), which is not directly used by the client. However, if this attribute is defined, it shall be copied into each QoE report, to facilitate network-side correlation.

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema targetNamespace="urn:3GPP:ns:PSS:DASH:QMC14"
    elementFormDefault="qualified"
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    xmlns:sv="urn:3gpp:metadata:2016:PSS:schemaVersion"
    xmlns="urn:3GPP:ns:PSS:DASH:QMC14">

    <xs:element name="QmcConfiguration" type="QmcConfigurationType"/>
```

```
    <xs:complexType name="QmcConfigurationType">
        <xs:sequence>
            <xs:element name="Range" type="RangeType" minOccurs="0" maxOccurs="unbounded"/>
            <xs:element name="LocationFilter" type="LocationFilterType" minOccurs="0"/>
            <xs:element name="StreamingSourceFilter" type="StreamingSourceFilterType"
minOccurs="0" maxOccurs="unbounded"/>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute name="metrics" type="xs:string" use="required"/>
        <xs:attribute name="samplePercentage" type="xs:double" use="optional"/>
        <xs:attribute name="reportingInterval" type="xs:unsignedInt" use="optional"/>
        <xs:attribute name="qoeReferenceId" type="xs:hexBinary" use="optional"/>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>

    <xs:complexType name="RangeType">
        <xs:sequence>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute name="startTime" type="xs:unsignedInt" use="optional"/>
        <xs:attribute name="duration" type="xs:duration" use="required"/>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>

<xs:complexType name="LocationFilterType">
        <xs:sequence>
            <xs:element name="cellID" type="xs:unsignedLong" minOccurs="0" maxOccurs="unbounded"/>
            <xs:element name="shape" type="ShapeType" minOccurs="0"/>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>

    <xs:complexType name="ShapeType">
        <xs:sequence>
            <xs:element name="PolygonList" type="PolygonListType" minOccurs="0"/>
            <xs:element name="CircularAreaList" type="CircularAreaListType" minOccurs="0"/>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>

    <xs:complexType name="PolygonListType">
        <xs:annotation>
            <xs:documentation> see [OMA MLP] </xs:documentation>
        </xs:annotation>
        <xs:sequence>
            <xs:element name="Polygon" minOccurs="0" maxOccurs="unbounded"/>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute name="ConfLevel" type="xs:unsignedInt" use="optional"/>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>

    <xs:complexType name="CircularAreaListType">
        <xs:annotation>
            <xs:documentation> see [OMA MLP] </xs:documentation>
        </xs:annotation>
        <xs:sequence>
            <xs:element name="CircularArea" minOccurs="0" maxOccurs="unbounded"/>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute name="ConfLevel" type="xs:unsignedInt" use="optional"/>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>

    <xs:complexType name="StreamingSourceFilterType">
        <xs:sequence>
            <xs:any namespace="##other" processContents="lax" minOccurs="0"
maxOccurs="unbounded"/>
        </xs:sequence>
```

```
        <xs:attribute name="streamingSource" type="xs:string" use="required"/>
        <xs:anyAttribute namespace="##other" processContents="lax"/>
    </xs:complexType>
</xs:schema>
```

Annex M (informative): Registrations information

M.1 3GPP registered URIs

The clause documents the registered URIs in the present document following the process in https://www.3gpp.org/3gpp-groups/core-network-terminals-ct/ct-wg1/uniform-resource-identifier-uri-list

Table A-1 lists all registered URN values as well as

- a brief description of its functionality;
- a reference to the specification or other publicly available document (if any) containing the definition;
- the name and email address of the person making the application; and
- any supplementary information considered necessary to support the application.

Table M-1: 3GPP Registered URNs

URN	Description	Reference	Contact	Remarks
urn:3GPP:ns:PSS:AdaptiveHTTPStreamingMPD:2009	DASH profile identifier for 3GPP Adaptive HTTP Streaming (Release-9 AHS) clause 7.3.2	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3GPP:PSS:profile:DASH10	DASH profile identifier for 3GP-DASH Release-10 Profile clause 7.3.3.1	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3GPP:PSS:profile:DASH11:MS3D	DASH profile identifier for 3GP-DASH Release 11 multiview stereoscopic 3D video profile clause 7.3.4	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3GPP:PSS:iop:DASH-enhanced	DASH profile identifier for 3GP-DASH Enhanced Interoperability Point clause 7.3.9	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3GPP:5GMS:iop:DASH	DASH profile identifier for 3GP-DASH Enhanced Interoperability Point clause 7.3.11	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3GPP:ns:DASH:MPD-ext:2011	DASH profile identifier for 3GP-DASH Enhanced Interoperability Point clause 8.2.2	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3GPP:ns:PSS:DASH:QM10	Quality reporting scheme identifier for 3GPP DASH clause 10.5	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none

urn:3gpp:dash:sand:messageset:pc:2016	SAND mode identifier for 'Proxy Caching' clause 13.4	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3gpp:dash:sand:messageset:na:2016	SAND mode identifier for 'Network Assistance' clause 13.4	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3gpp:dash:sand:messageset:qoe:2016	SAND mode identifier for 'Consistent QoE/QoS' clause 13.4	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3gpp:dash:sand:messageset:sand4m:2018	SAND mode identifier for 'SAND for Multi-Network Access' clause 13.4	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3gpp:dash:schema:sandmessageextension:2017	SAND mode identifier for 'SAND Message Extensions' clause 13.6	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3GPP:ns:PSS:DASH:IU15	Interactivity Usage reporting scheme identifier for 3GPP DASH clause 10.5	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3GPP:dash:3dVideoDisparityRange:2013	3D Video Disparity Range Descriptor Scheme Identifier clause C.4	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3GPP:dash:3dVideoDisplayInformation:2013	3D Video Disparity Range Descriptor Scheme Identifier clause C.5	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none
urn:3gpp:metadata:2011:HSD:receptionreport	Reception reporting scheme identifier for 3GPP DASH clause J.1	TS 26.247	Thomas Stockhammer tsto@qti.qualcomm.com	none

Annex N (informative): Change history

Change history							
Date	TSG #	TSG Doc.	CR	Rev	Subject/Comment	Old	New
2011-06	52	SP-110305			Version 10.0.0 approved at TSG SA#52		10.0.0
2011-11	54	SP-110794	0003		Alignment with MPEG DASH	10.0.0	10.1.0
2011-11	54	SP-110794	0005	3	QoE Updates for Correction, Clarification and MPEG DASH Alignment	10.0.0	10.1.0
2011-11	54	SP-110794	0006	2	QoS Support for 3GP-DASH Services	10.0.0	10.1.0
2012-06	56	SP-120221	0007	3	Alignment with MPEG DASH	10.1.0	10.2.0
2012-06	56	SP-120221	0010	1	Correction of Table Reference for Change Commands in MPD Deltas	10.1.0	10.2.0
2012-06	56	SP-120221	0012	2	ContentProtection element update to signal version of DRM system	10.1.0	10.2.0
2012-09	57	SP-120504	0009	3	QoE Reporting for DASH over Combined MBMS Download and HTTP-based Delivery	10.2.0	11.0.0
2012-09	57	SP-120509	0014	6	Inclusion of MVC support for DASH	10.2.0	11.0.0
2012-09	57	SP-120509	0015	4	Inclusion of 3D Video Format Information in DASH MPD	10.2.0	11.0.0
2012-12	58	SP-120761	0016	7	Supporting HTTP Partial Response	11.0.0	11.1.0
2013-03	59	SP-130019	0022	1	DASH QoE Reporting Schema bug fix	11.1.0	11.2.0
2013-03	59	SP-130015	0024	1	DASH QoE Reporting Example bug fix	11.1.0	11.2.0
2013-06	60	SP-130184	0026	2	DASH Profiles correction	11.2.0	11.3.0
2013-06	60	SP-130184	0028	1	Correction of Cardinality of DeltaSupport element in table 8-5	11.2.0	11.3.0
2013-06	60	SP-130184	0030	2	Correction of MPD Delta MIME type information	11.2.0	11.3.0
2013-06	60	SP-130184	0031		General Corrections to DASH	11.2.0	11.3.0
2013-09	61	SP-130350	0037		Corrections to DASH	11.3.0	11.4.0
2013-09	61	SP-130354	0035		3D Video Disparity Range and 3D Video Display Information Descriptor Schemes	11.4.0	12.0.0
2013-12	62	SP-130565	0041	1	Registration of MIME Type for QoE Reports	12.0.0	12.1.0
2013-12	62	SP-130566	0043		Correction on Playlist Metric Attribute	12.0.0	12.1.0
2013-12	62	SP-130578	0044		Update of MPD Delta MIME type information	12.0.0	12.1.0
2013-12	62	SP-130564	0047	1	3GP-DASH MPD Schema Fix	12.0.0	12.1.0
2013-12	62	SP-130565	0050		Corrections to MPD Delta example	12.0.0	12.1.0
2014-03	63	SP-140006	0053	1	Corrections to 3GP-DASH	12.1.0	12.2.0
2014-03	63	SP-140013	0054	1	Missing media decoders specification for 3GP-DASH in MBMS	12.1.0	12.2.0
2014-03	63	SP-140009	0055	1	HEVC support	12.1.0	12.2.0
2014-06	64	SP-140205	0058	2	Registration of MIME Type for QoE Reports	12.2.0	12.3.0
2014-06	64	SP-140205	0061	2	Correction of Cardinality of the SegmentList Element	12.2.0	12.3.0
2014-12	66	SP-140720	0067	1	Correction on Registration of MIME Type for QoE Reports	12.3.0	12.4.0
2014-12	66	SP-140732	0075	1	MI-EMO DASH Robustness	12.3.0	12.4.0
2014-12	66	SP-140737	0074	2	A new QoE metric of playout delay for media start-up	12.4.0	13.0.0
2015-09	69	SP-150449	0082	3	Alignment with ISO/IEC 23009-1:2014 (MPEG-DASH)	13.0.0	13.1.0
2015-09	69	SP-150449	0083	3	Partial File Handling	13.0.0	13.1.0
2015-12	70	SP-150656	0084	3	Event mechanism support for eDASH	13.1.0	13.2.0
2015-12	70	SP-150656	0085	1	Proposed Guidelines on DASH	13.1.0	13.2.0
2015-12	70	SP-150656	0087		DASH enhancements	13.1.0	13.2.0

Change history							
Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
2016-06	SA#72	SP-160262	0088	2	F	eDASH: $Time$ needs to go	13.3.0
2016-12	SA#74	SP-160773	0090	4	B	DASH Audio-video MOS support	14.0.0
2016-12	SA#74	SP-160945	0091	3	B	Location based filter and certain streaming program filter support for DASH	14.0.0
2017-03	SA#75	SP-170027	0093	1	C	Remove the HTTP metrics from the QoE reporting	14.1.0
2017-03	SA#75	SP-170027	0094	1	C	QoE Measurement Collection functionalities for QoE reporting	14.1.0
2017-06	SA#76	SP-170323	0095	-	F	Correction for QoE Session Configuration	14.2.0
2017-06	SA#76	SP-170323	0096	-	F	Correction for QoE Device Information	14.2.0
2017-06	SA#76	SP-170323	0097	-	F	Correction for QoE Geographical and Source Filtering	14.2.0
2017-06	SA#76	SP-170323	0098	1	F	Correction for QoE Measurement Collection	14.2.0
2017-09	SA#77	SP-170601	0100	2	F	Correction for Clause Quotation of the Quality Reporting Schemes	14.3.0
2017-09	SA#77	SP-170601	0106	-	F	QoE XML and OMA-DM corrections	14.3.0
2017-09	SA#77	SP-170601	0107	-	F	Clarification for the PlayList Metric	14.3.0
2017-09	SA#77	SP-170602	0099	3	B	SAND Support in 3GPP DASH	15.0.0
2017-12	SA#78	SP-170823	0112	2	F	SAND Network Assistance mode	15.1.0
2017-12	SA#78	SP-170828	0114	2	F	Clarifications for Description of Duration	15.1.0
2017-12	SA#78	SP-170828	0115	1	F	Clarifications for Measurement Unit of Duration	15.1.0
2017-12	SA#78	SP-170822	0119	-	A	Add missing QoE metric to annex K	15.1.0
2017-12	SA#78	SP-170821	0122	1	A	TV Service Support in DASH	15.1.0
2017-12	SA#78	SP-170823	0123	3	F	Corrections to SAND	15.1.0
2017-12	SA#78	SP-170821	0127	-	A	Correction to QoE Reporting	15.1.0
2018-03	SA#79	SP-180022	0129	-	F	Correction of QMC Configuration	15.2.0
2018-03	SA#79	SP-180022	0132	3	A	Correction on segment duration information	15.2.0
2018-03	SA#79	SP-180026	0133	-	F	Clarifications on QoE Reporting	15.2.0
2018-06	SA#80	SP-180266	0134	1	C	DASH-IF Alignment of SAND	15.3.0
2018-06	SA#80	SP-180264	0138	-	A	Correction for QoE StreamingSource filtering	15.3.0
2018-06	SA#80	SP-180272	0139	1	B	SAND support for MBMS and Multi-Network Modes	15.3.0
2018-09	SA#81	SP-180652	0155	-	F	Corrections to QMC signalling	15.4.0
2018-09	SA#81	SP-180657	0142	1	F	Clarification for rebuffer	16.0.0
2018-09	SA#81	SP-180655	0145	1	B	Signalling and Reporting of Interactivity Usage in 3GP-DASH	16.0.0
2018-09	SA#81	SP-180657	0149	2	F	Correction to the mapping of buffering event start	16.0.0
2018-12	SA#82	SP-180981	0156	2	F	Correction to the description of Mobile	16.1.0
2018-12	SA#82	SP-180981	0159	1	F	Corrections for attribute type	16.1.0
2018-12	SA#82	SP-180981	0160	-	F	Quotation correction	16.1.0
2018-12	SA#82	SP-180968	0162	-	A	SAND Network Assistance mode clarifications	16.1.0
2019-12	SA#86	SP-190997	0163	1	B	Support for Service Interactivity via Event Signalling and DASH APIs	16.2.0
2020-06	SA#88-e	SP-200390	0165	2	F	Correction on Partial File Handling	16.3.0
2020-06	SA#88-e	SP-200595	0166	-	C	QoE Measurement Collection	16.3.0
2020-09	SA#89-e	SP-200668	0167		B	DASH Profile for CMAF	16.4.0
2020-10	Post SA#89-e					Editorial: Update of the Change History Table	16.4.1
2021-09	SA#93-e	SP-211026	0168	4	F	QoE configuration release	16.5.0
2021-12	SA#93-e					ToC Update	16.5.1
2022-03	SA#95-e	SP-220033	0169	1	F	Add support of per-slice QoE measurement	17.0.0
2022-06	SA#96	SP-220596	0172		F	Registration of URNs	17.1.0
2022-06	SA#96	SP-220600	0173	1	F	Add DNN/Slice in the QoE report schema	17.1.0
2022-06	SA#96	SP-220600	0174	1	F	Support of NR QoE features	17.1.0
2022-06	SA#96	SP-220602	0175		F	Requirement on UE Behaviour regarding QoE Measurement and Reporting	17.1.0
2022-12	SA#98-e	SP-221054	0176	2	F	CR to TS 26.247 Add slice scope into the metrics configuration	17.2.0
2023-03	SA#99	SP-230249	0178	-	F	Corrections to DASH quality metric and QoE configuration and reporting	17.3.0
2023-03	SA#99	SP-230262	0179	1	F	Corrections to RAN visible QoE configuration and reporting	17.3.0
2023-09	SA#101	CP-230912	0182	3	A	Removal of HTTPList from 3GPP Metrics	17.4.0
2023-09						Editorial corrections.	17.4.1

2024-03	SA#103	SP-240040	0184		C	[5GMS_Pro_Ph2] Carriage of media delivery session ID in QoE report	18.0.0
2024-06	SA#104	SP-240689	0186	1	F	HTTP Reference Uplift	18.1.0
2024-06	SA#104	SP-240908	0185	3	F	Additional QM10 reporting schema fields for alignment	18.1.0
2024-09	SA#105	SP-241112	0187	1	F	[iQoE, TEI18] Declaration of schema version compliance	18.2.0
2025-01	SA#106	SP-241751	0189	2	F	QMC support for application session delivered via the MBS	18.3.0
2025-04						Fixed editorial corruption of XML listings	18.3.1
2025-06	SA#108	SP-250699	0192	4	F	Correction to MBS communication service type for QMC	18.4.0
2025-10	-	-	-	-	-	Update to Rel-19 version (MCC)	**19.0.0**

History

Document history		
V19.0.0	November 2025	Publication